

A Prelude to Practical Quantum Computing

A brief overview

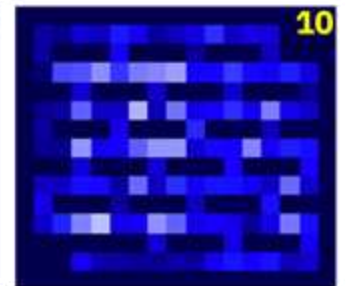
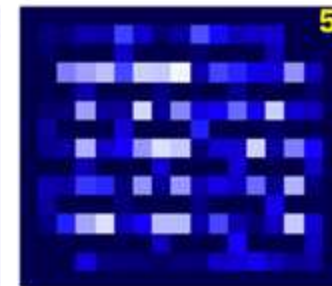
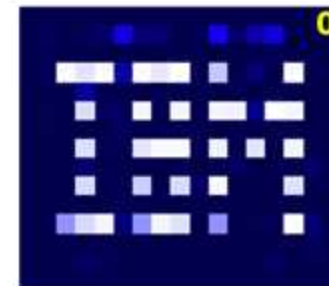
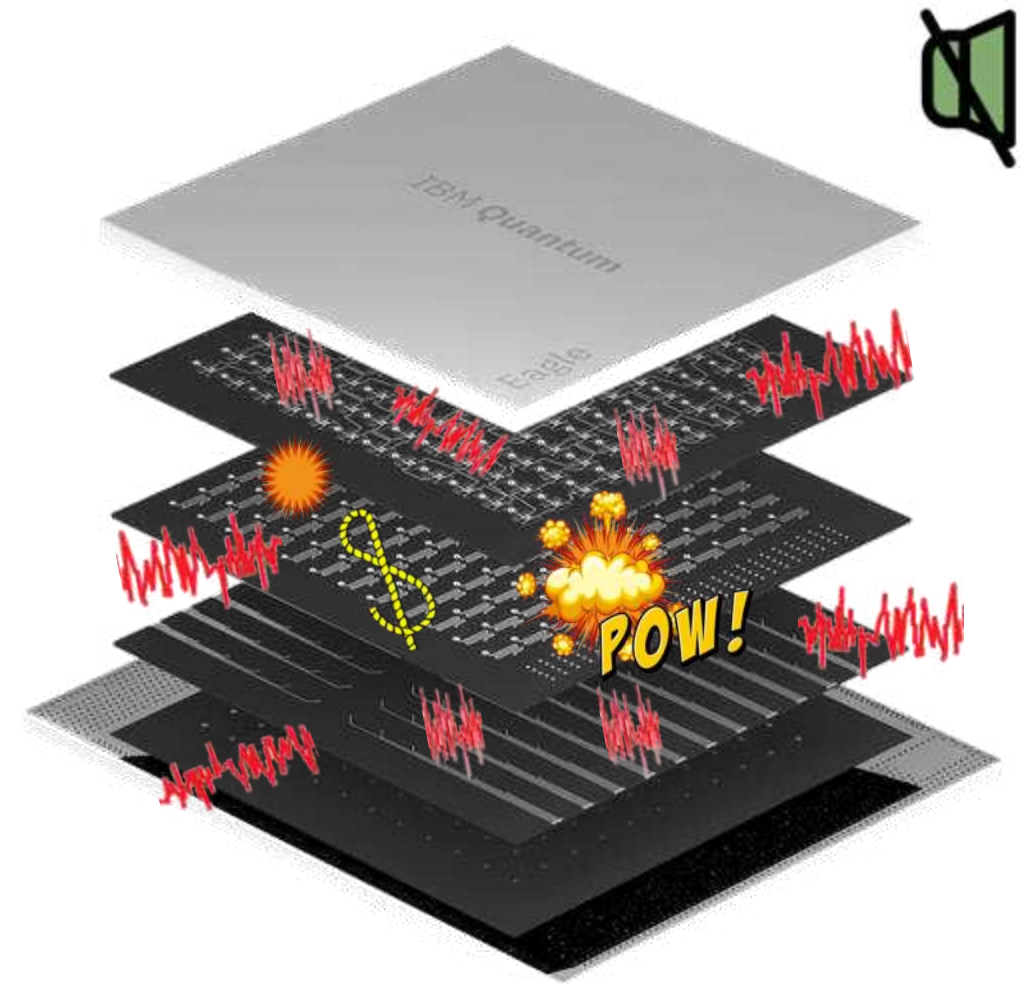
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Have you used
a quantum computer?



A Prelude to Practical Quantum Computing

I. Big picture today?

Quantum Computing Status

II. What is a qubit?

Physics & Hardware

III. Let's run a real quantum algorithm

Design, Noise & Mitigation

Slides from lecture

Posted here
Tutorials

zlatko-minev.com/education
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IBM Weekly Seminar

[Qiskit YouTube](#)



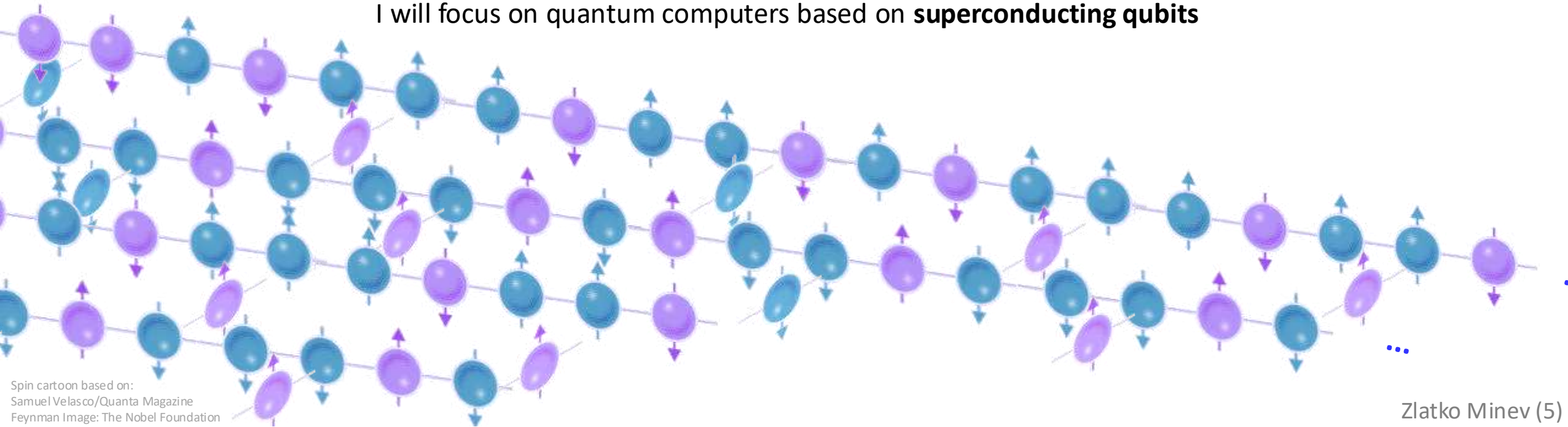
Part I

What is the big picture today?

How is it going for quantum computers?

The last 40 years and looking ahead to 10 more
1980 – . . . – 2010 – 2020 – 2023 in 60 seconds

I will focus on quantum computers based on **superconducting qubits**





“Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy.”

- R.P. Feynman 1981

International Journal of Theoretical Physics, Vol. 21, Nos. 6/7, 1982

Simulating Physics with Computers

Richard P. Feynman

Department of Physics, California Institute of Technology, Pasadena, California 91107

Received May 7, 1981

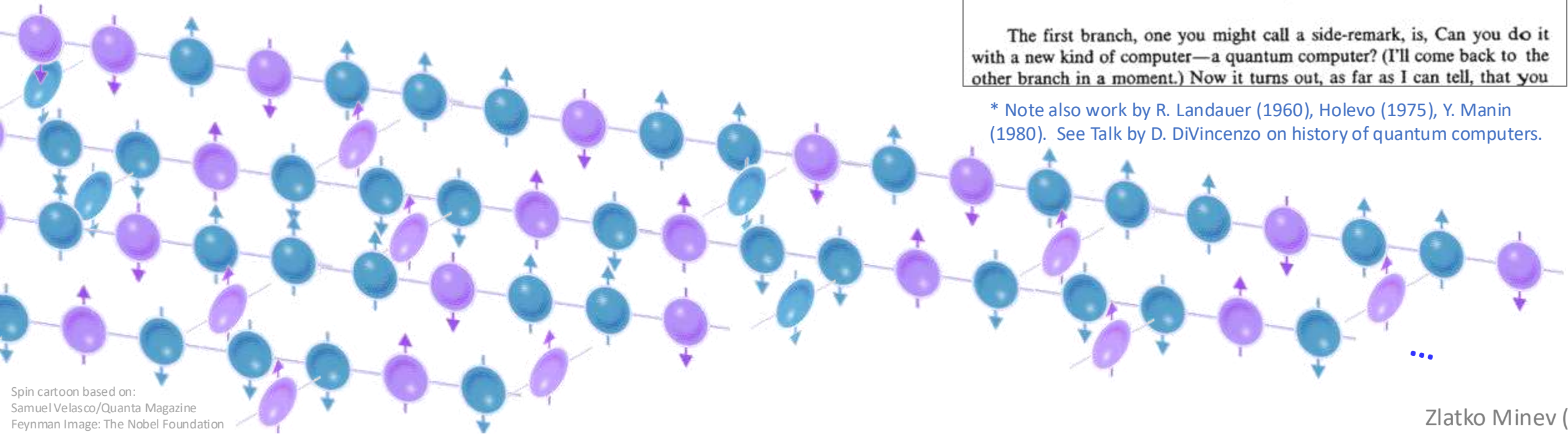
1. INTRODUCTION

On the program it says this is a keynote speech—and I don't know that we thought of before—a logical, universal automaton, can we imitate this situation? And I'm going to separate my talk here, for it branches into two parts.

4. QUANTUM COMPUTERS—UNIVERSAL QUANTUM SIMULATORS*

The first branch, one you might call a side-remark, is, Can you do it with a new kind of computer—a quantum computer? (I'll come back to the other branch in a moment.) Now it turns out, as far as I can tell, that you

* Note also work by R. Landauer (1960), Holevo (1975), Y. Manin (1980). See Talk by D. DiVincenzo on history of quantum computers.



My experience circa 2010

Maybe 1 or 2 qubits
working some small
fraction of the time
in select labs

Photo with dilution fridge called
Sunshine from Michel Devoret's
lab at Yale during my Ph.D.



Hopes for a
working
qubit in
here

Today

A 127-qubit quantum computer installed in the the lobby cafeteria of a research building dutifully executing jobs almost all the time.



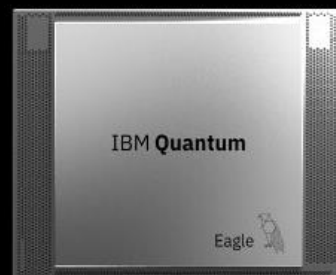




2019
Falcon
27 Qubits



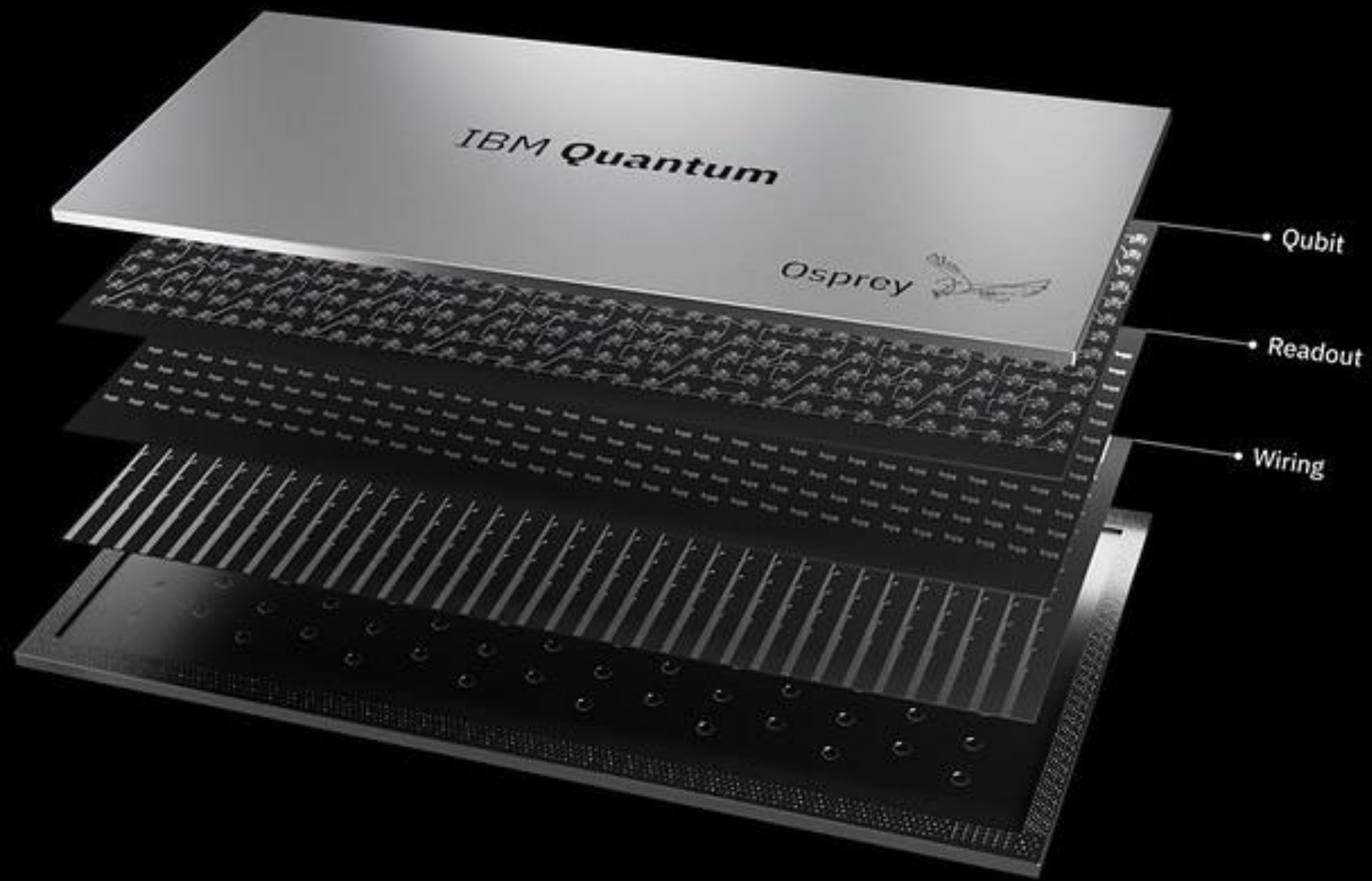
2020
Hummingbird
65 Qubits



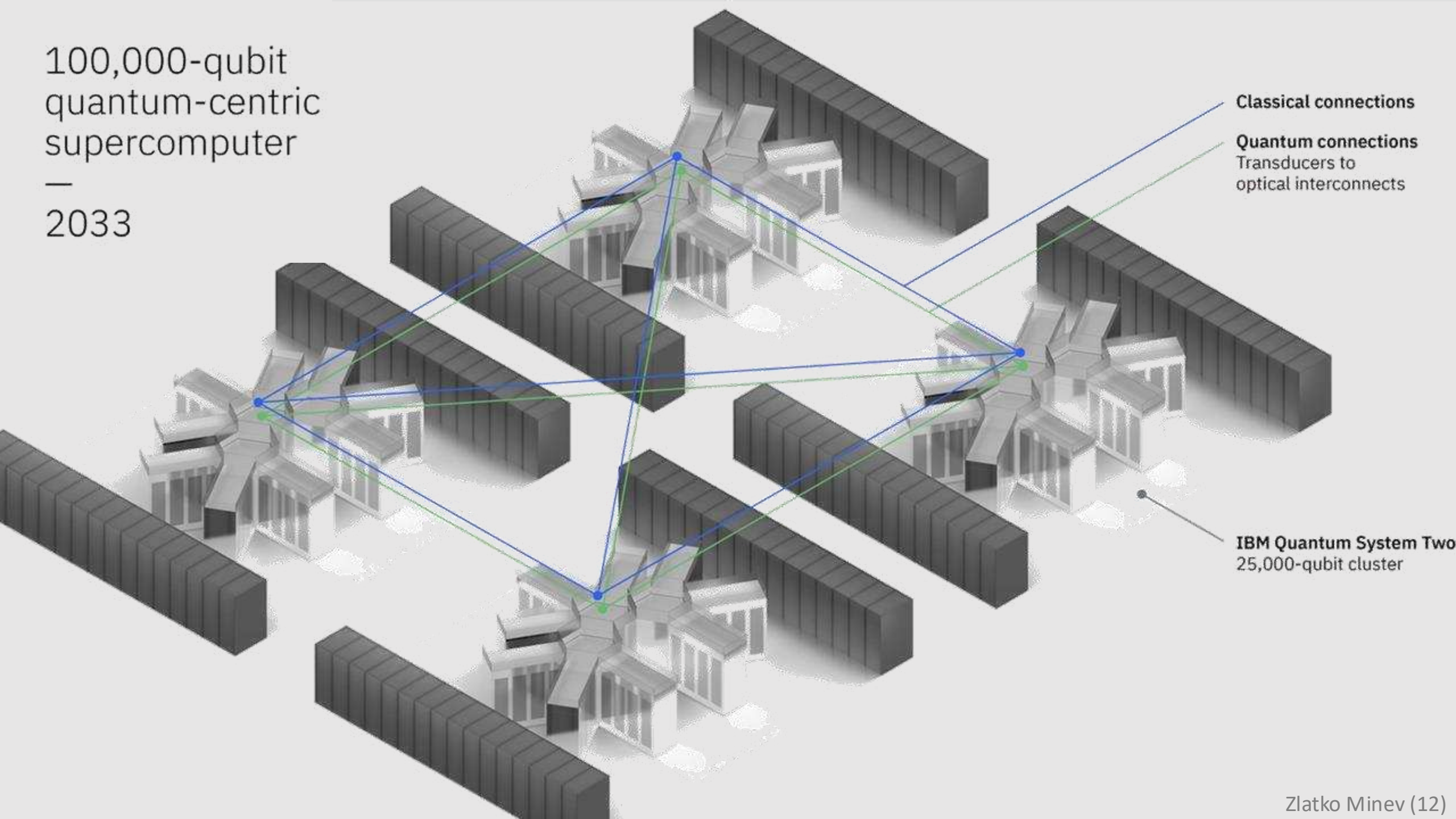
2021
Eagle
127 Qubits



2022
Osprey
433 Qubits



100,000-qubit
quantum-centric
supercomputer
—
2033

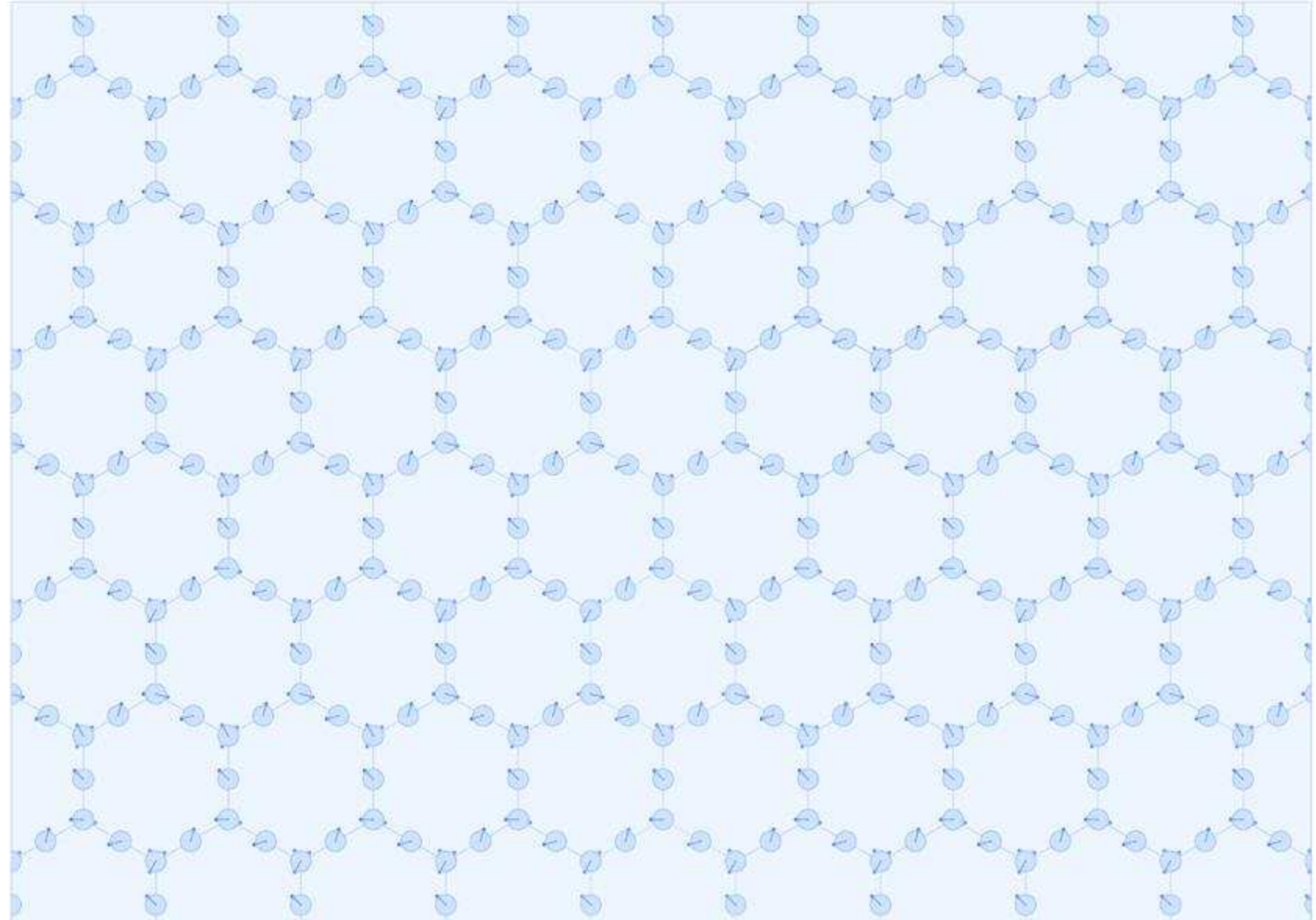


Back to today ...

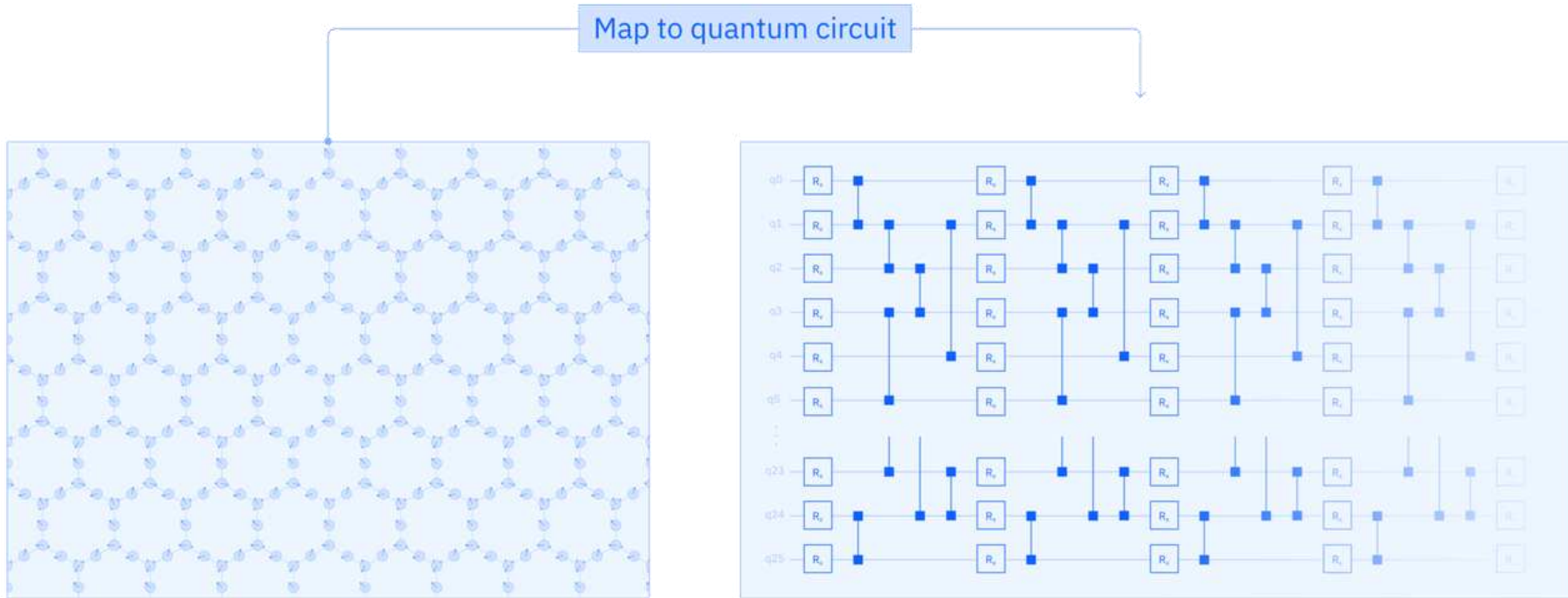
Example: Quantum simulation

Example task:

Simulate the out-of-equilibrium quantum dynamics of a 2D spin chain lattice to find the evolution of the global and local magnetization.

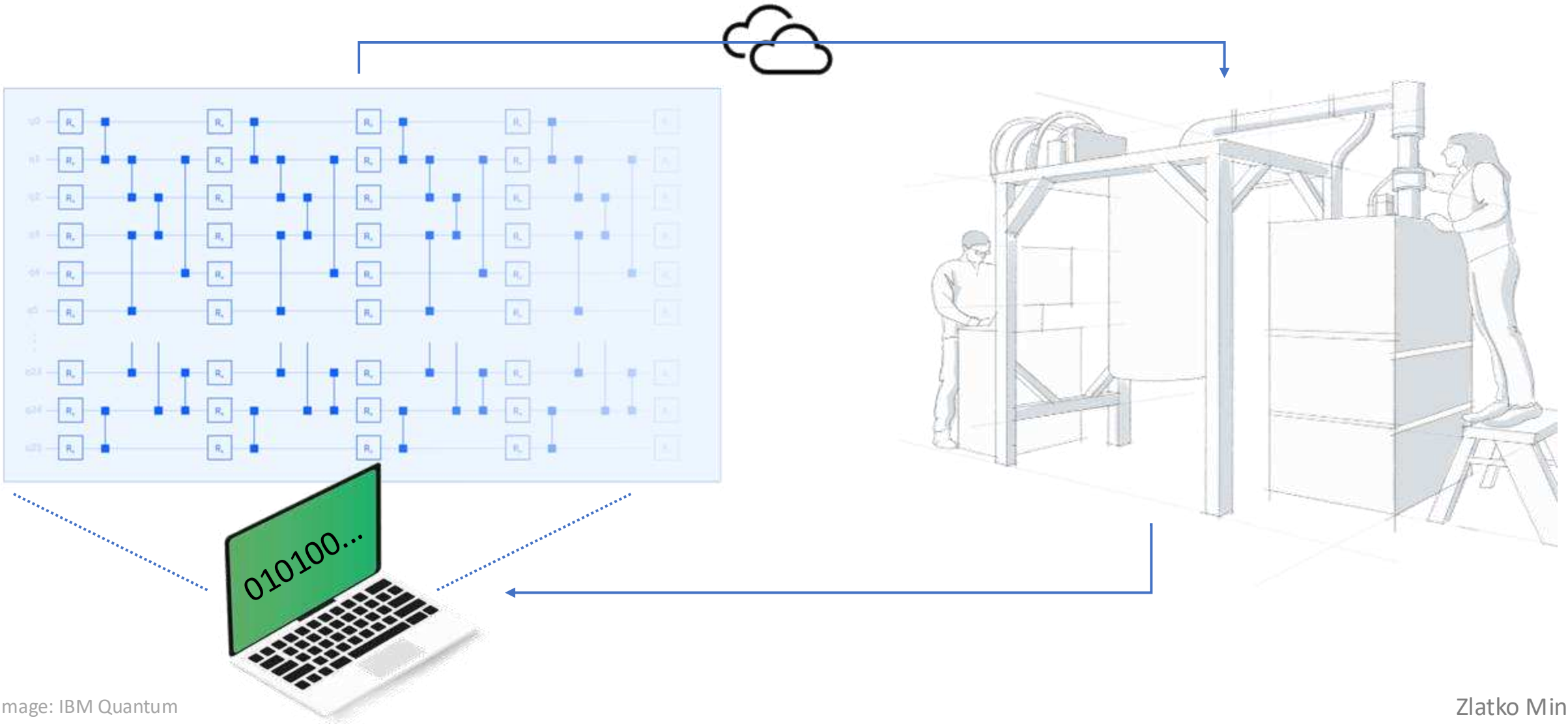


Quantum simulation on a quantum computer

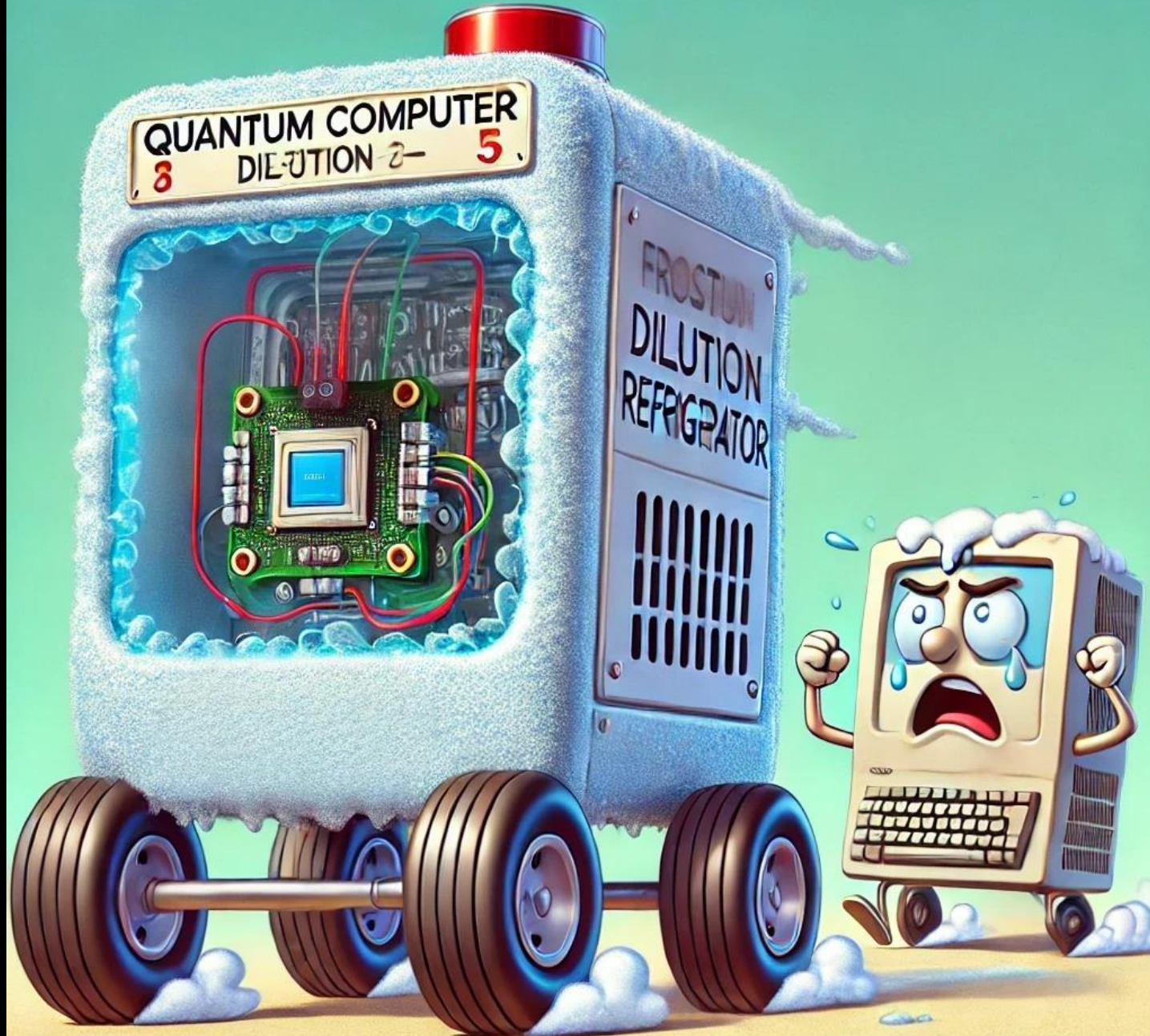


Quantum simulation on a quantum computer

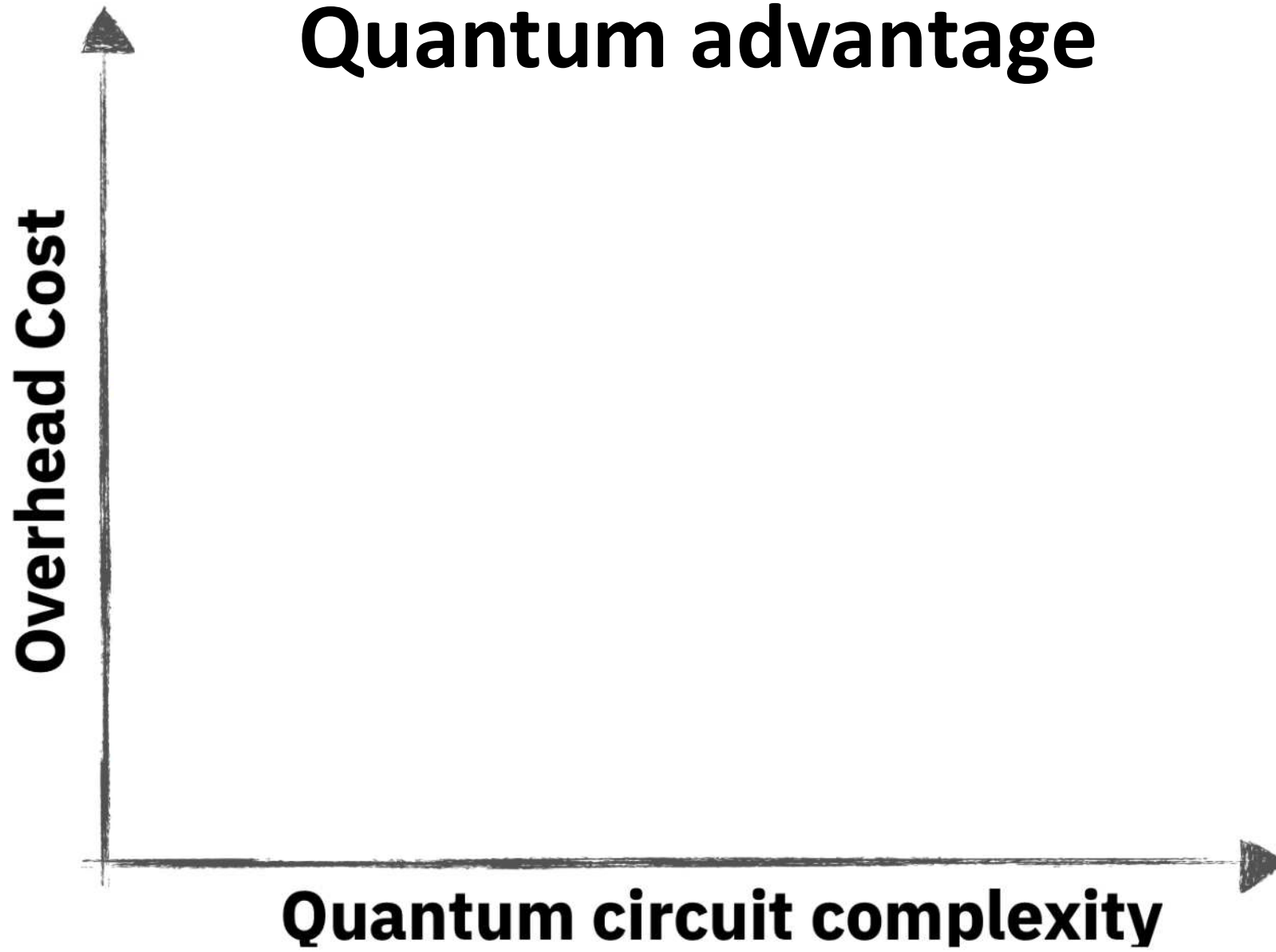
Execute on a real quantum computer device and obtain results as classical data

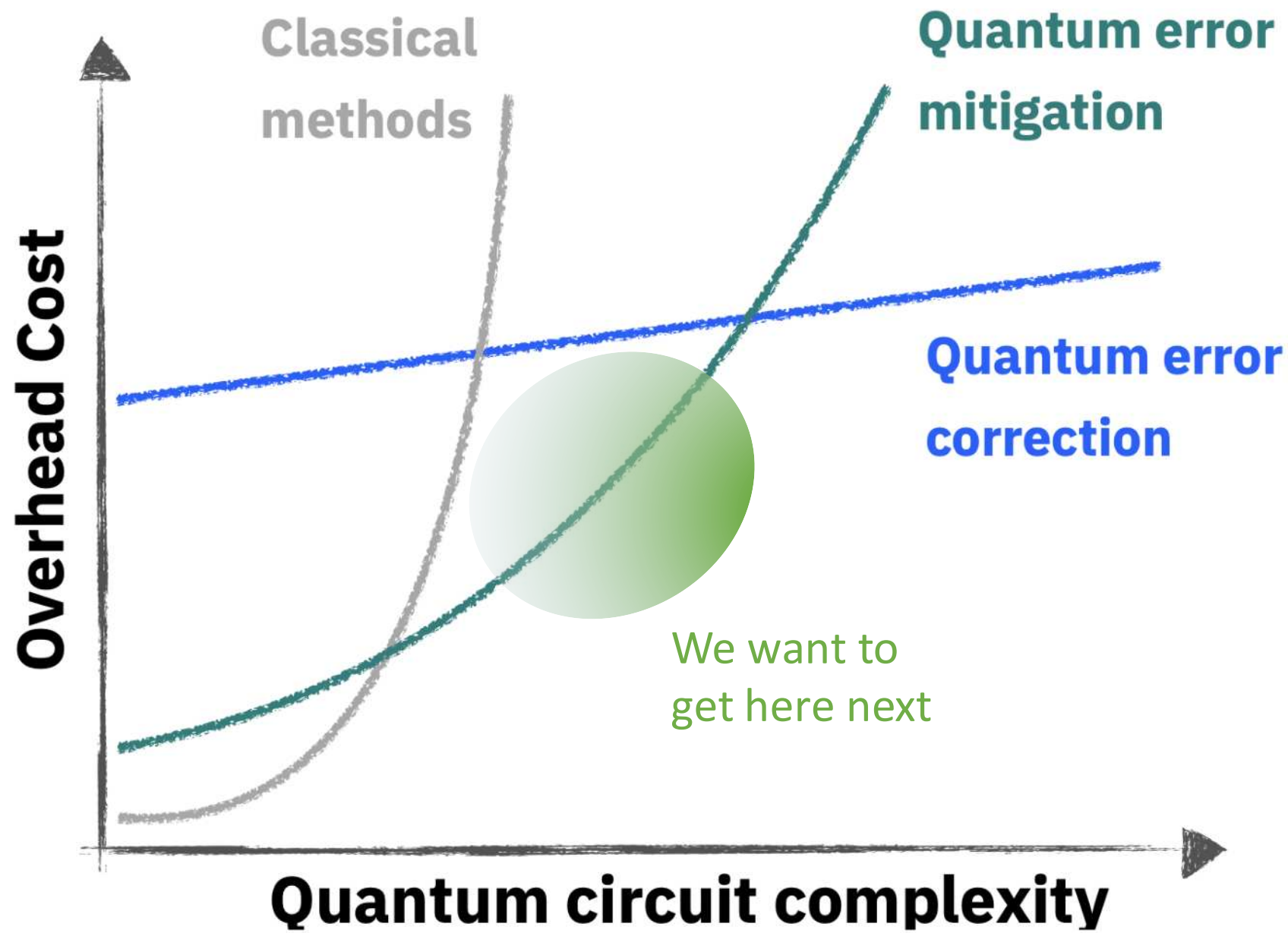


Quantum advantage

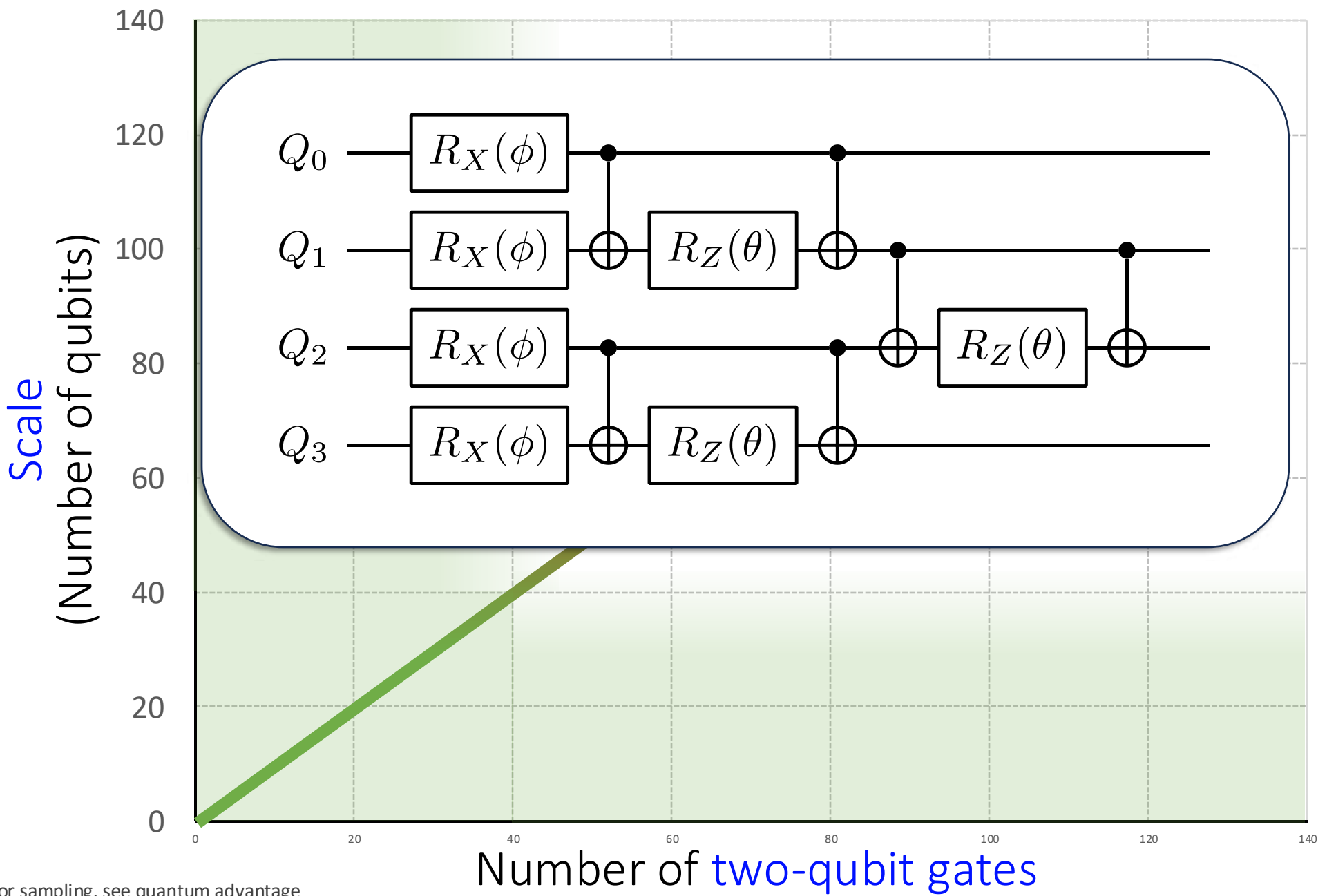


Quantum advantage

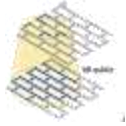




Landscape of quantum circuits



Some early utility-scale experiments



[0] Kim, Eddins, ..., Temme, Kandala.
Nature 618, 500–505 (2023)



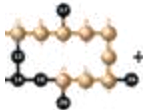
[1] Yu, Zhao, Wei.
arXiv: 2207.09994 (2022)



[2] Shtanko, Wang, Zhang, Harle, Seif,
Movassagh, Mineev.
arXiv: 2307.07552 (2023)



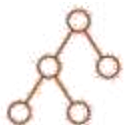
[3] Farrell, Illa, Ciavarella, Savage.
arXiv: 2308.04481 (2023)



[4] Bäumer, Tripathi, Seif,
Minev.
arXiv: 2308.13065 (2023)



[5] Chen, Zhu, Verresen, Seif,
Baümer, ... Trebst, Kandala.
arXiv: 2309.02863 (2023)

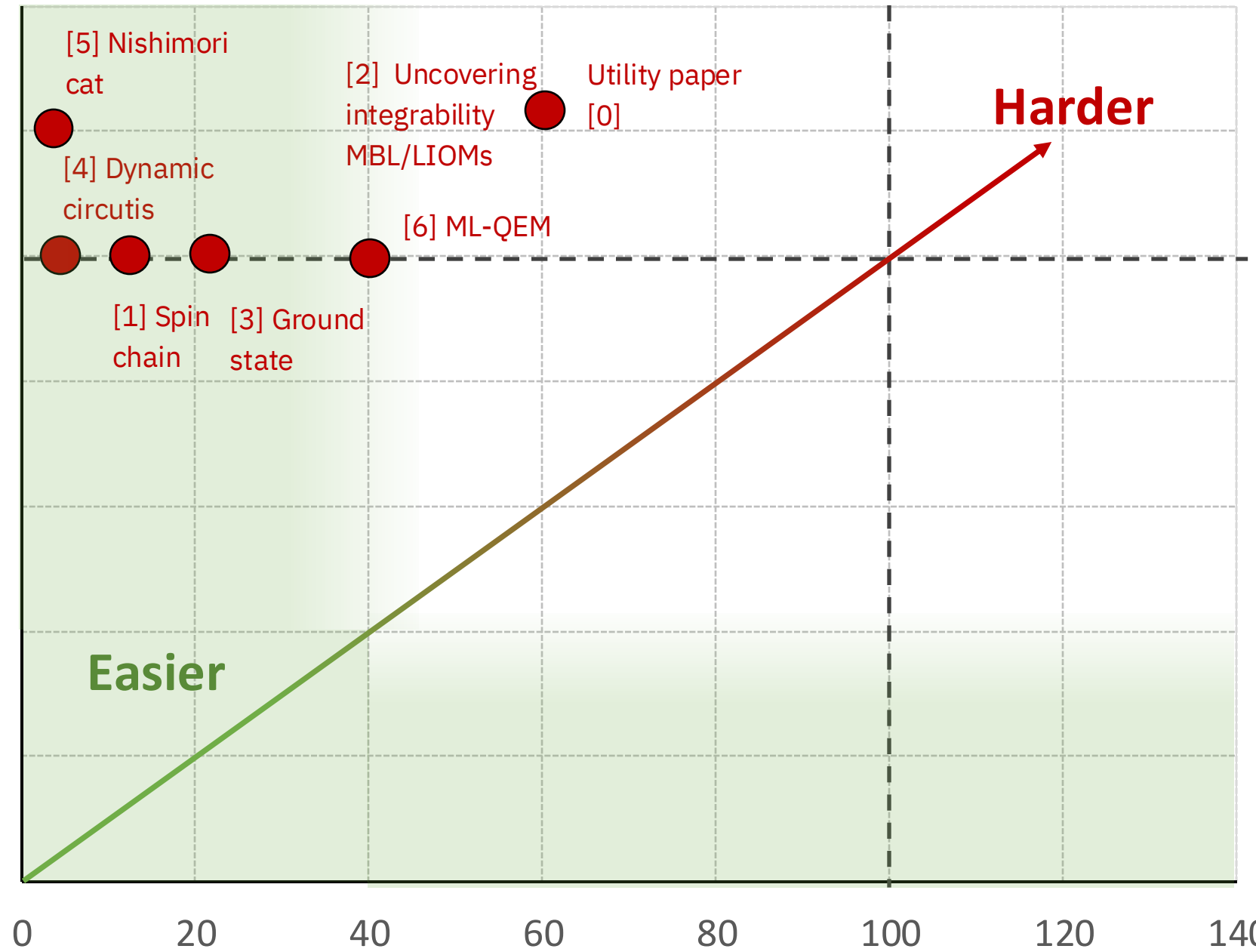


[6] Liao, Wang, Sitdikov, Salcedo,
Seif, Minev.
arXiv: 2308.13065 (2023)

... see next slide

Scale
(Number of qubits)

140
120
100
80
60
40
20
0



Number of two-qubit gate layers (depth)

Zlatko Minev
IBM Quantum

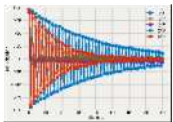
On superconducting qubit platforms
Signal stops or decays more than 50% beyond this depth

Utility-scale demos: More recent examples



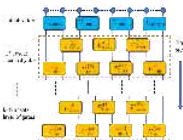
Evidence for the utility of quantum computing before fault tolerance
Nature, 618, 500 (2023)
127 qubits / 2880 CX gates

simulation



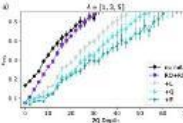
Characterizing quantum processors using discrete time crystals
arXiv:2301.07625
80 qubits / 7900 CX gates

simulation



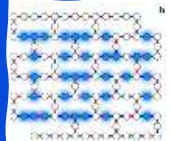
Simulating large-size quantum spin chains on cloud-based superconducting quantum computers
Phys. Rev. Research 5, 013183 (2023)
102 qubits / 3186 CX gates

simulation



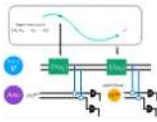
Best practices for quantum error mitigation with digital zero-noise extrapolation
arXiv:2307.05203
104 qubits / 3605 ECR gates

tools



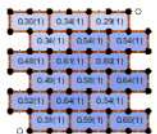
Uncovering Local Integrability in Quantum Many-Body Dynamics
arXiv:2307.07552
124 qubits / 2641 CX gates

simulation



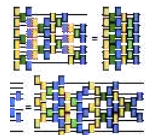
Quantum reservoir computing with repeated Measurements on superconducting devices
arXiv:2310.06706
120 qubits / 49470 ECR gates + meas.

QML



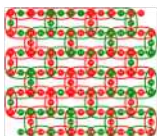
Realizing the Nishimori transition across the error threshold for constant-depth quantum circuits
arXiv:2309.02863
125 qubits / 429 gates + meas.

simulation



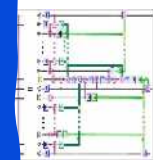
Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits
PRX Quantum 5, 020315 (2024)
100 qubits / 788 CX gates

simulation



Scaling Whole-Chip QAOA for Higher-Order Ising Spin Glass Models on Heavy-Hex Graphs
arXiv:2312.00997
127 qubits / 420 CX gates

optimization



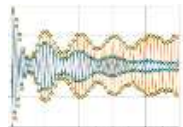
Efficient Long-Range Entanglement using Dynamic Circuits
arXiv:2308.13065
101 qubits / 504 ECR gates + meas

tools



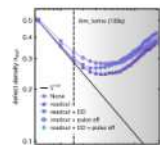
Quantum Simulations of Hadron Dynamics in the Schwinger Model using 112 Qubits
arXiv:2401.08044
112 qubits / 13,858 CZ gates

simulation



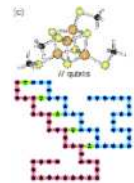
Unveiling clean two-dimensional discrete time quasicrystals on a digital quantum computer
arXiv:2403.16718
133 qubits / 15,000 CZ gates

simulation



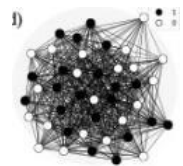
Benchmarking digital quantum simulations and optimization above hundreds of qubits using quantum critical dynamics
arXiv:2404.08053
133 qubits / 1440 CX gates

simulation



Chemistry Beyond Exact Solutions on a Quantum-Centric Supercomputer
arXiv:2405.05068
77 qubits / 3590 CZ gates

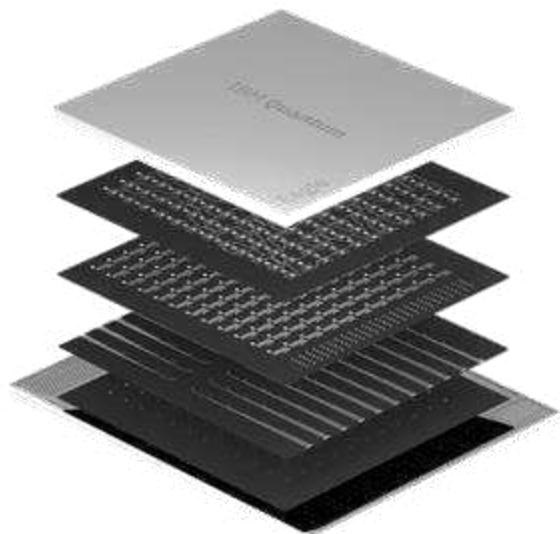
simulation



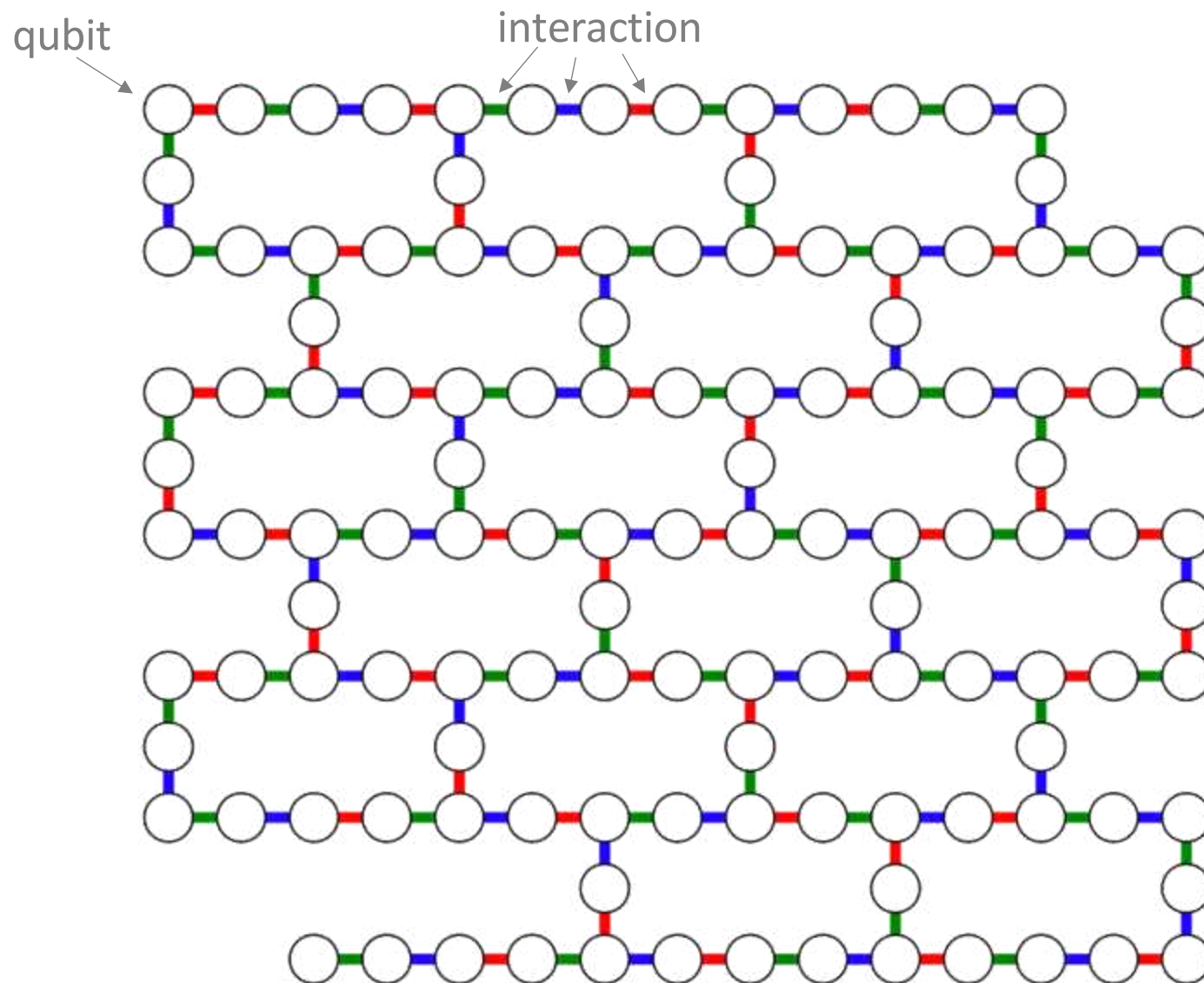
Towards a universal QAOA protocol: Evidence of quantum advantage in solving combinatorial optimization problems
arXiv:2405.09169
109 qubits / 21,200 ECR gates

optimization

Many-body quantum dynamics at utility scale

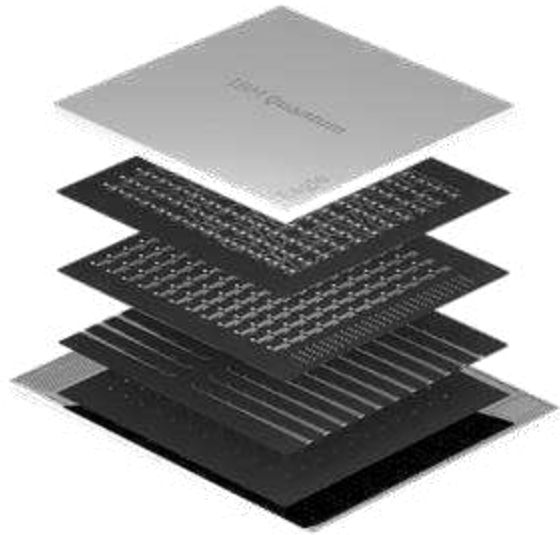


Number of qubits	124Q
Connectivity	2D h-hex
Depth in cX gates	60
Floquet steps	20
Total number of cXs	2,641

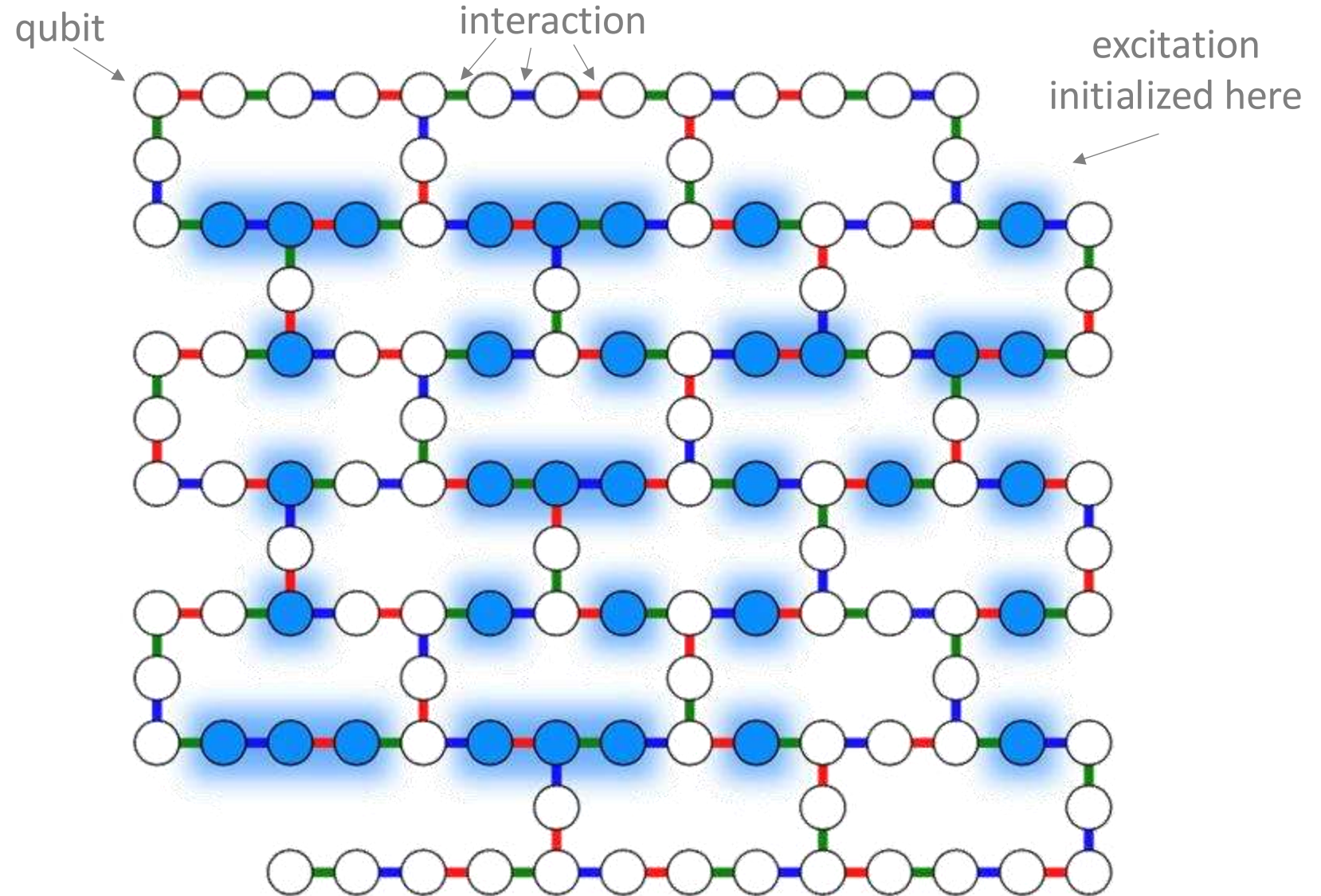


Uncovering Local Integrability in Quantum Many-Body Dynamics

Initialize lattice in fun states

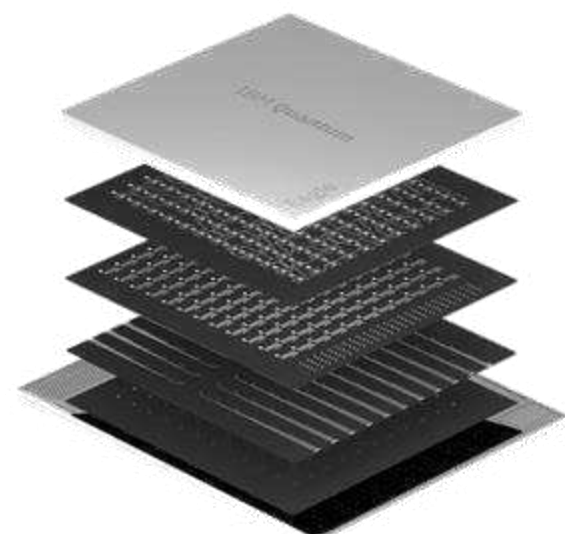


Number of qubits	124Q
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Total number of cXs	2,641

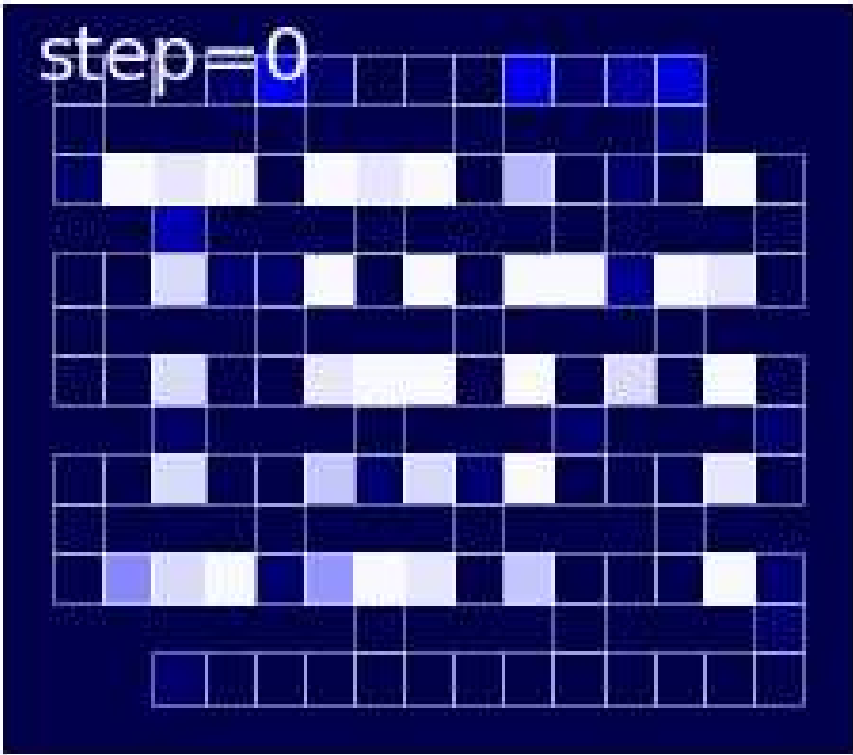


Color represent spin polarization

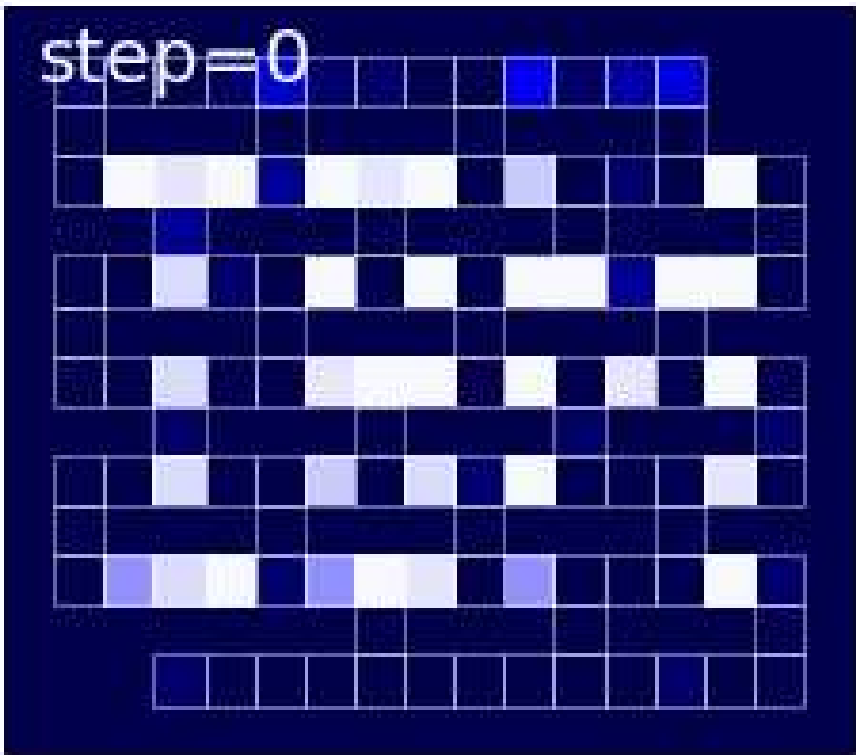
Quantum dynamics in different regimes



Number of qubits	124Q
Connectivity	2D h-hex
Depth in cX gates	60
Floquet steps	20
Total number of cXs	2,641

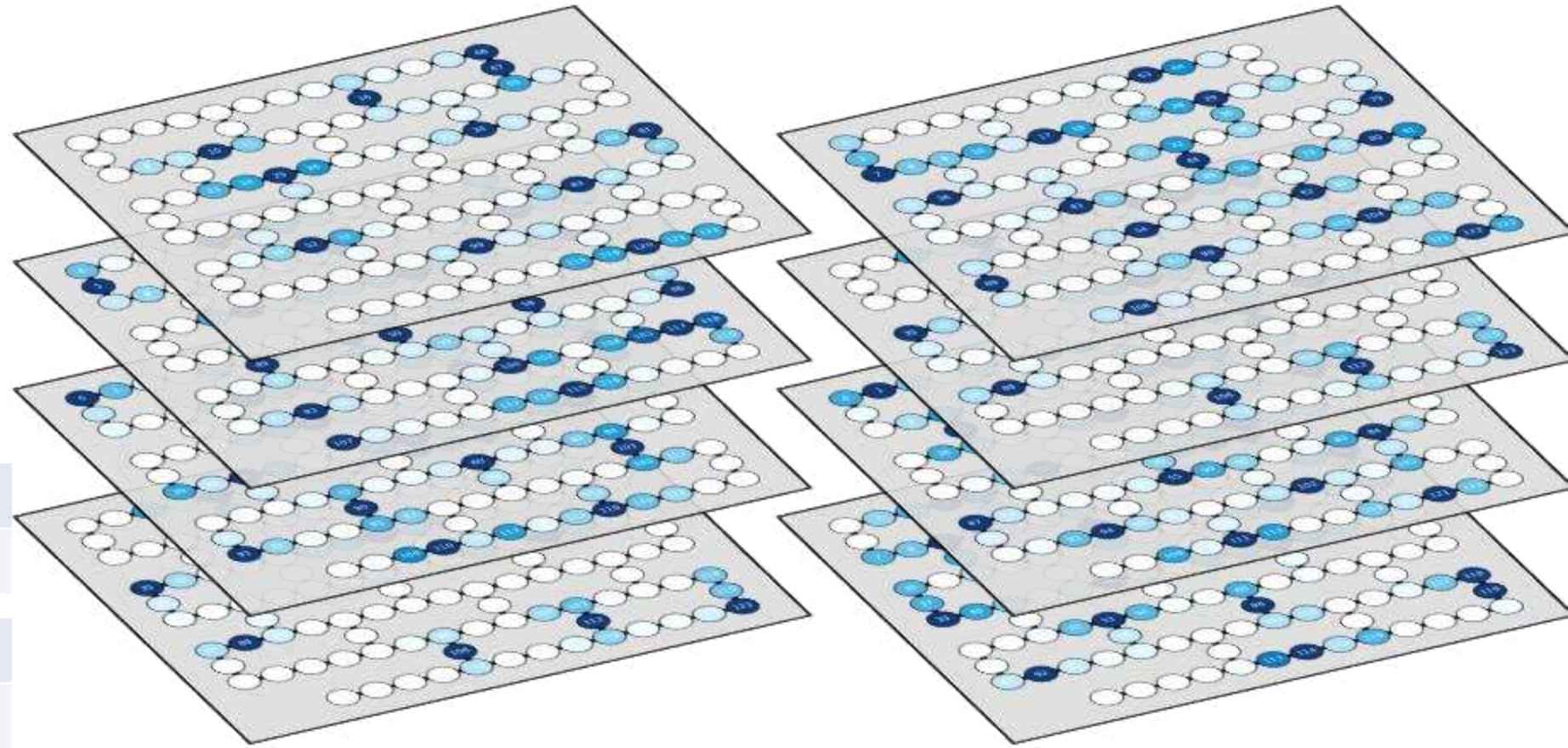
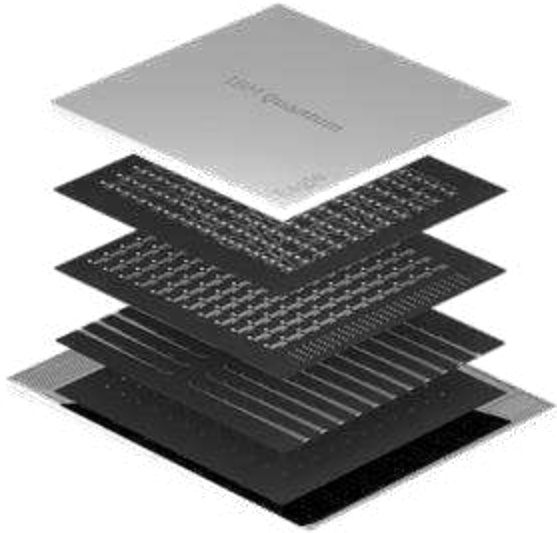


Thermalizing regime



Prethermal regime

All 124 local integrals of motion (LIOMs) extracted



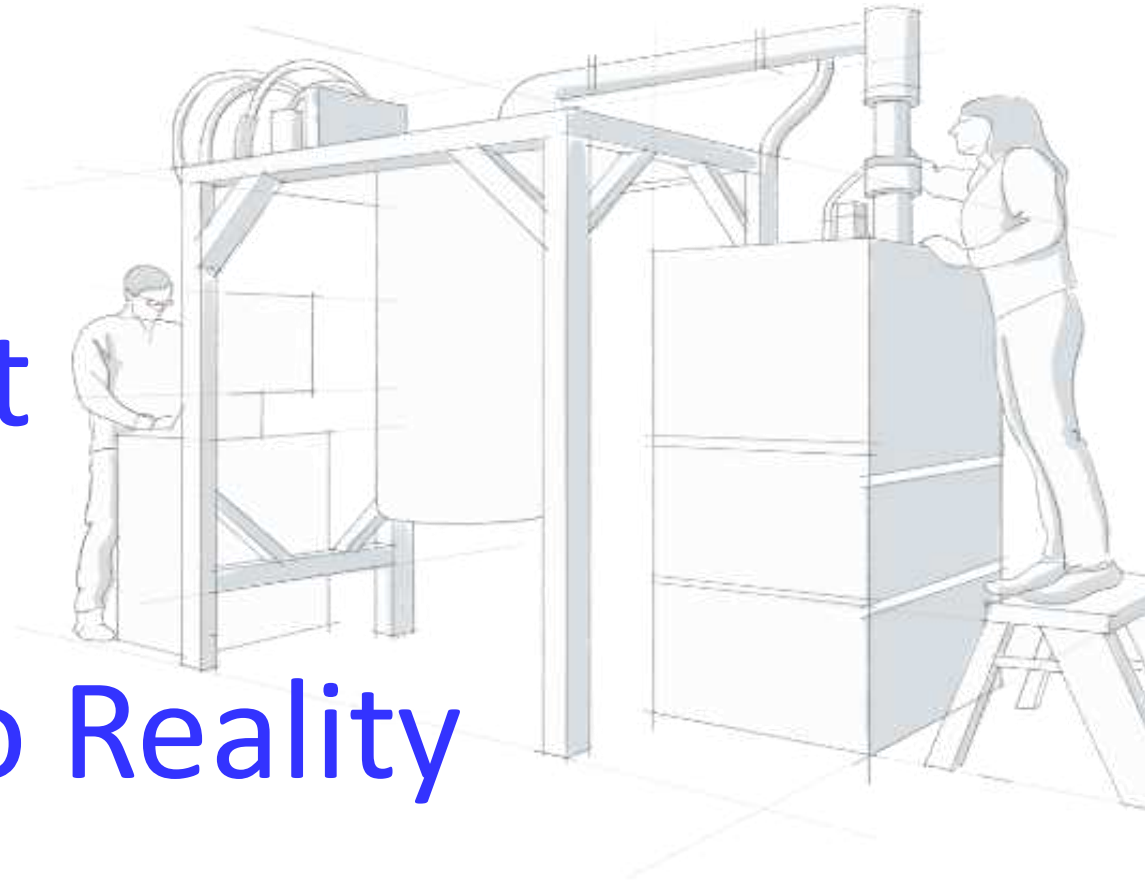
Number of qubits	124Q
Connectivity	2D h-hex
Depth in cX gates	60
Floquet steps	20
Total number of cXs	2,641

Part II

What is a qubit?

Qubit

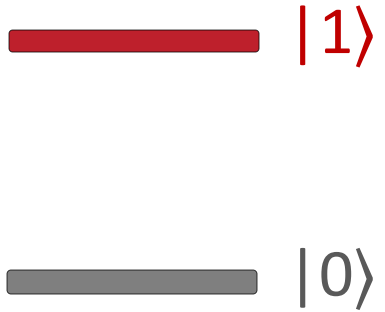
From Idea to Reality



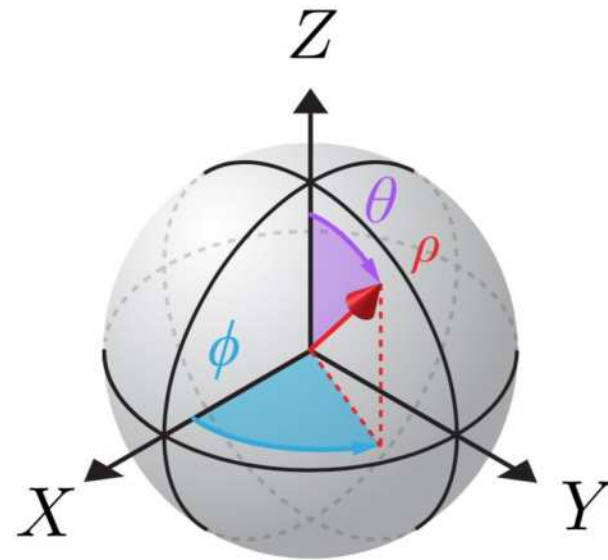
THE BIG PICTURE
before calculations

Qubit: idea

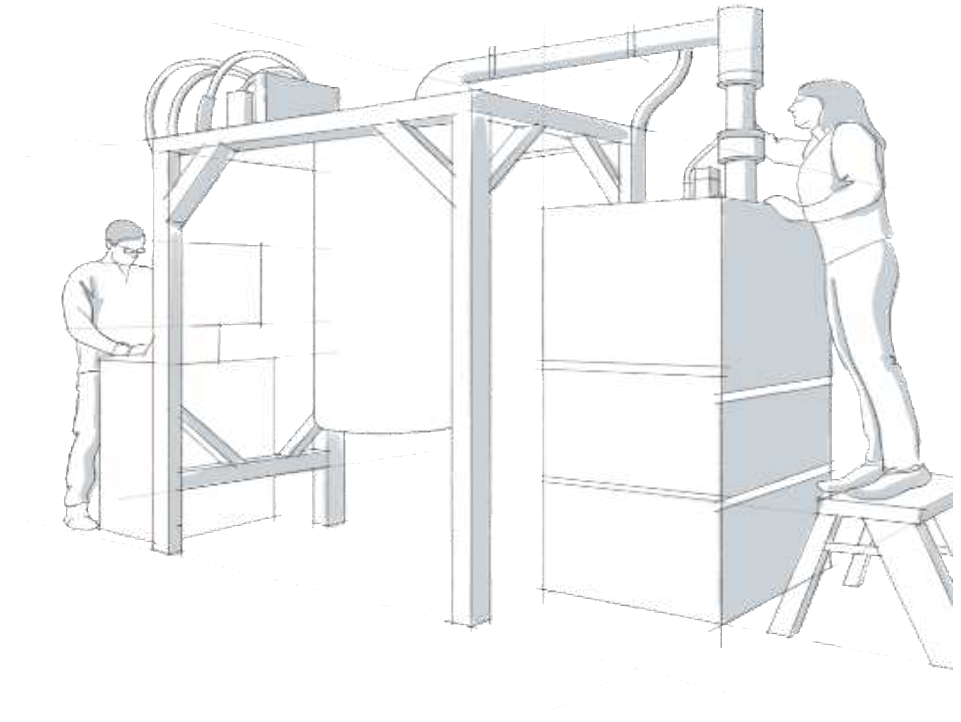
Energy levels



Hilbert space

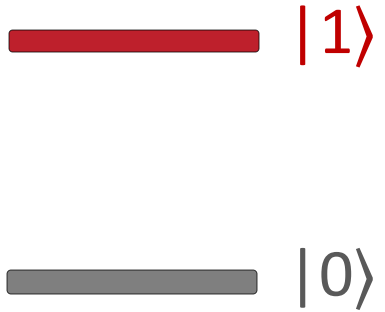


Quantum cloud?

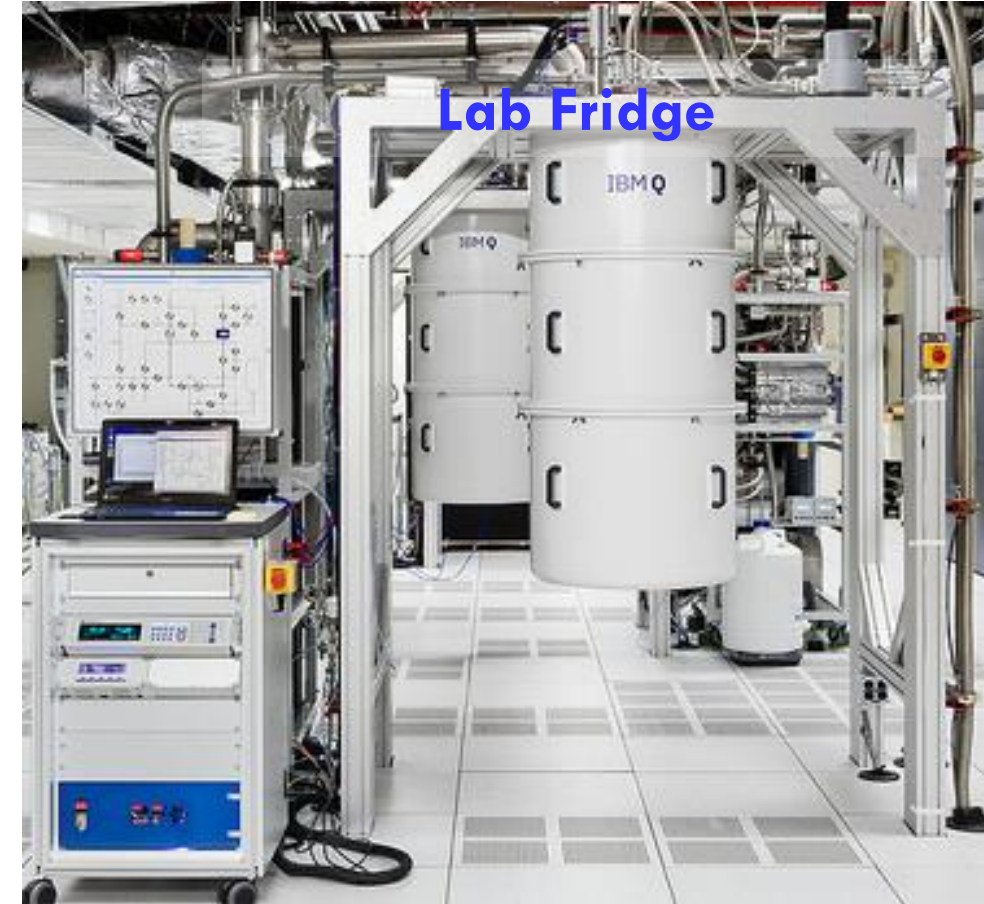
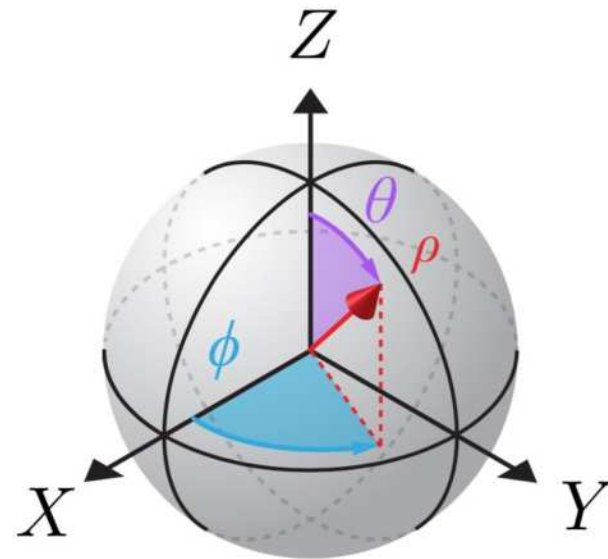


Qubit: idea and reality

Energy levels



Hilbert space



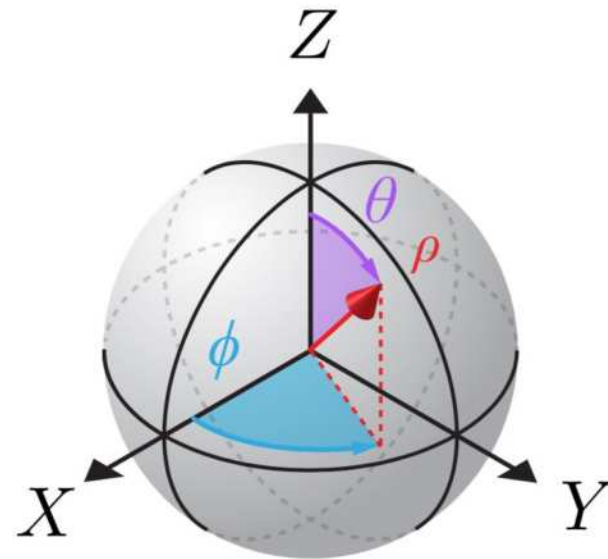
Qubit: idea and reality

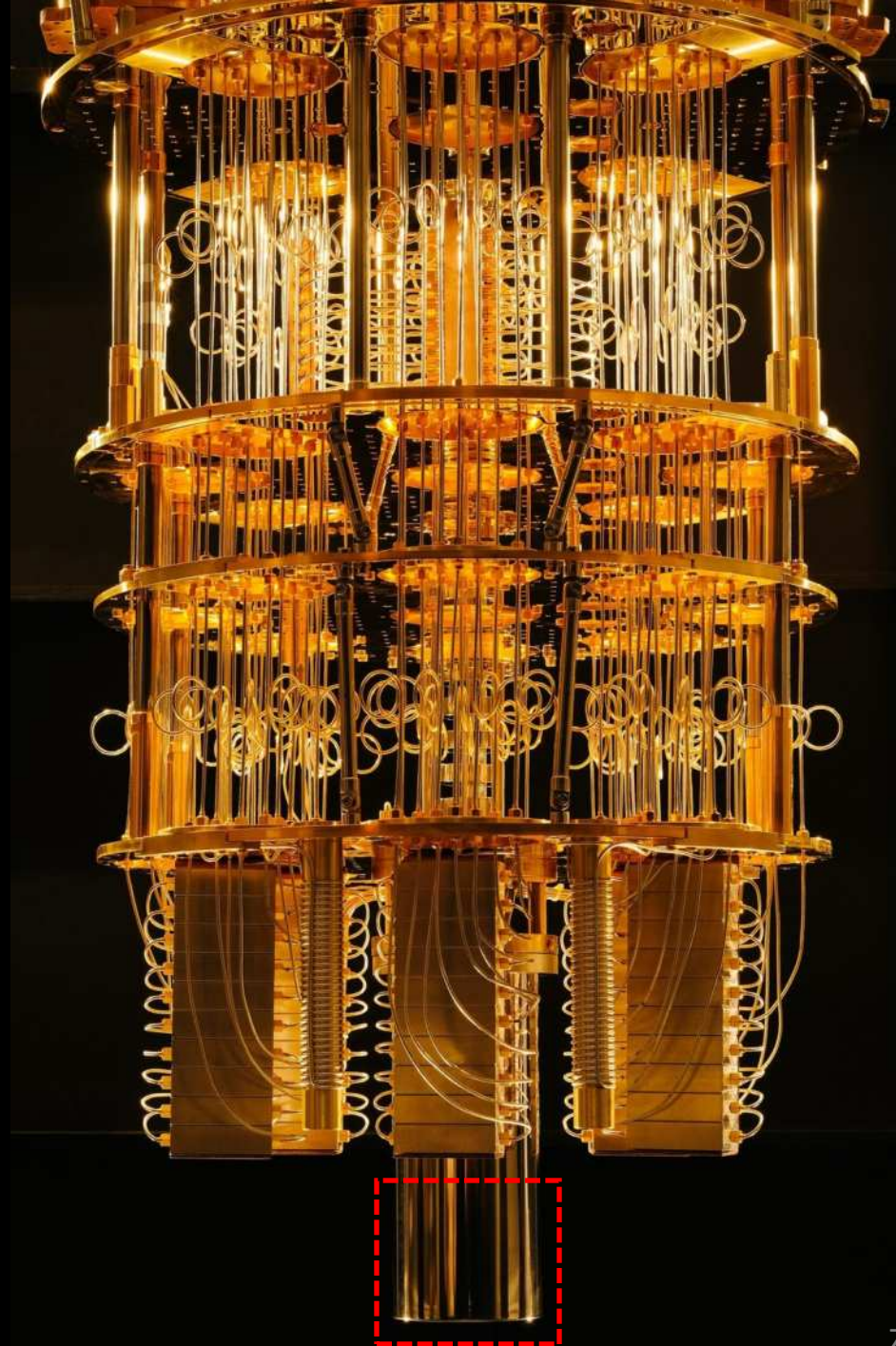
Energy levels

— $|1\rangle$

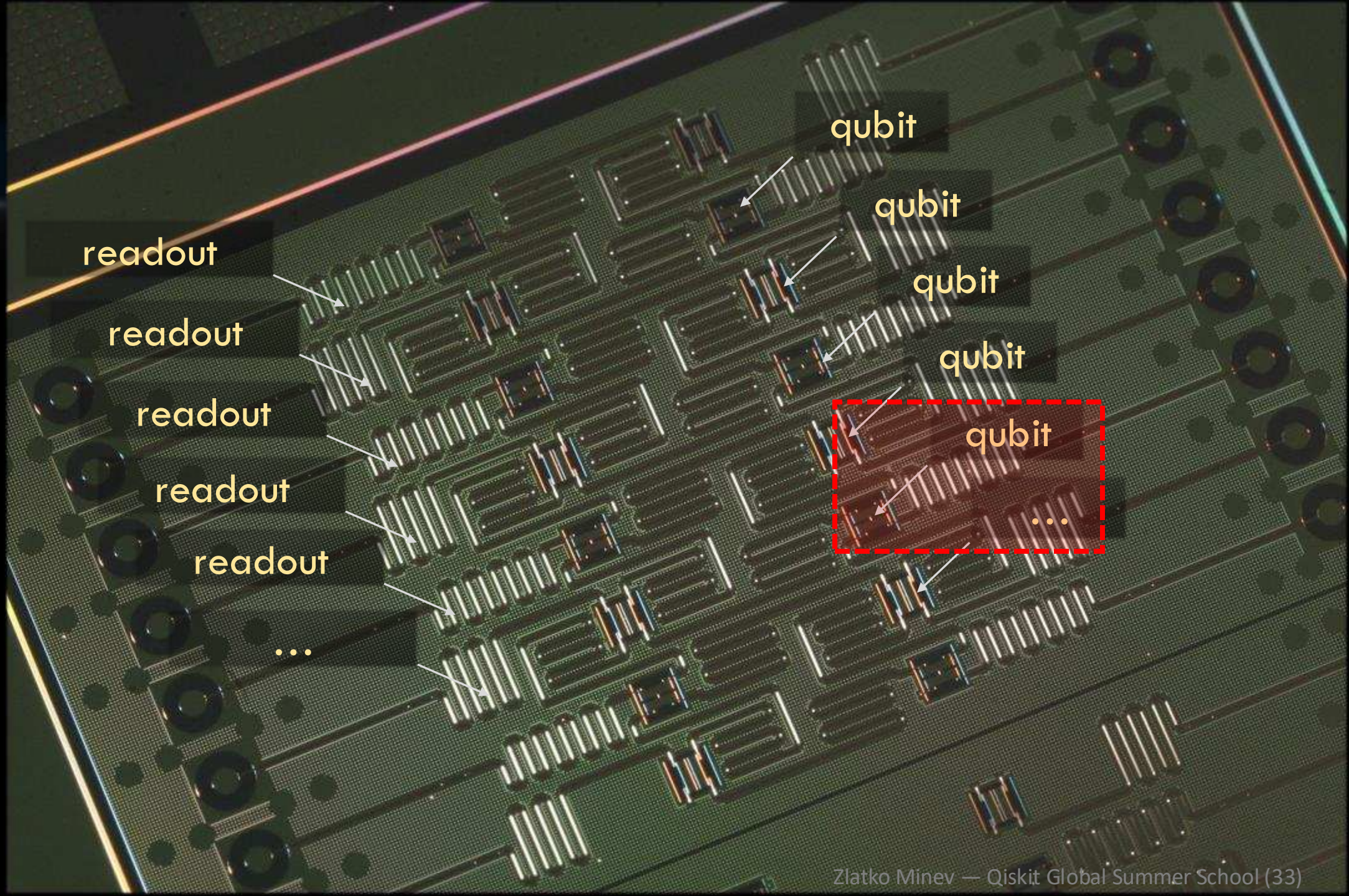
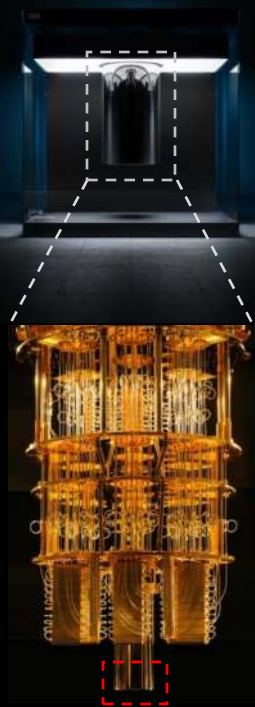
— $|0\rangle$

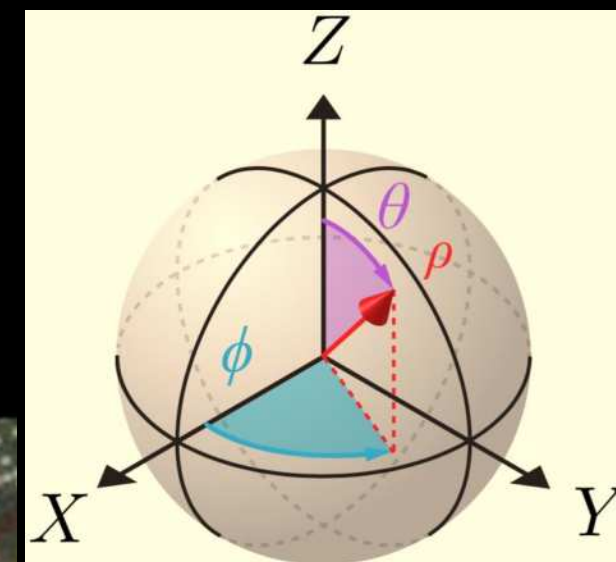
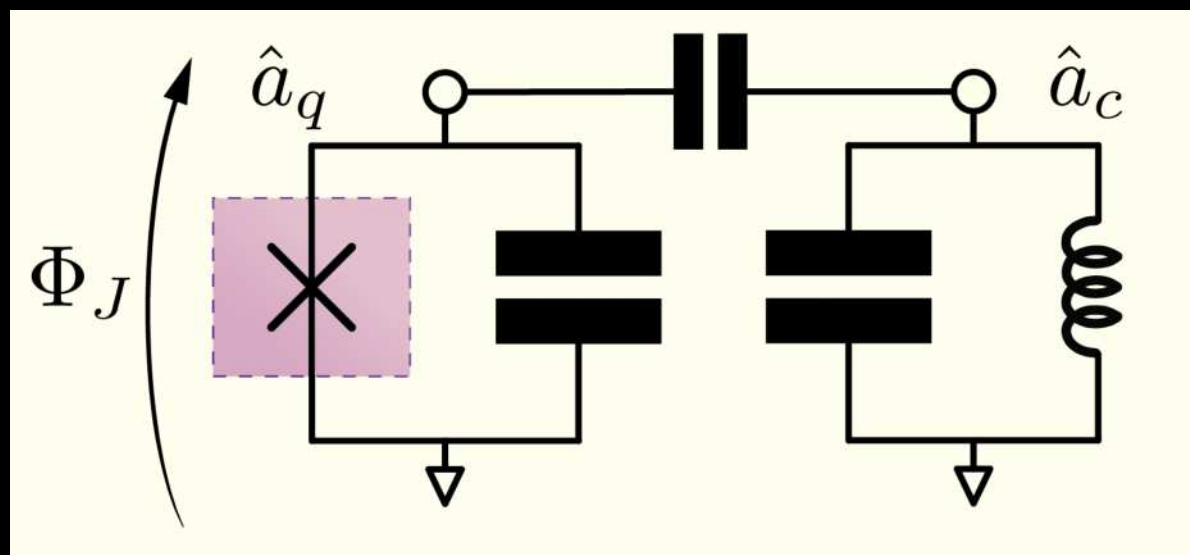
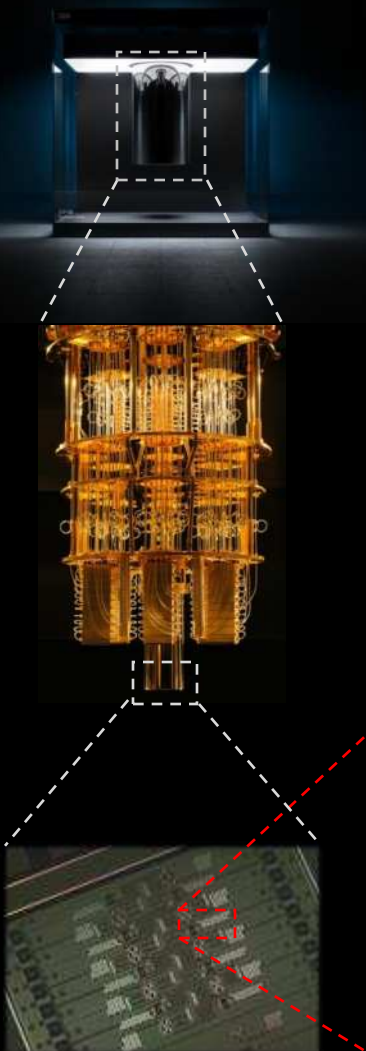
Hilbert space



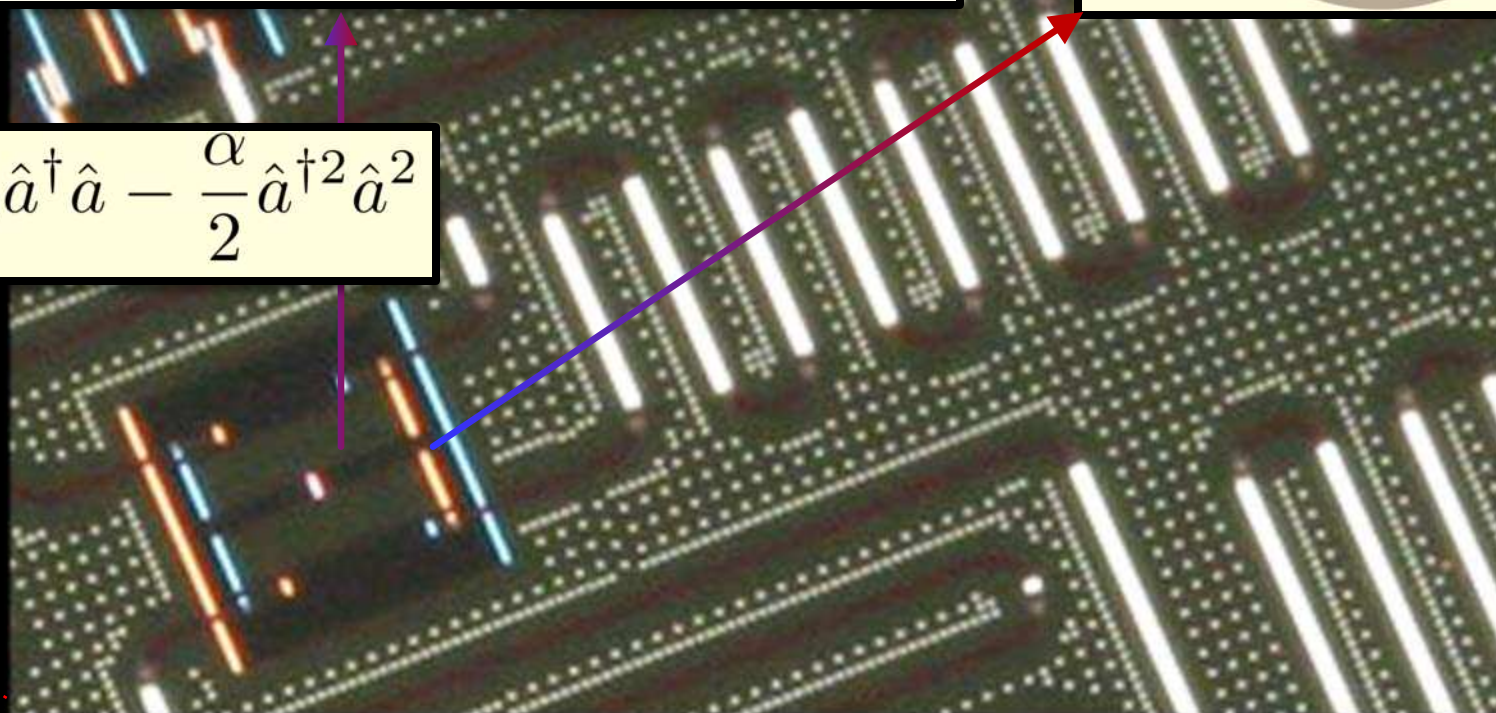


Operation at
15 mK (-273.13 °C)

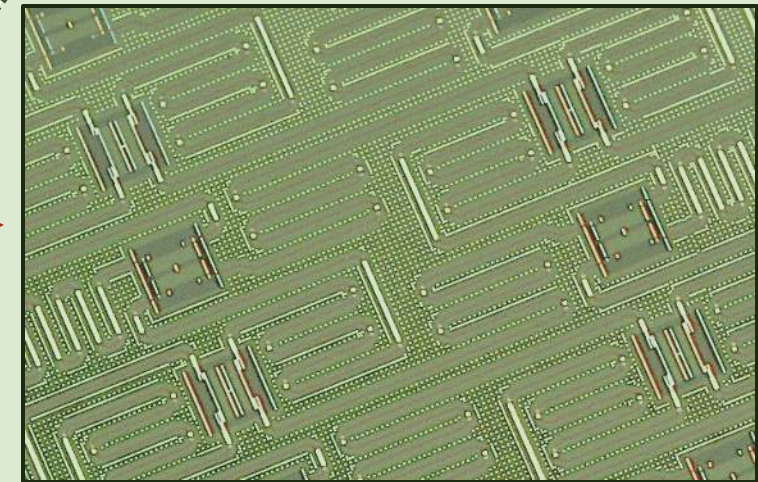
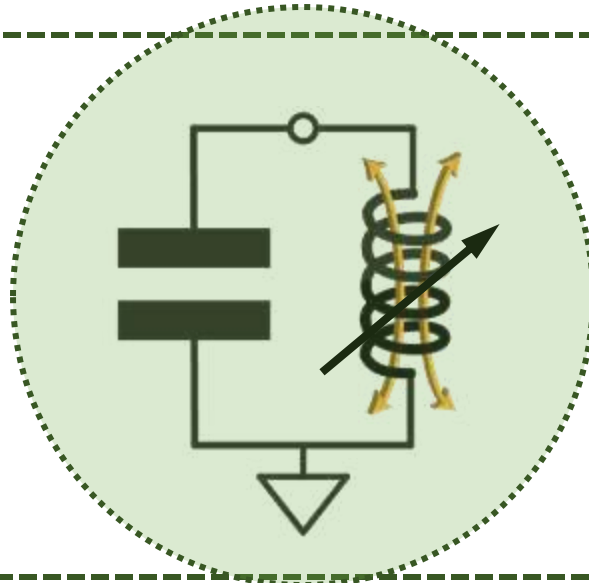
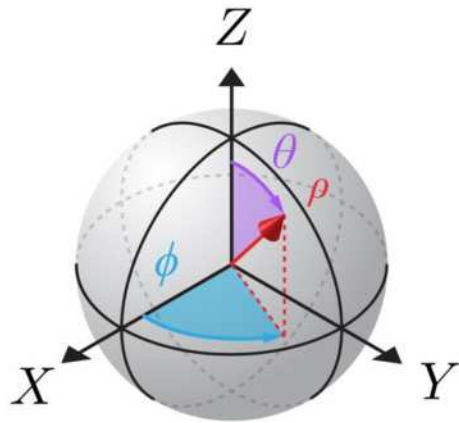
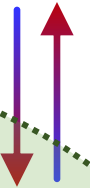
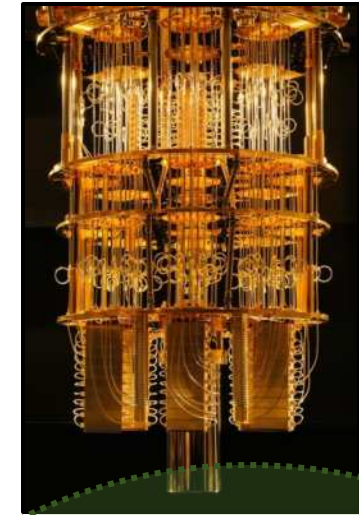


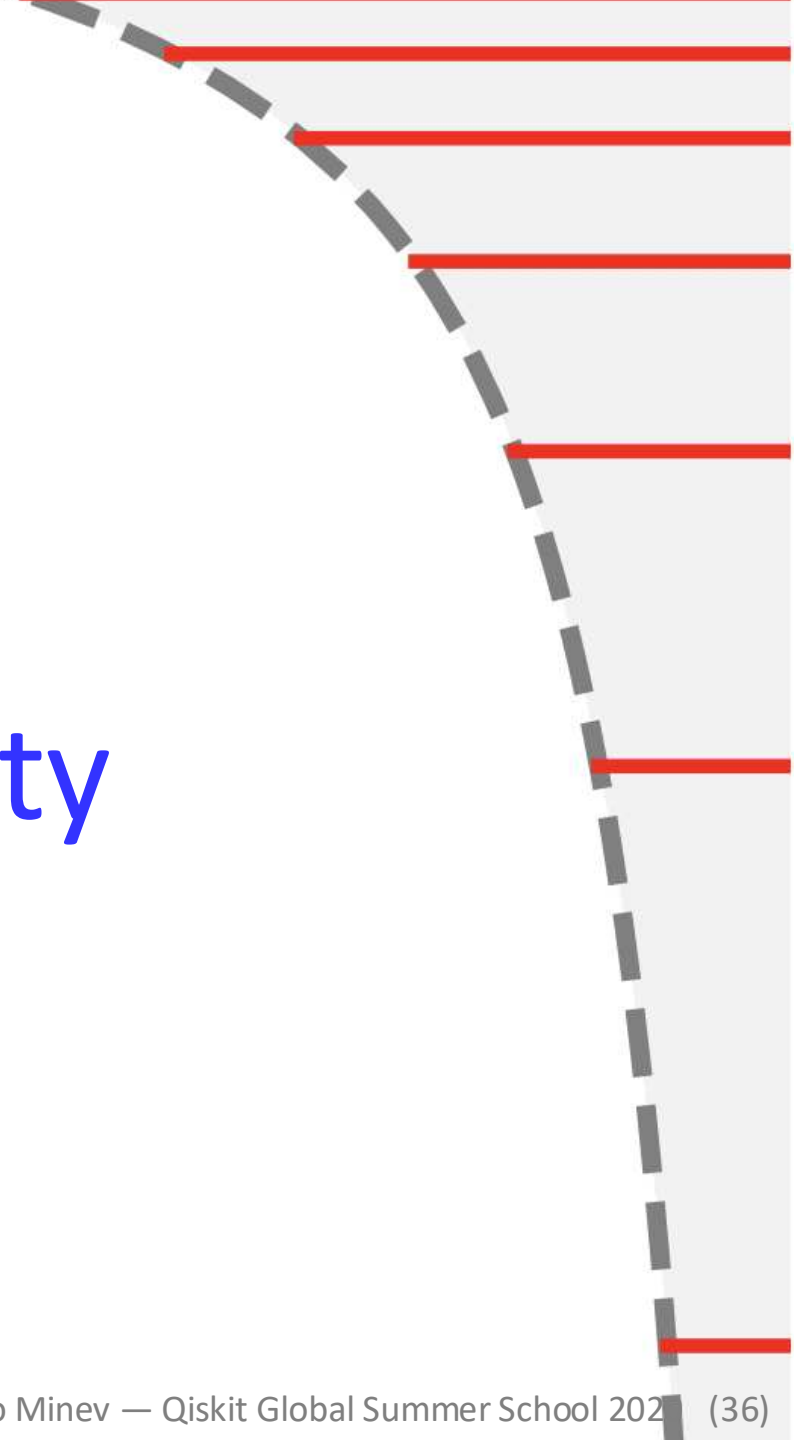


$$\hat{H} \approx \omega_0 \hat{a}^\dagger \hat{a} - \frac{\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$



cQED qubit in the cloud: Summary of flow





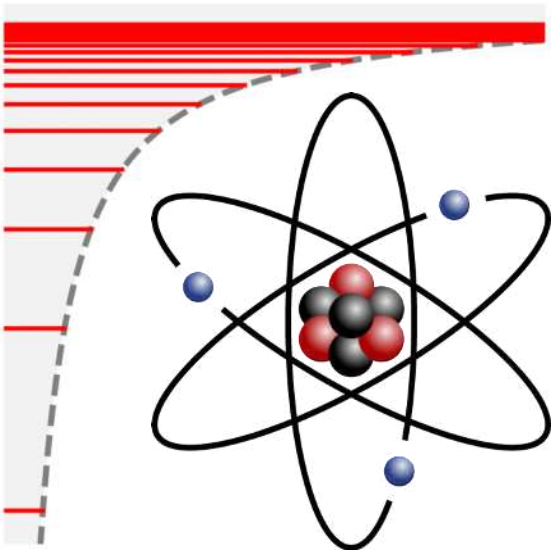
Qubit

From Idea to Reality

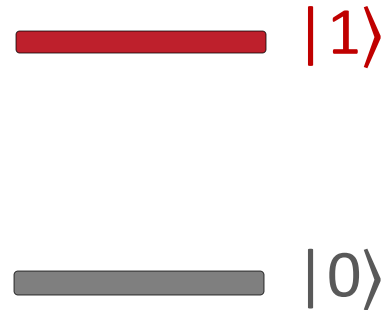
Concepts

From qubit representation to reality

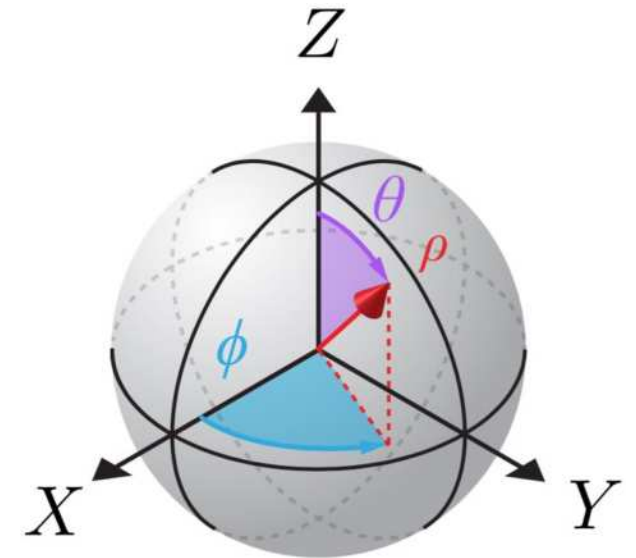
Realization



Energy levels



Hilbert space*



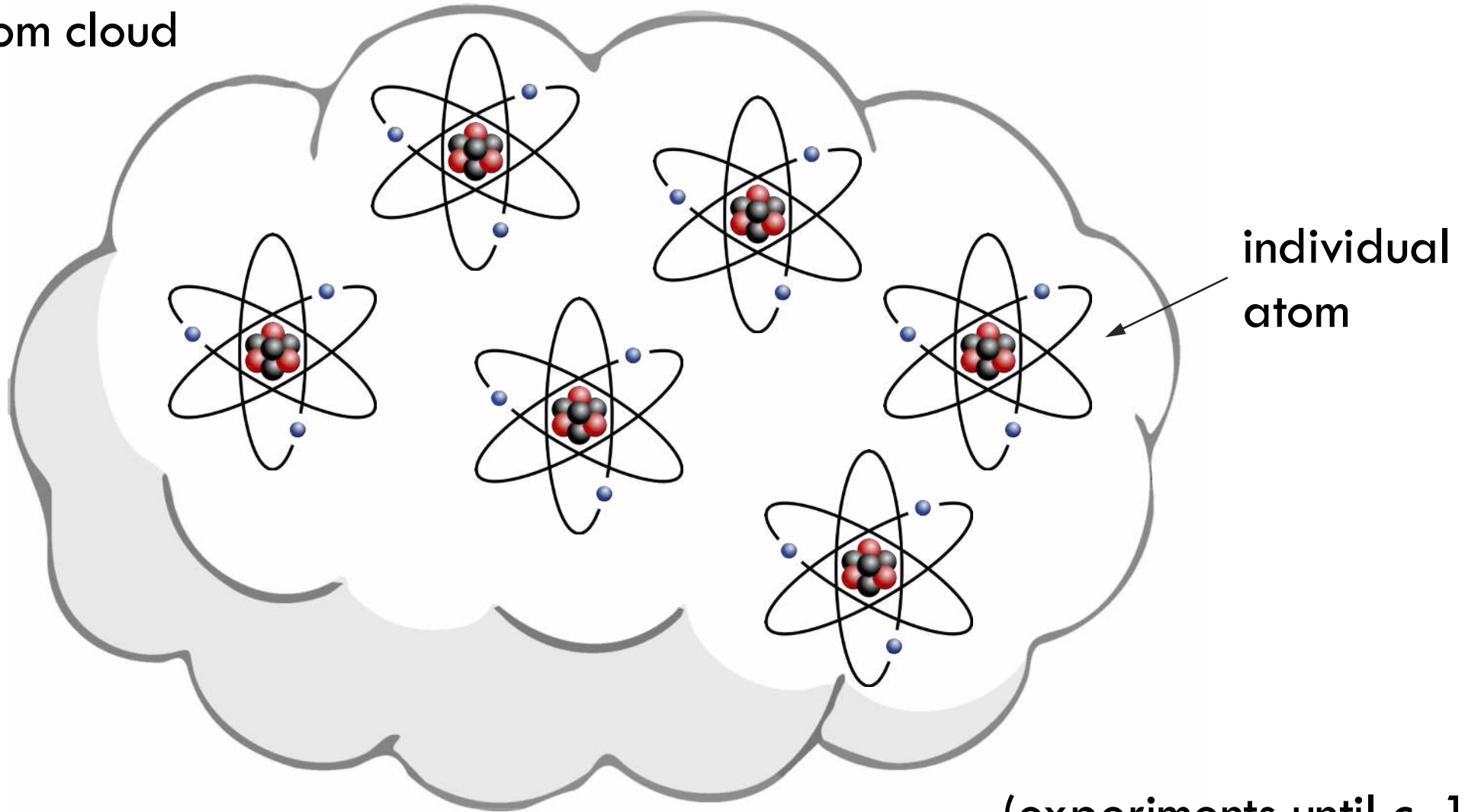
* Bloch sphere is a mere geometrical representation of $SO(3)$, but the density matrix ρ is in $SU(2)$, a double cover of $SO(3)$.

* A density matrix operator lives not in the Hilbert space H but in the Liouville space $H \otimes H$.

Images: Minev, arXiv:1902.10355; atom art: Indoleces.

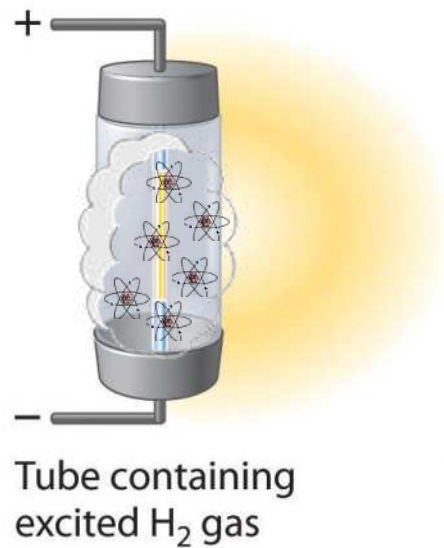
Origins of quantum

e.g., atom cloud

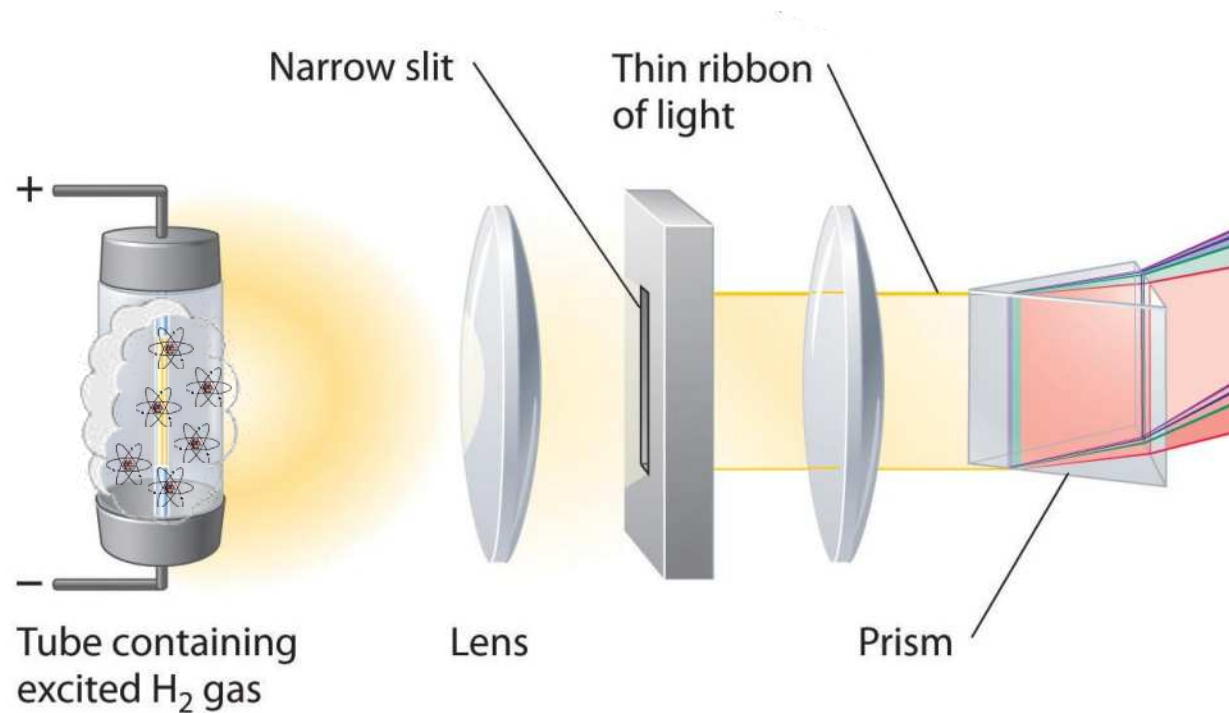


(experiments until c. 1980)

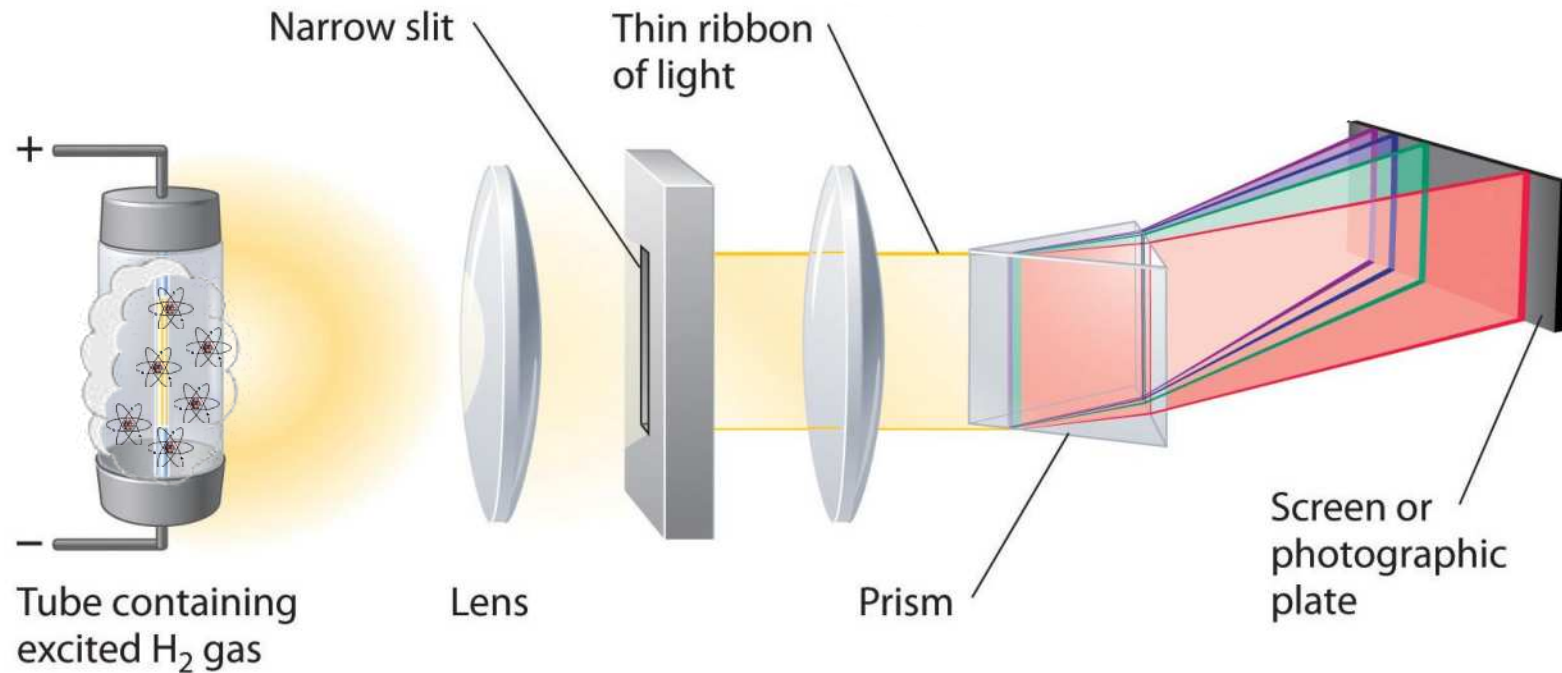
Atomic emission of light



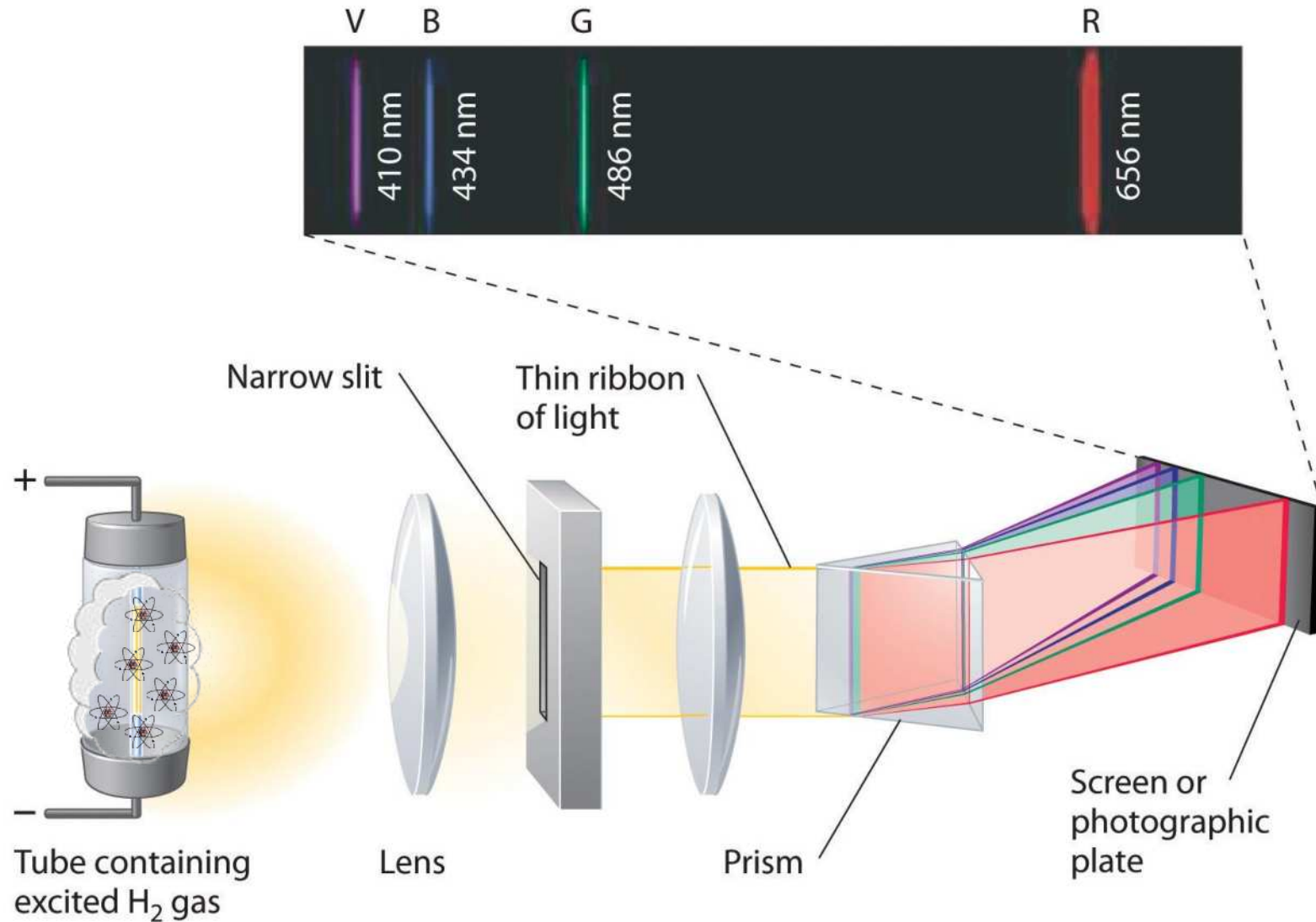
Atomic emission of light



Atomic emission of light



Atomic emission of light



Quantized levels

Atomic emission spectra

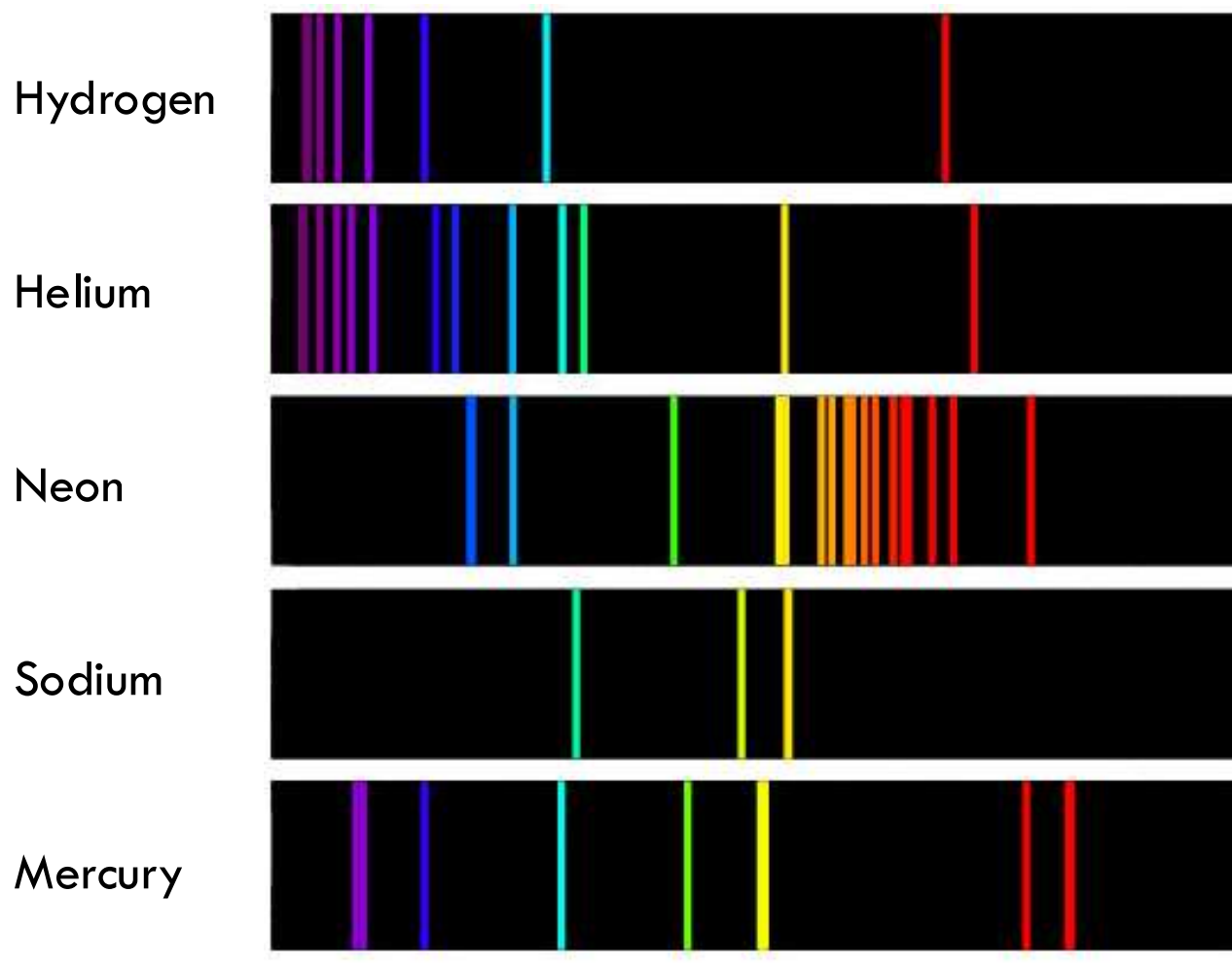
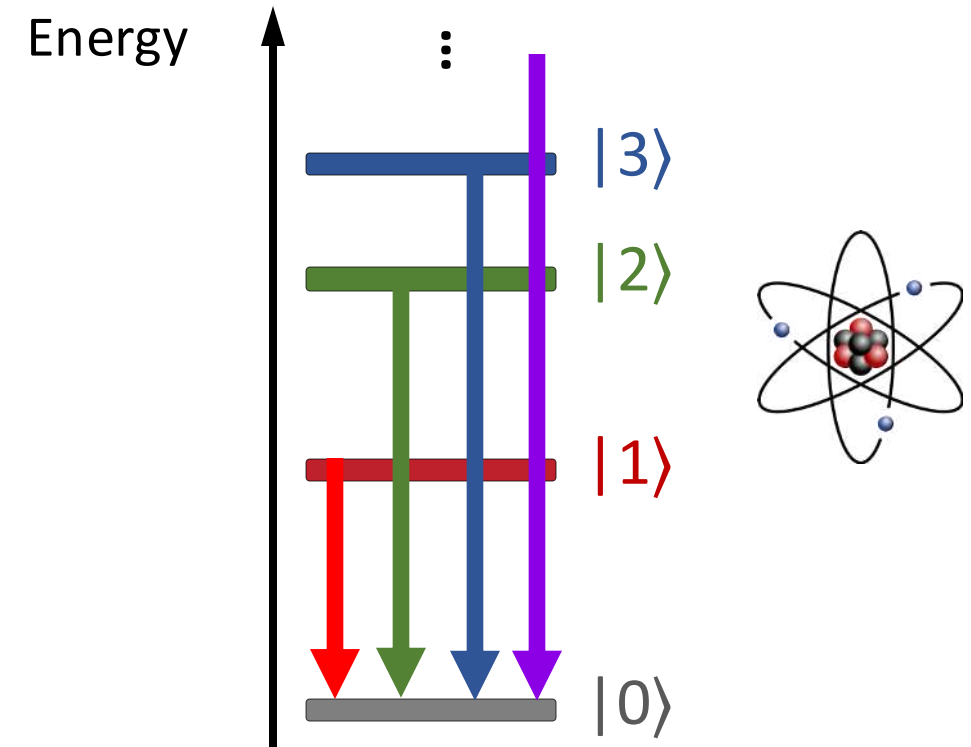


Image credit: NMSU, N. Vogt

Atoms are quantum: discrete intrinsic energy levels*



Quantization of energy

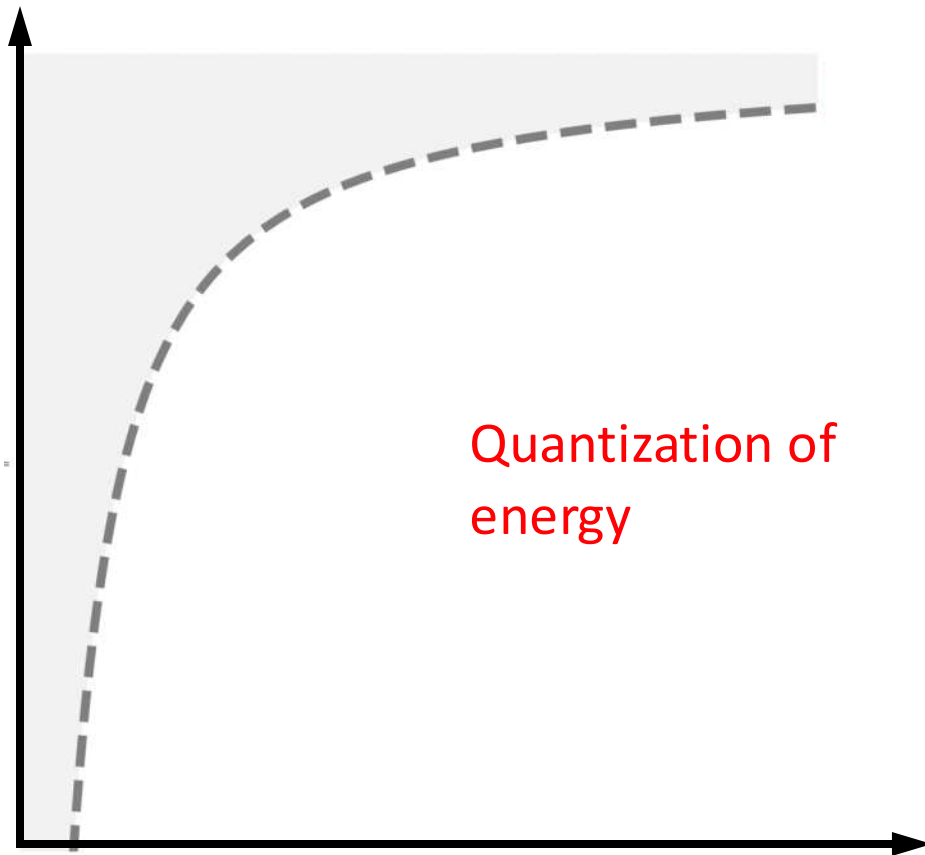
* The notion of an energy level was proposed by Bohr in 1913.
Zlatko Minev — Qiskit Global Summer School 2020 (43)

The light of atoms



Atomic energy levels and transitions

Electron potential-energy landscape



Radial distance from nucleus

Qubit from Atom

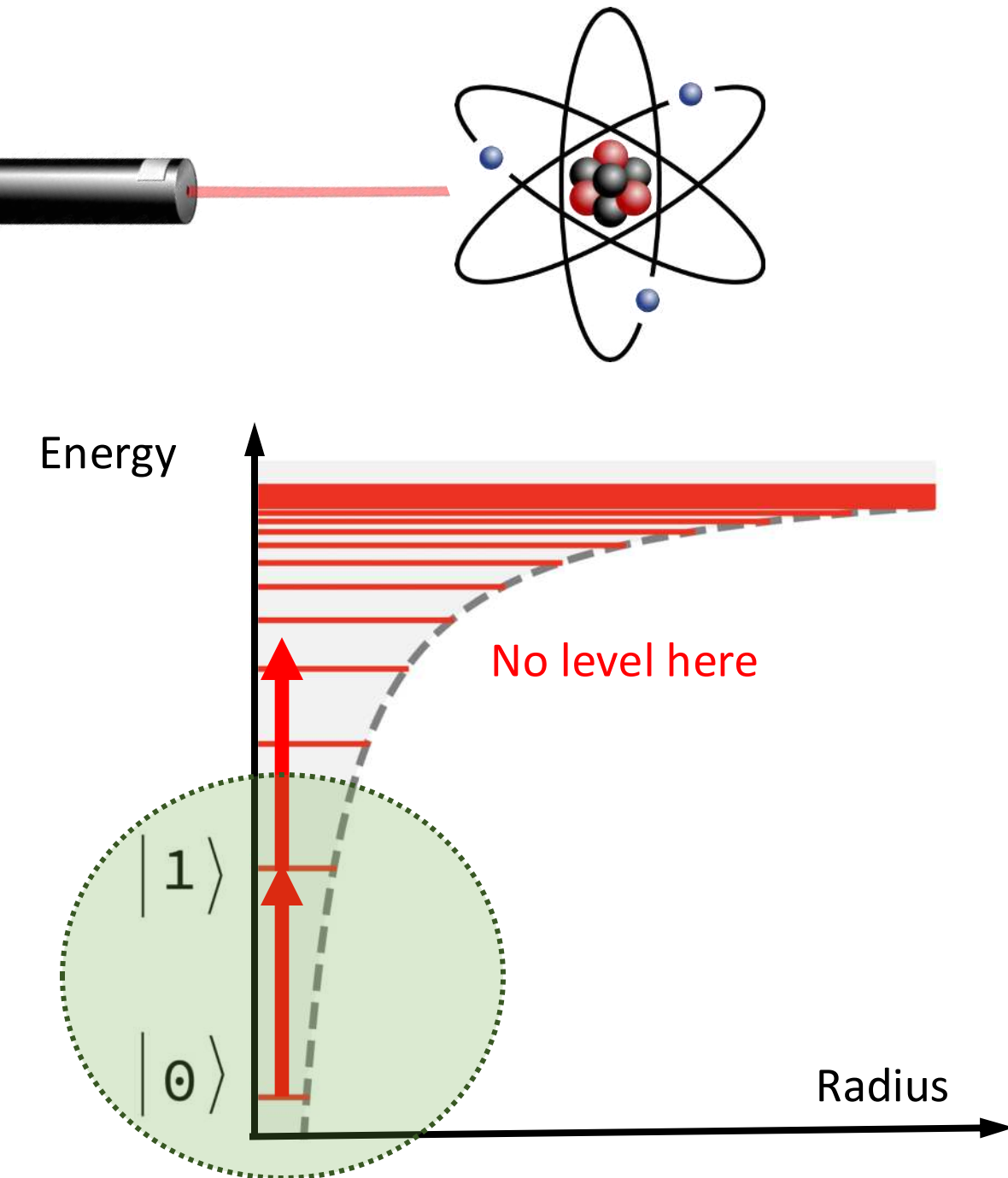
**Anharmonic degree of
freedom and spin**

**Isolated from environment and
thermal bath**

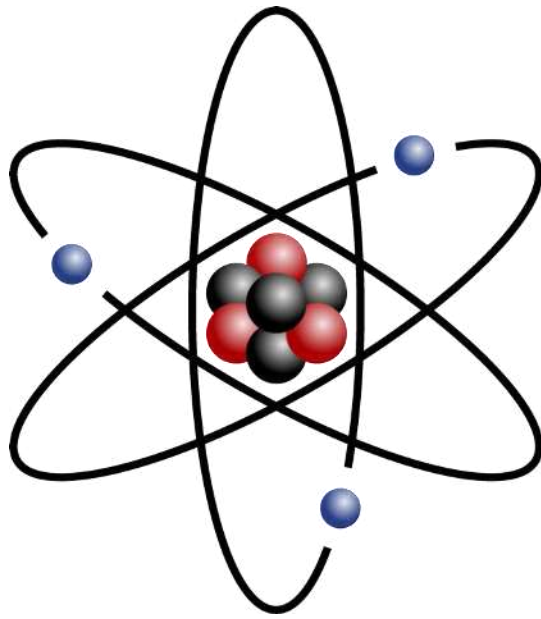
Low-loss

**Level diagram allows for qubit-
specific control and readout**

**There are always more than
two levels**

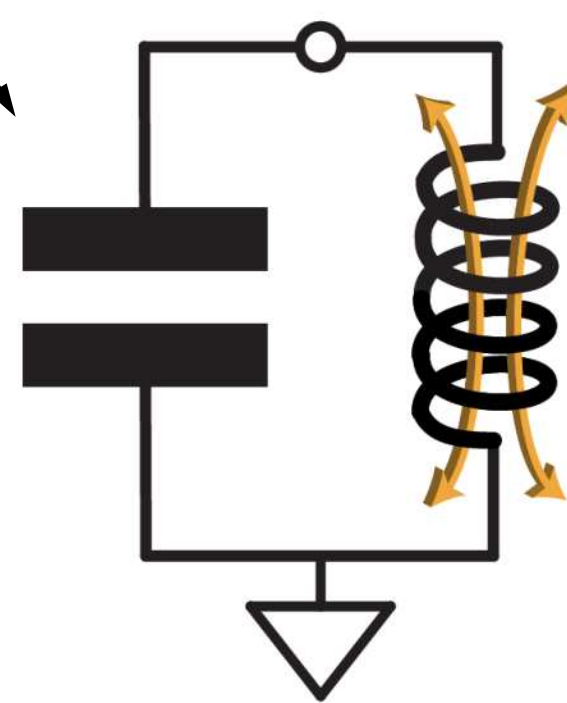


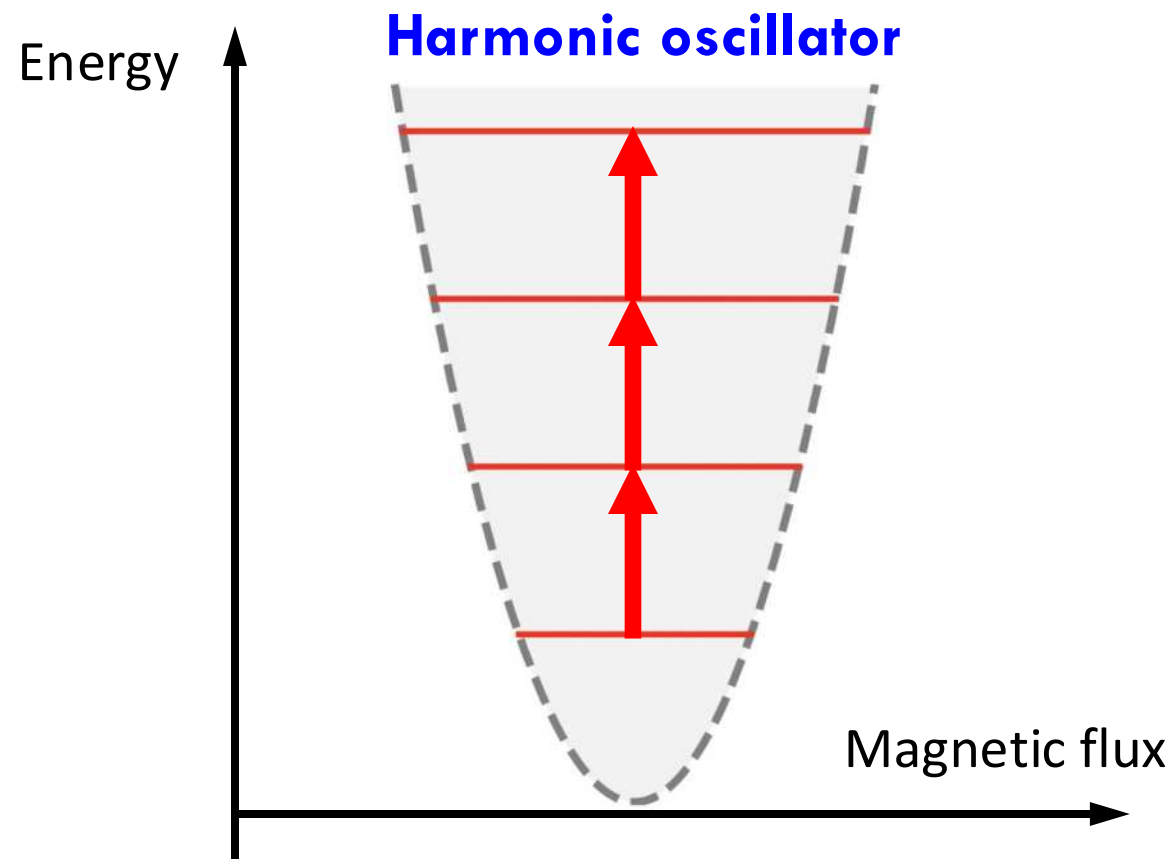
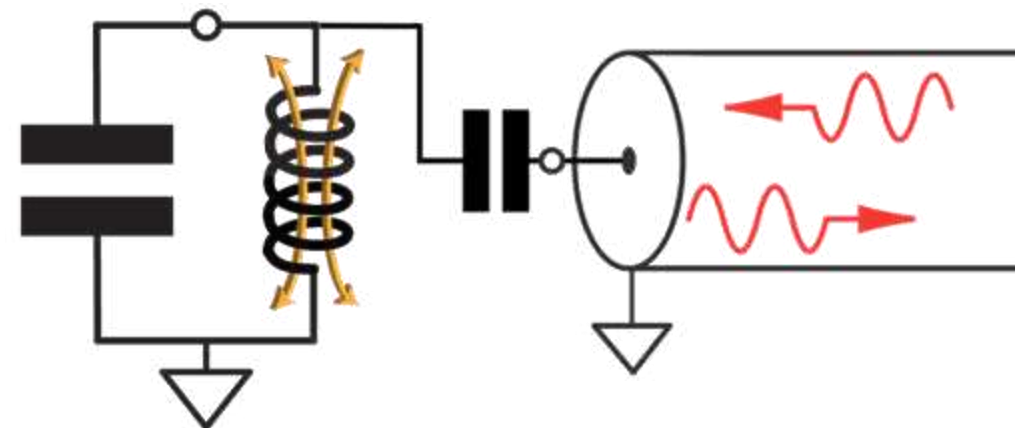
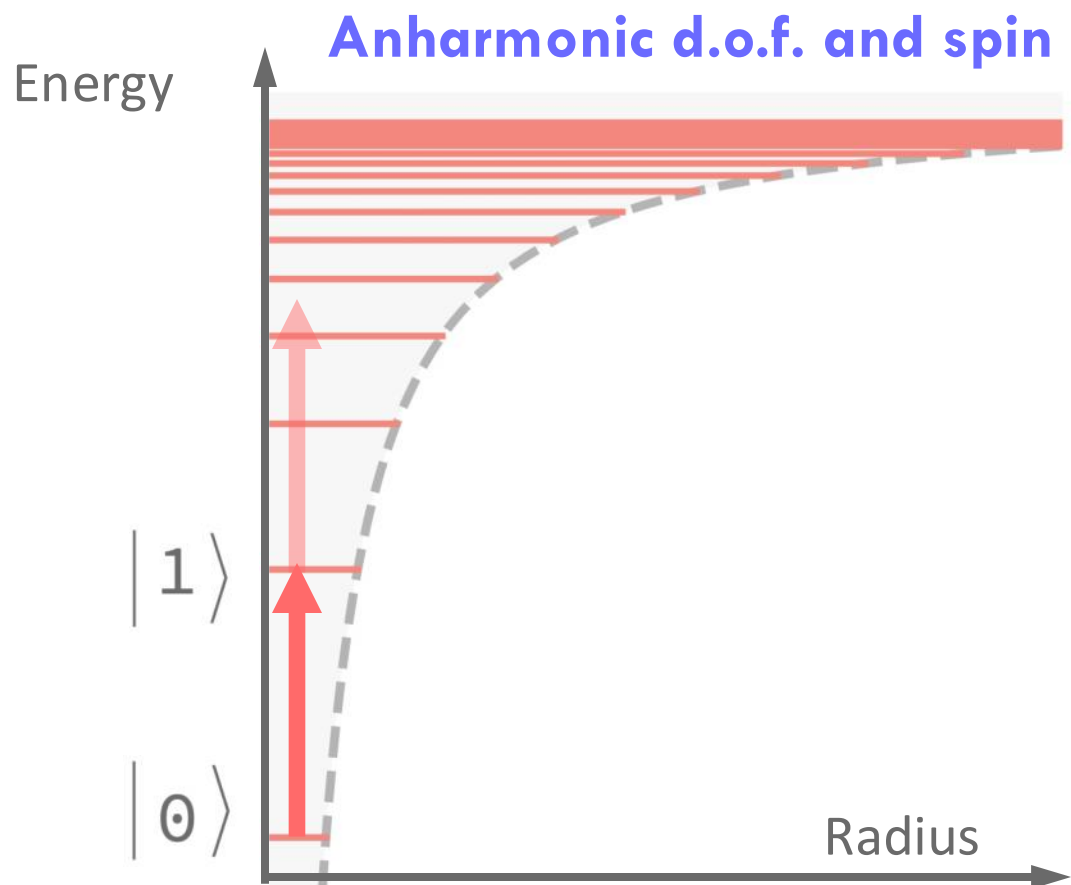
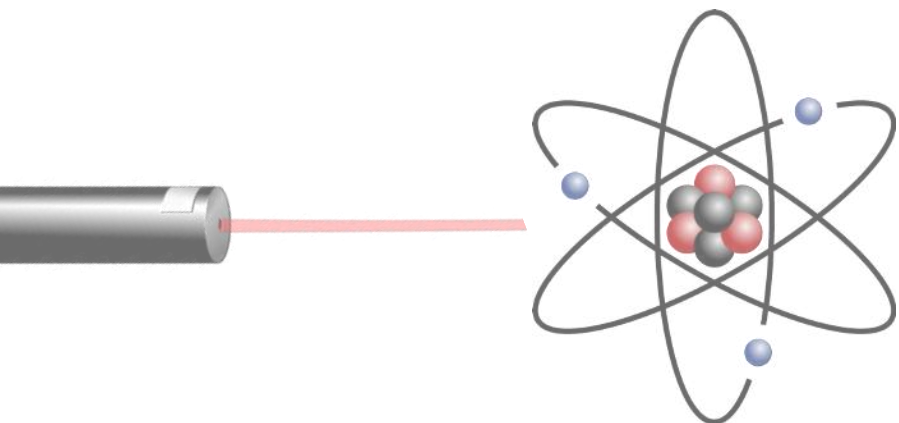
Artificial atoms

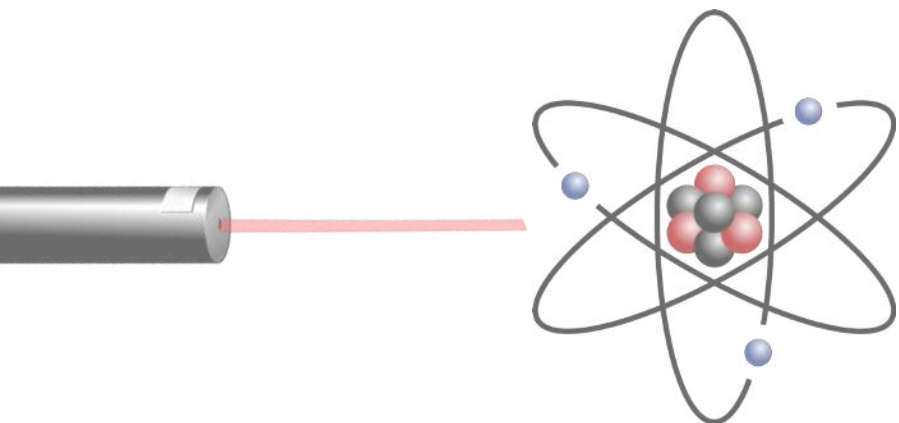


capacitor

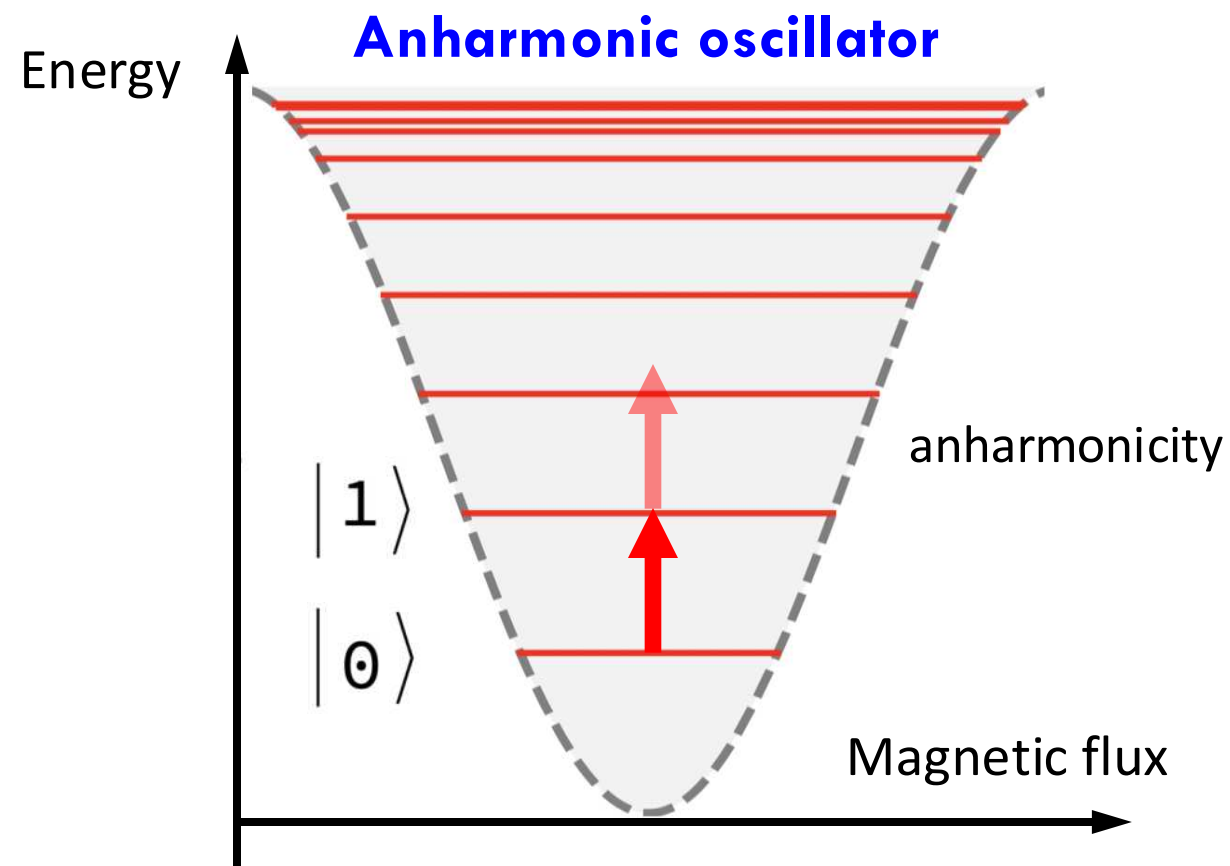
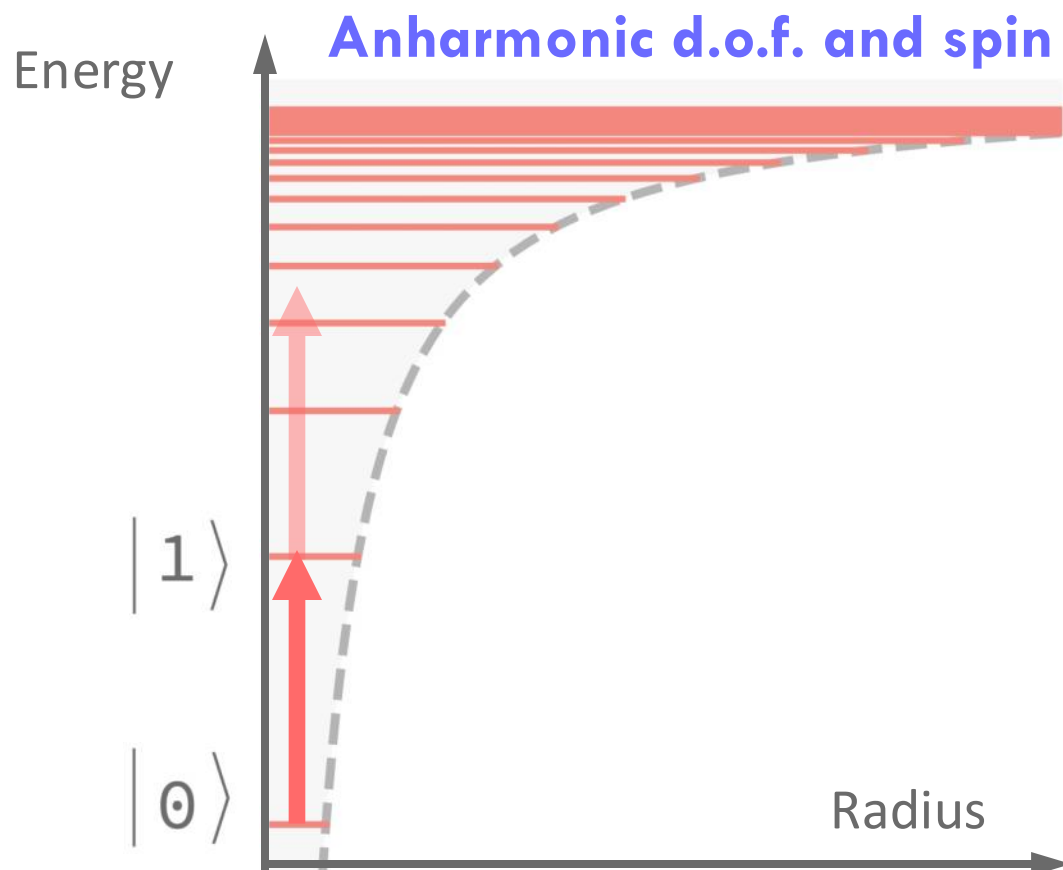
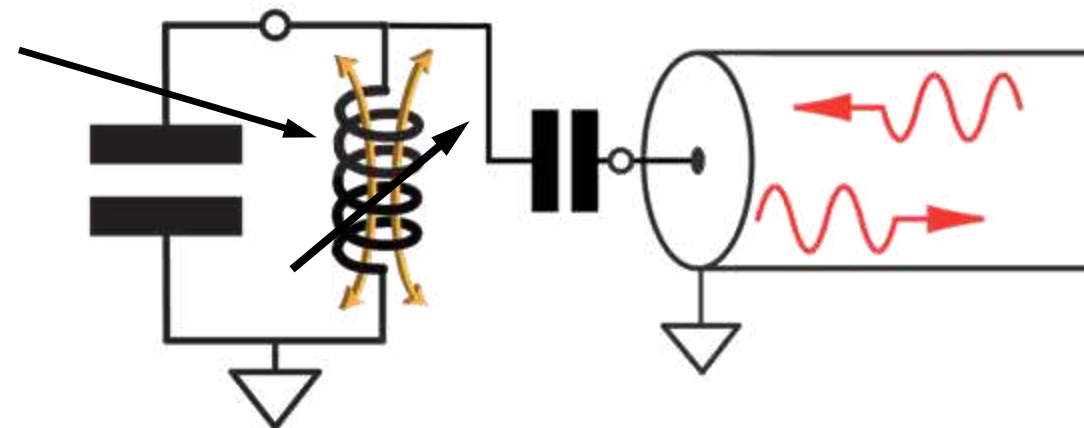
inductor







non-linear
inductor



Big-picture connections

Idealization of
qubit

— $|1\rangle$

— $|0\rangle$

Anharmonic

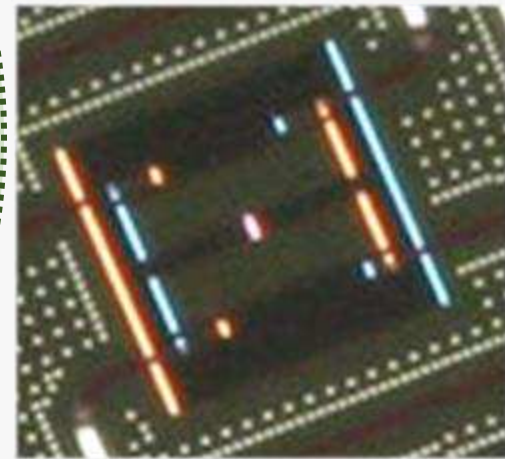
oscillator

Physical circuit

model

Physical layout

Circuit Quantum
Electrodynamics
(cQED)



Idealization

Physical reality

There are two kinds of physicists:

Those who believe all of physics is *spins*.

Those who believe all of physics is *oscillators*.

A few introductory reviews

And many more... check online or ask us for specific topic

Qiskit Textbook (2020; more chapters coming)

Blais, A., Grimsmo, A. L., Girvin, S. M., & Wallraff, A. (2020)
Circuit Quantum Electrodynamics (arXiv:2005.12667)

Kjaergaard, M., Schwartz, ... Oliver, W. D. (2020)
Superconducting Qubits: Current State of Play
Annual Reviews of Condensed Matter Physics 11, 369-395

Krantz, P., Kjaergaard, M., Yan, F., ... & Oliver, W. D. (2019)
A quantum engineer's guide to superconducting qubits
Applied Physics Reviews, 6(2), 021318

Corcoles, A. D., Kandala, A., ... Gambetta, J. M. (2019)
Challenges and Opportunities of Near-Term Quantum Computing Systems. *Proceedings of the IEEE*, 1–15.

Wendin, G. (2017)
Quantum information processing with superconducting circuits. *Reports on Progress in Physics*, 80(10), 106001

Gambetta, J. M., Chow, J. M., & Steffen, M. (2017)
Building logical qubits in a superconducting quantum computing system. *Npj Quantum Information*, 3(1), 2

Girvin, S. M. (2011) Circuit QED: superconducting qubits coupled to microwave photons. *Quantum machines: measurement and control of engineered quantum systems*, 113, 2.

Clerk, A. A., Girvin, S. M., Marquardt, F., & Schoelkopf, R. J. (2010)
Introduction to quantum noise, measurement, and amplification
Reviews of Modern Physics, 82(2), 1155–1208

Clarke, J., & Wilhelm, F. K. (2008)
Superconducting quantum bits. *Nature*, 453(7198), 1031–1042

Devoret, M. H. (1997)
Quantum Fluctuations in Electrical Circuits.
In *Fluctuations Quantiques/Quantum Fluctuations* (p. 351)

...

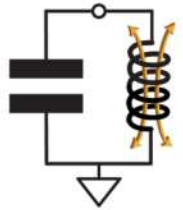
Qiskit Global Summer School – Minev lectures on cQED

Superconducting Qubits I:

Making Your First Qubit From an Oscillator

Introduction to Circuit
Quantum Electrodynamics (cQED)

Zlatko K. Minev



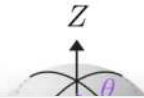
IBM Quantum
IBM T.J. Watson Research Center, Yorktown Heights, NY



@zlatko_minev

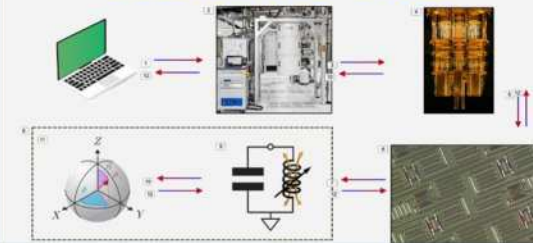


zlatko-minev.com

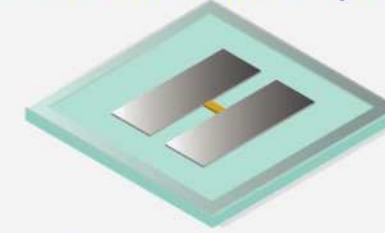


On the road ahead

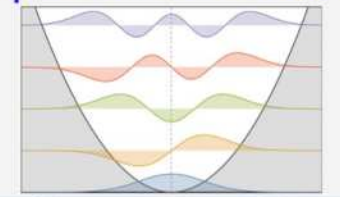
Qubit in the cloud



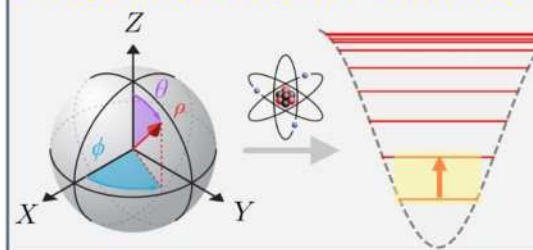
cQED: Transmon qubit



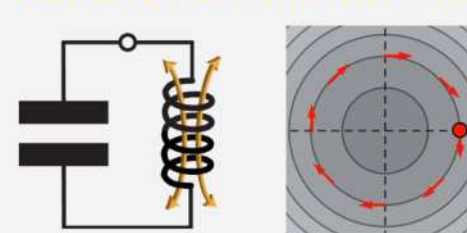
Unveiling the quantum oscillator



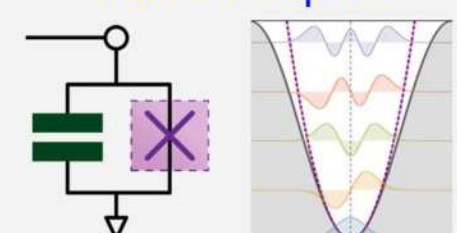
Qubit from atom / oscillator



Classical circuits & the LC



Transmon qubit



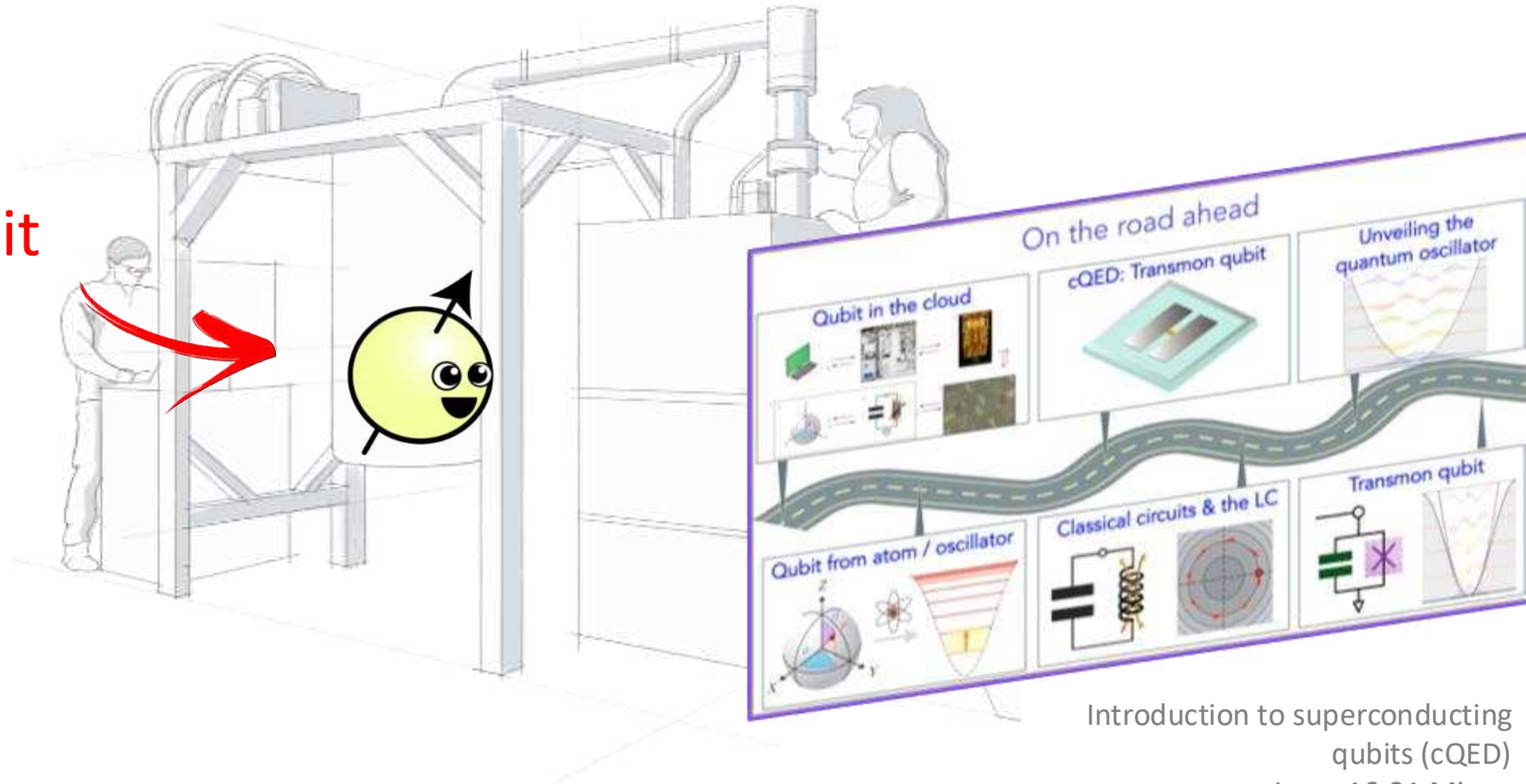
Find on the internet

Part III

Let's design and run a real quantum circuit

Hello World with a real experiment!

A qubit

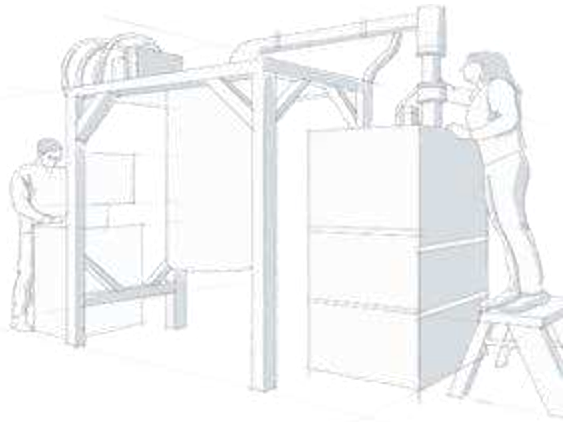


Introduction to superconducting
qubits (cQED)

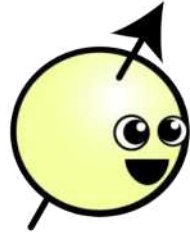
Lecs. 16-21 Minev

QGSS 2020 at qiskit.org/learn

Hello World! building blocks

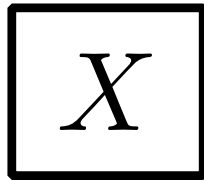


A qubit

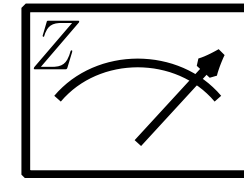
 $|1\rangle$ $|0\rangle$

Computational
basis states

Operations: qubit gate



Measurements: qubit observable



refresher:

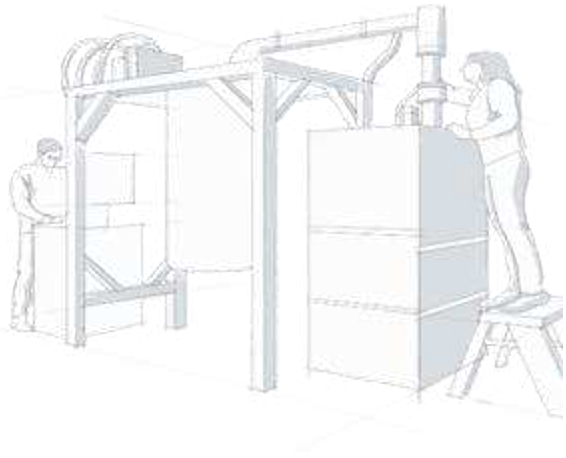
$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

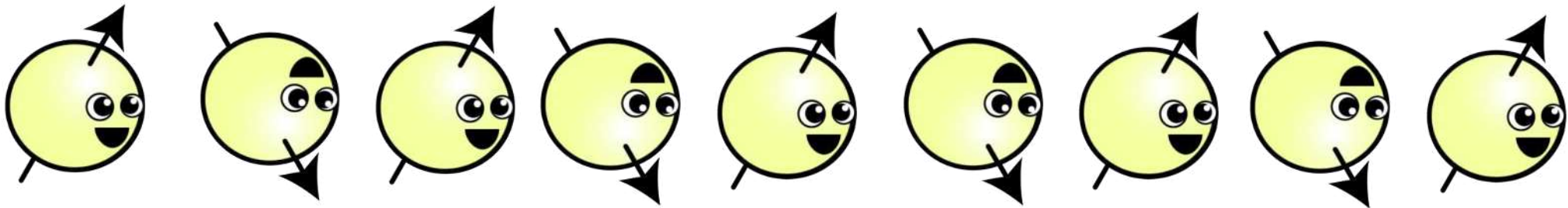
$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! Even-odd algo: qubit flipper



Task: Classify or report if a classical positive integer d is even or odd.



flip spin d times, measure polarization

refresher:

$$X |0\rangle = |1\rangle$$

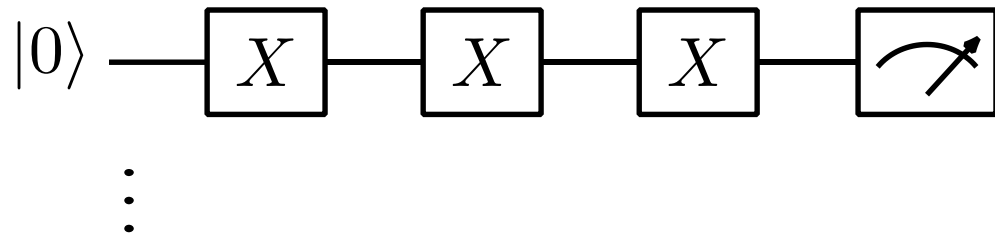
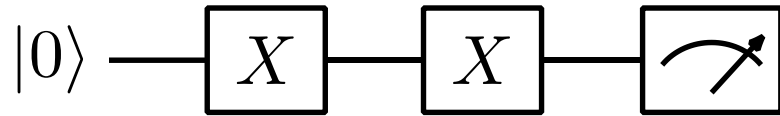
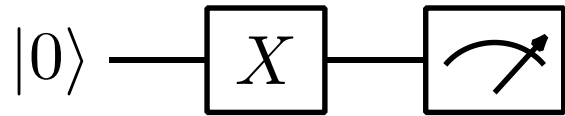
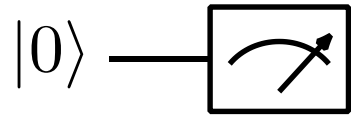
$$X |1\rangle = |0\rangle$$

$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! qubit flipper quantum circuits

depth



refresher:

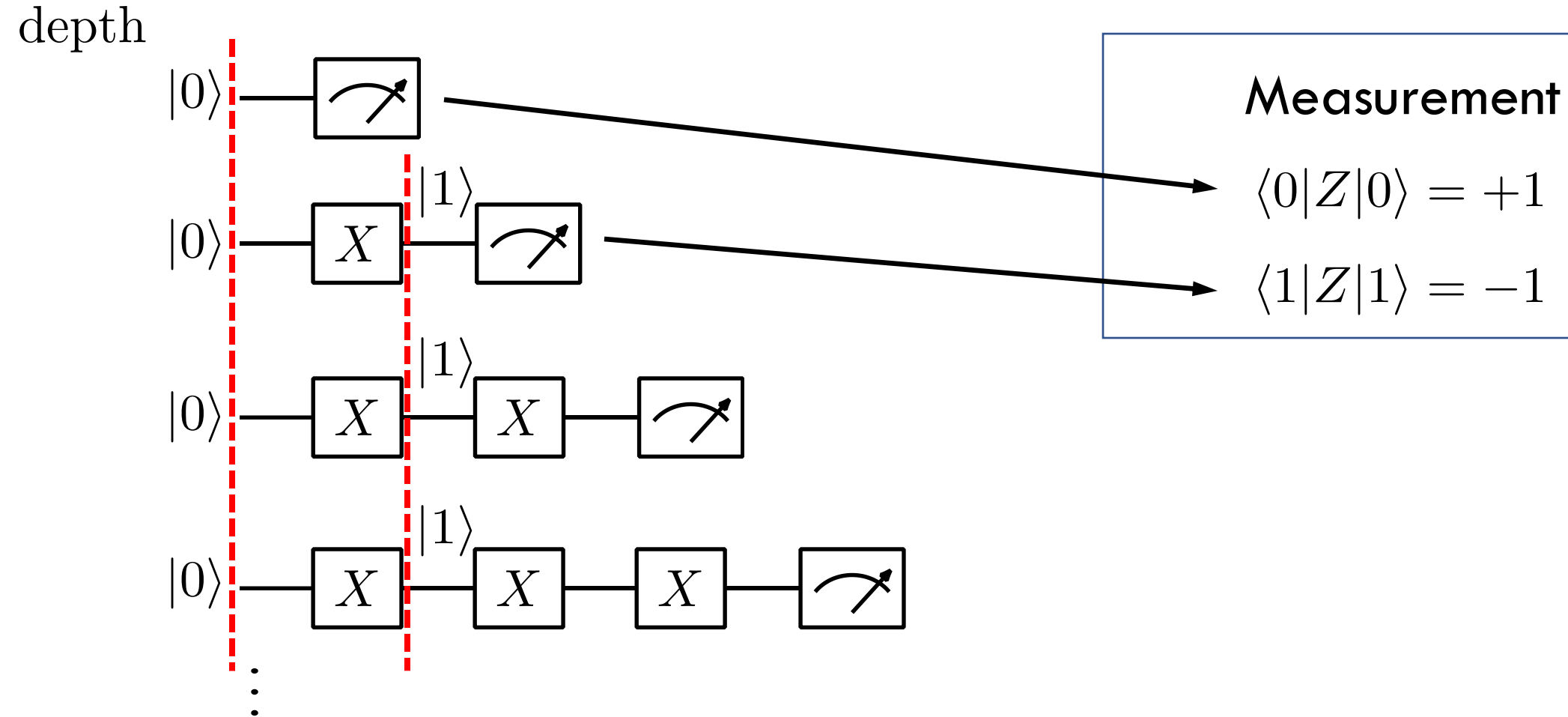
$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! “debugger” step through



refresher:

$$X |0\rangle = |1\rangle$$

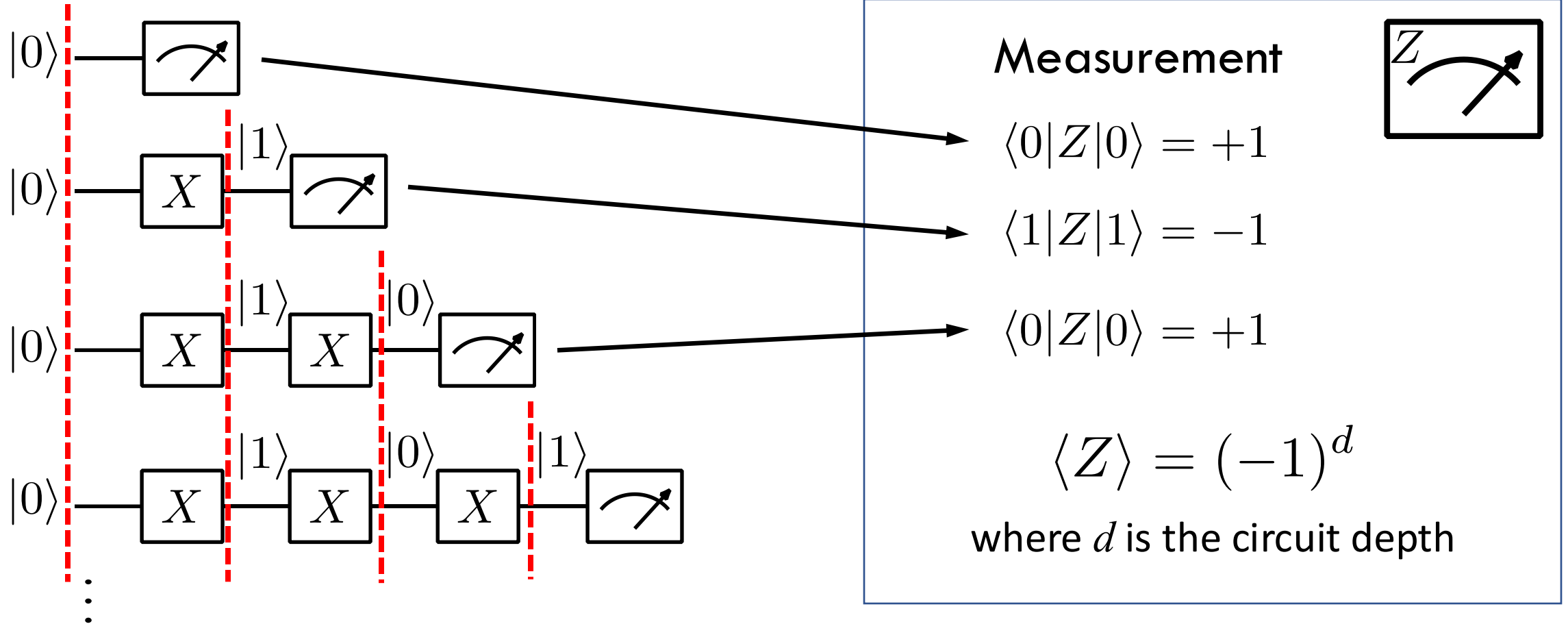
$$X |1\rangle = |0\rangle$$

$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

Hello World! “debugger” step through

depth



refresher:

$$X |0\rangle = |1\rangle$$

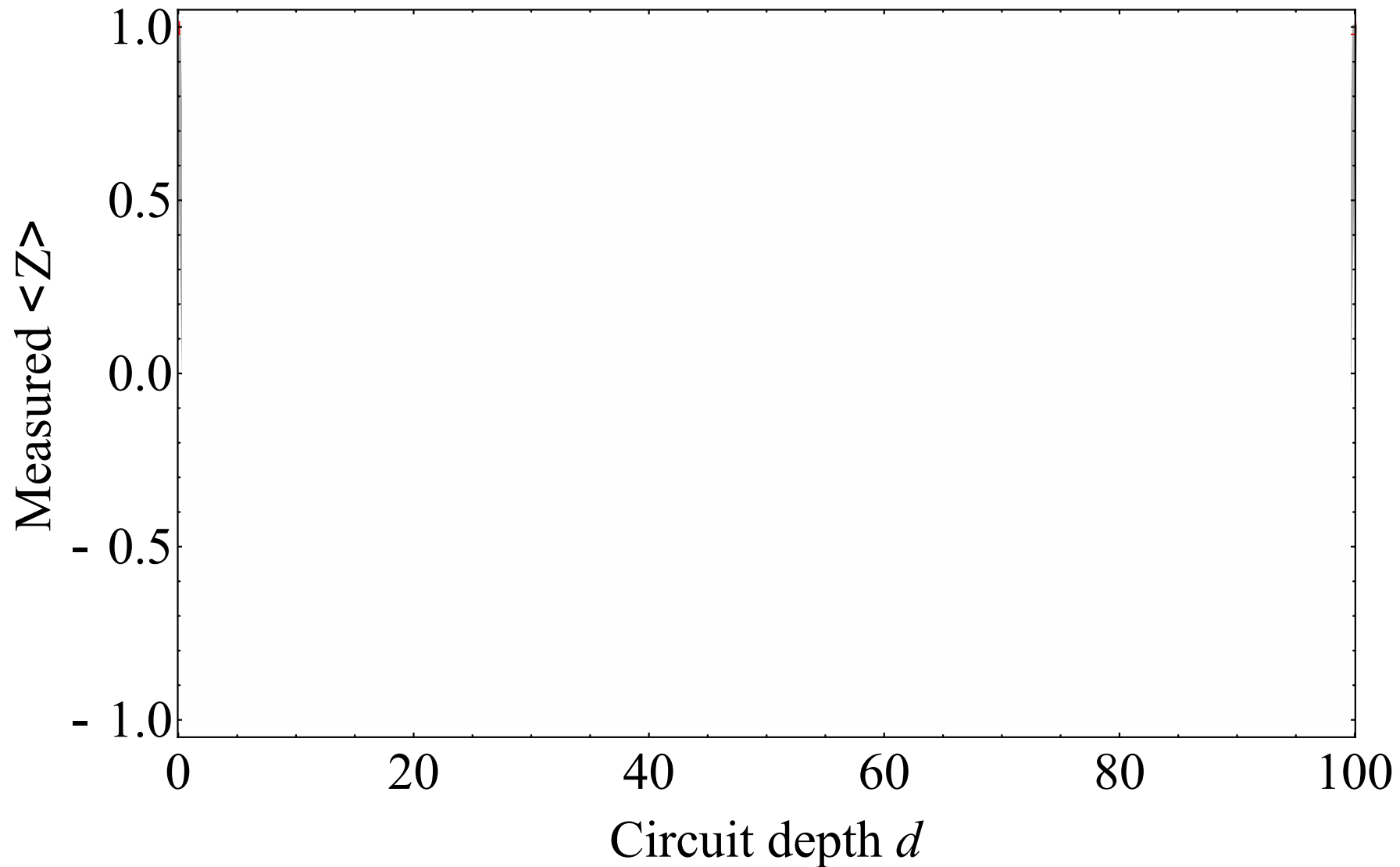
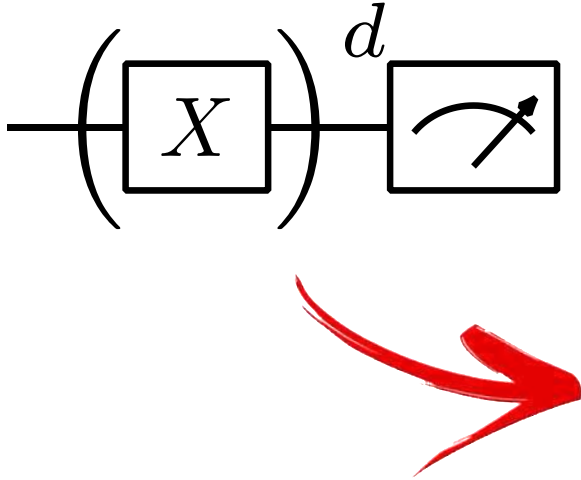
$$X |1\rangle = |0\rangle$$

$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

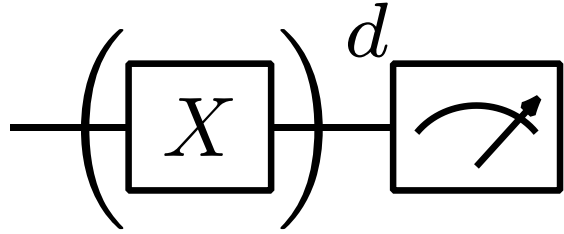
Hello World! Ideal expectation results

$$\langle Z \rangle = (-1)^d$$

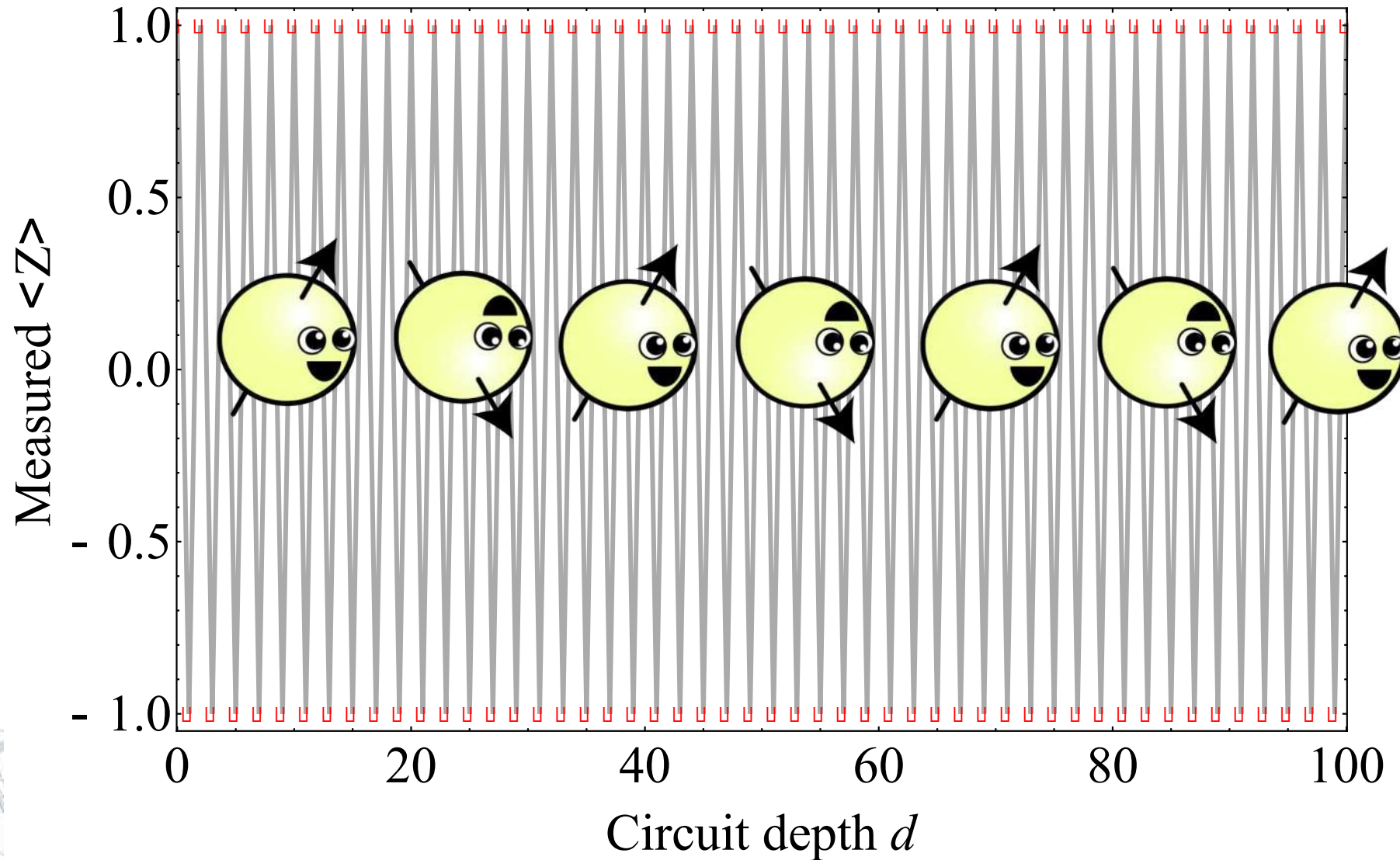
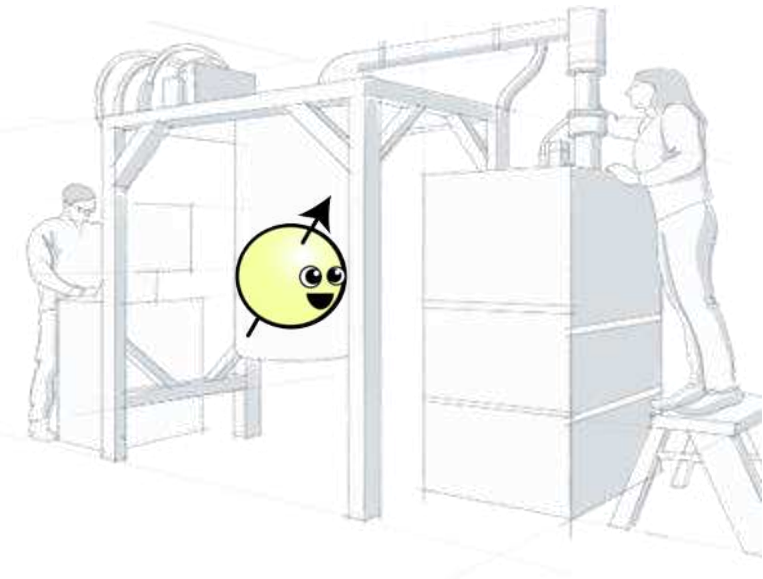


Hello World! Ideal expectation results

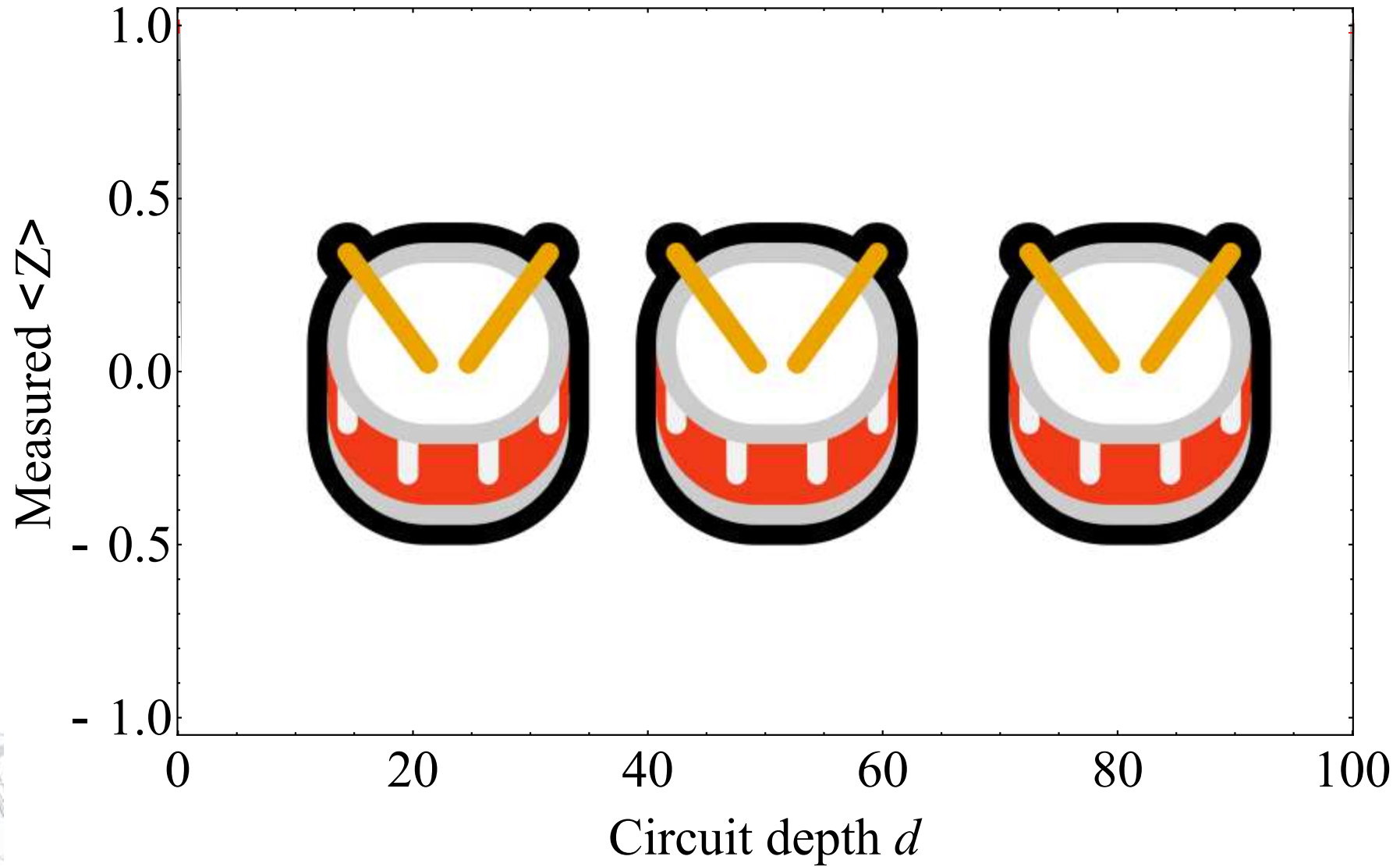
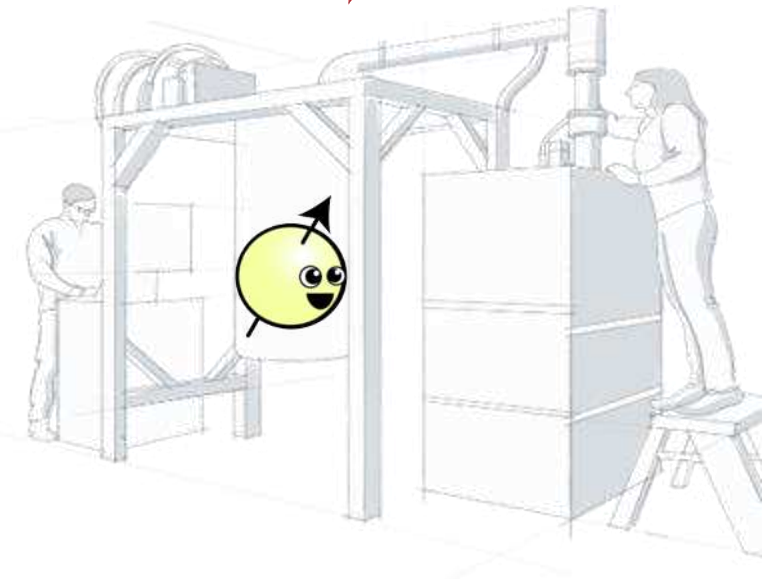
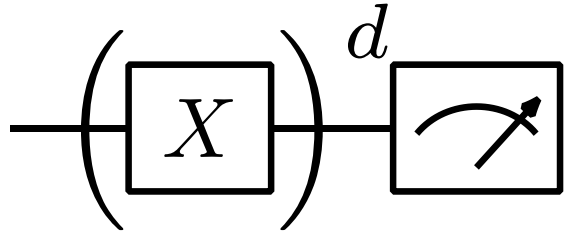
$$\langle Z \rangle = (-1)^d$$



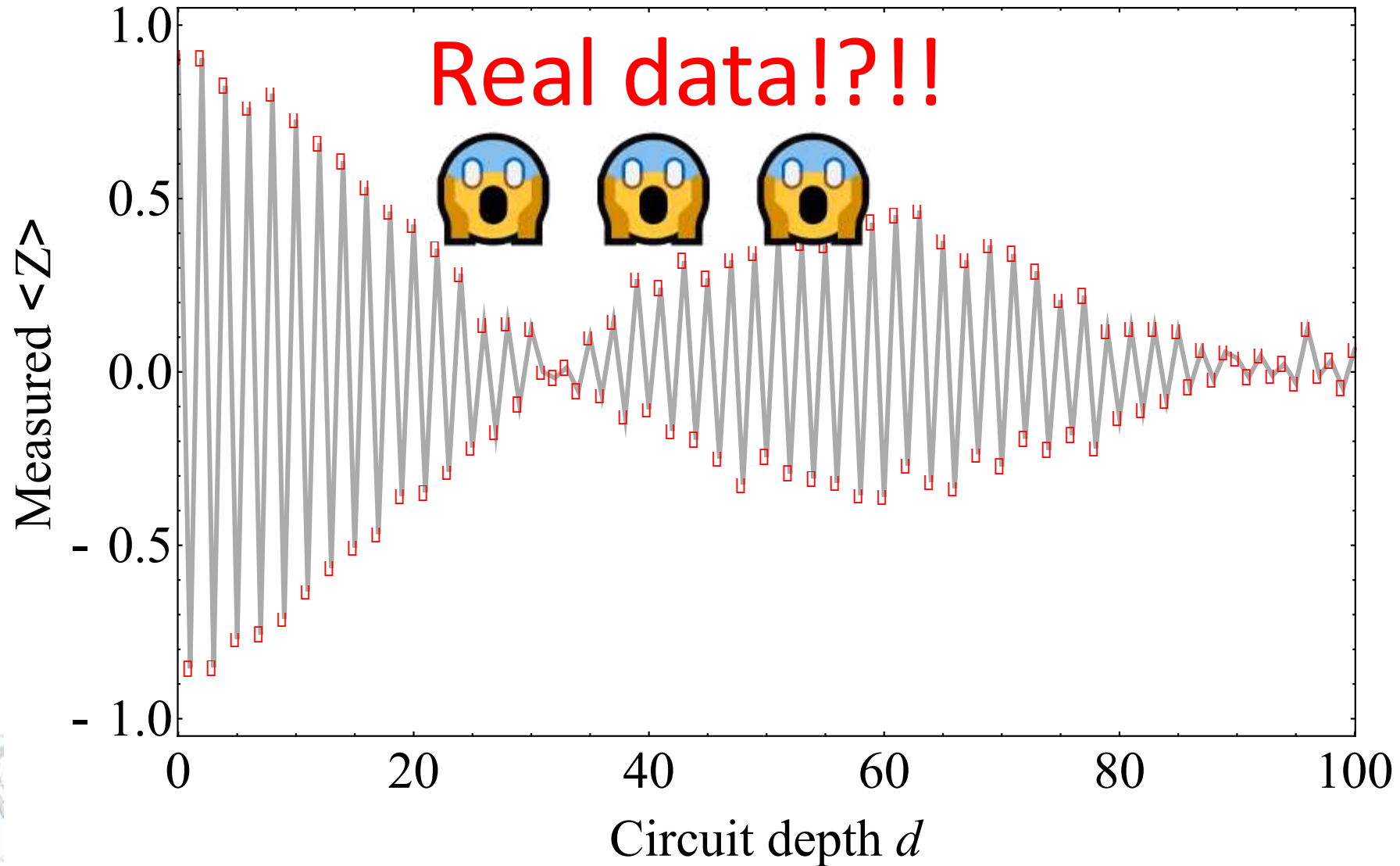
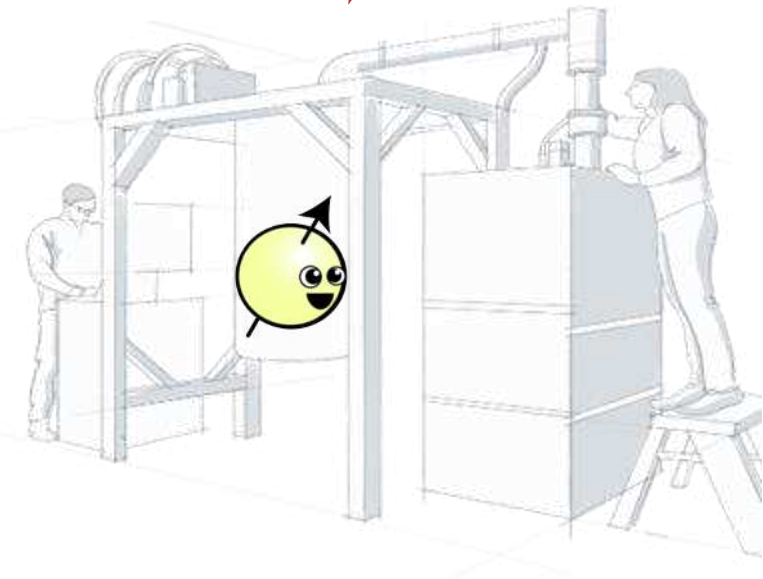
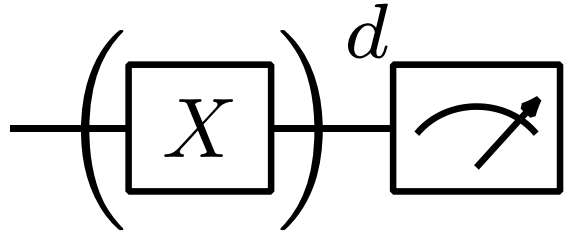
Let's run on a real device!



Hello World! Running



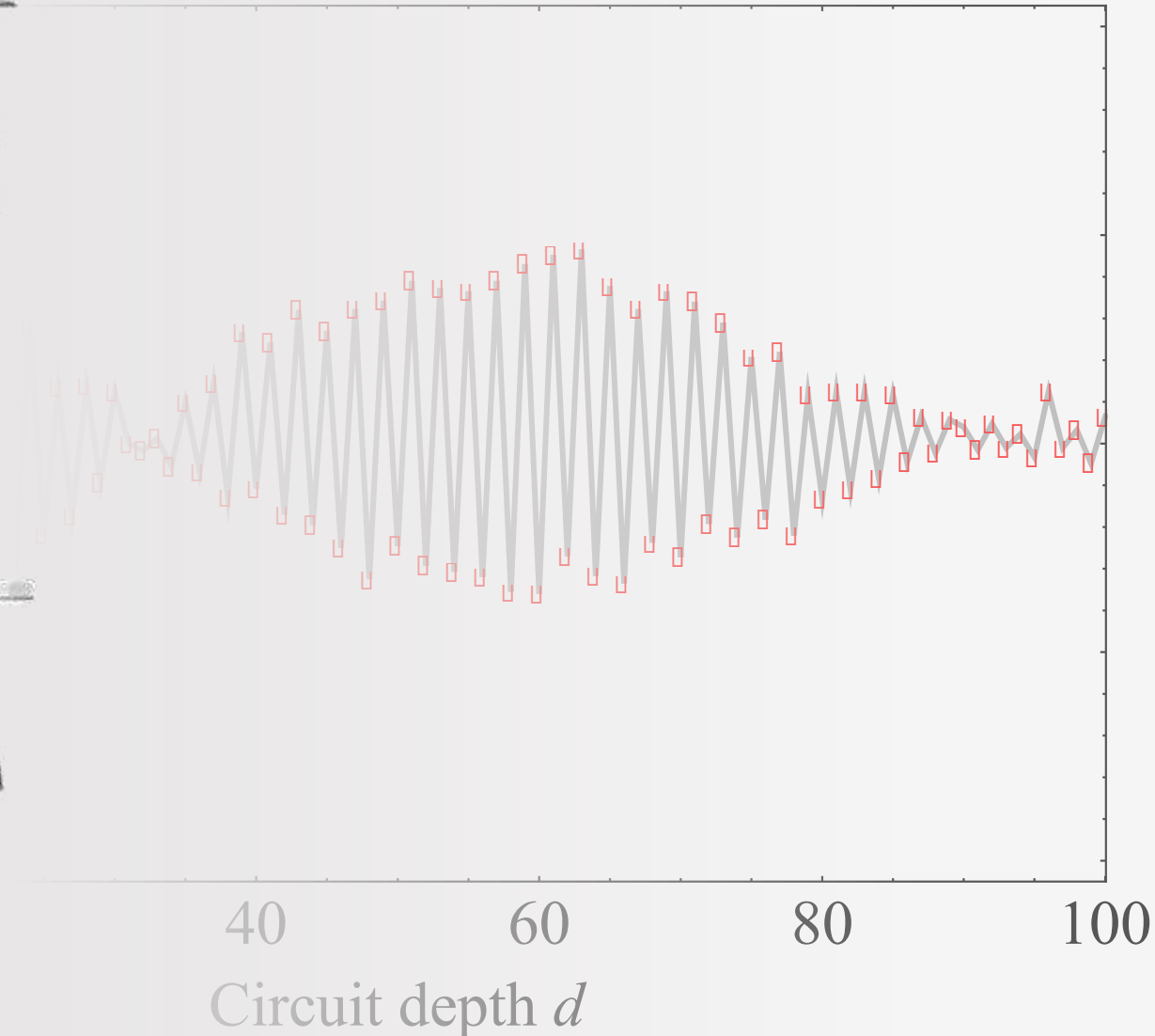
Hello World! Real expectation results

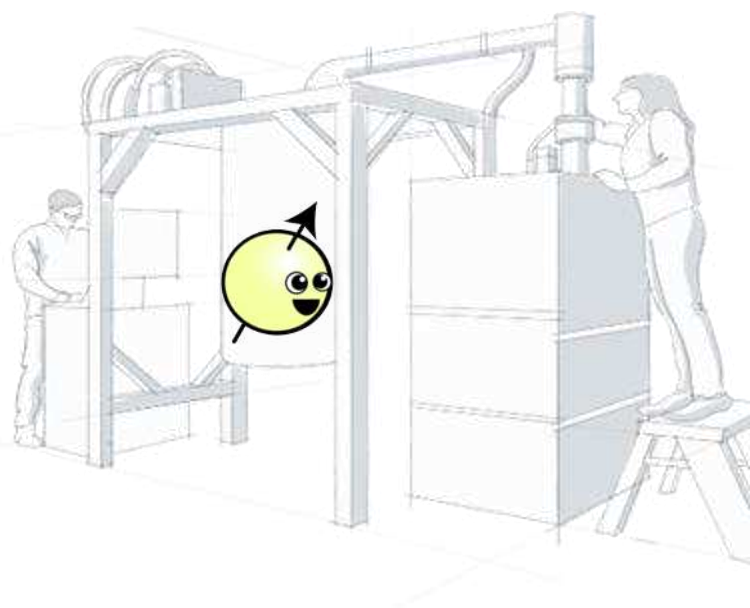


Real & noisy quantum processors: Why study noise?



“Well, your quantum computer is broken in every way possible simultaneously.”

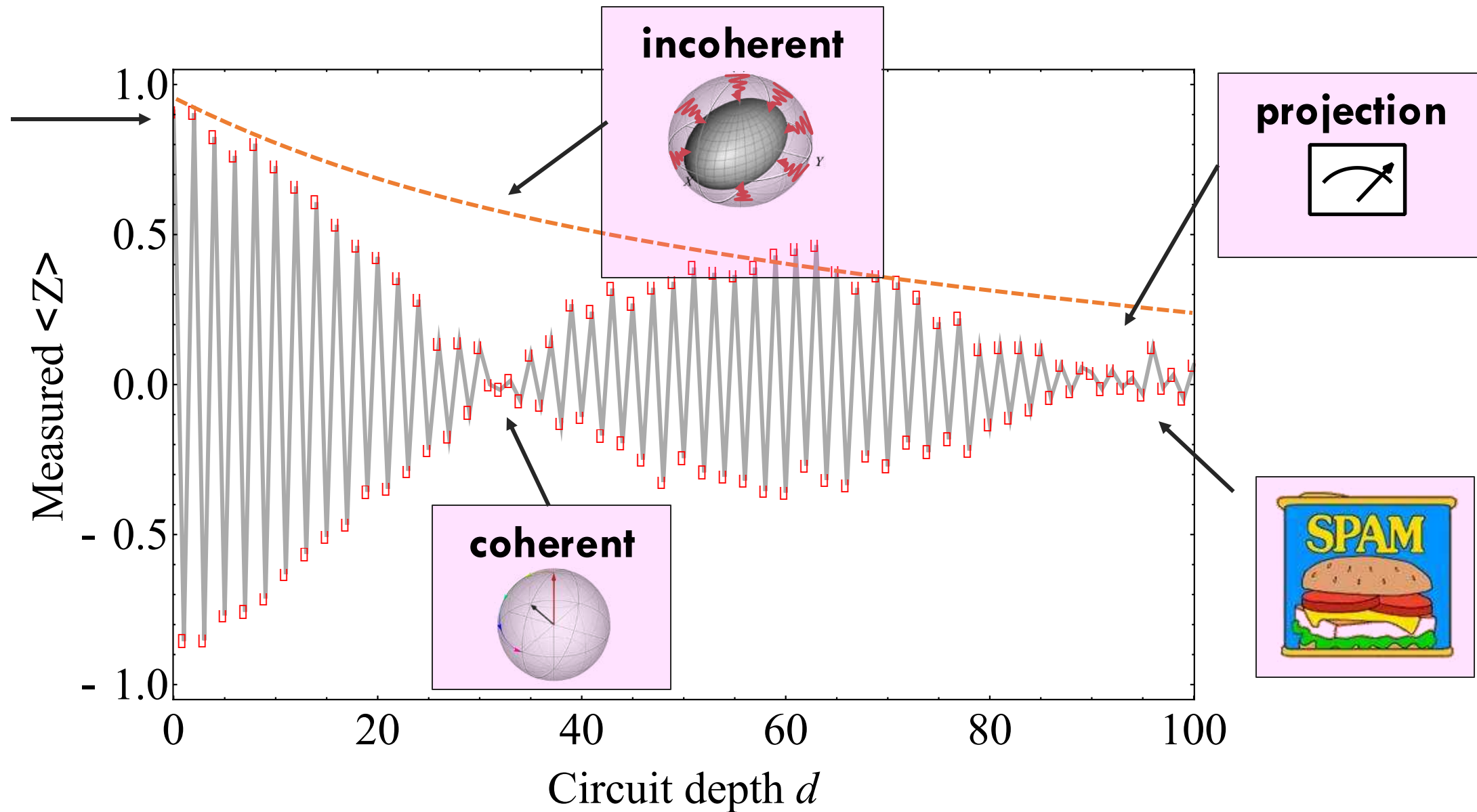




“Quantum phenomena
do *not* occur in a Hilbert space,
they occur in a laboratory.”

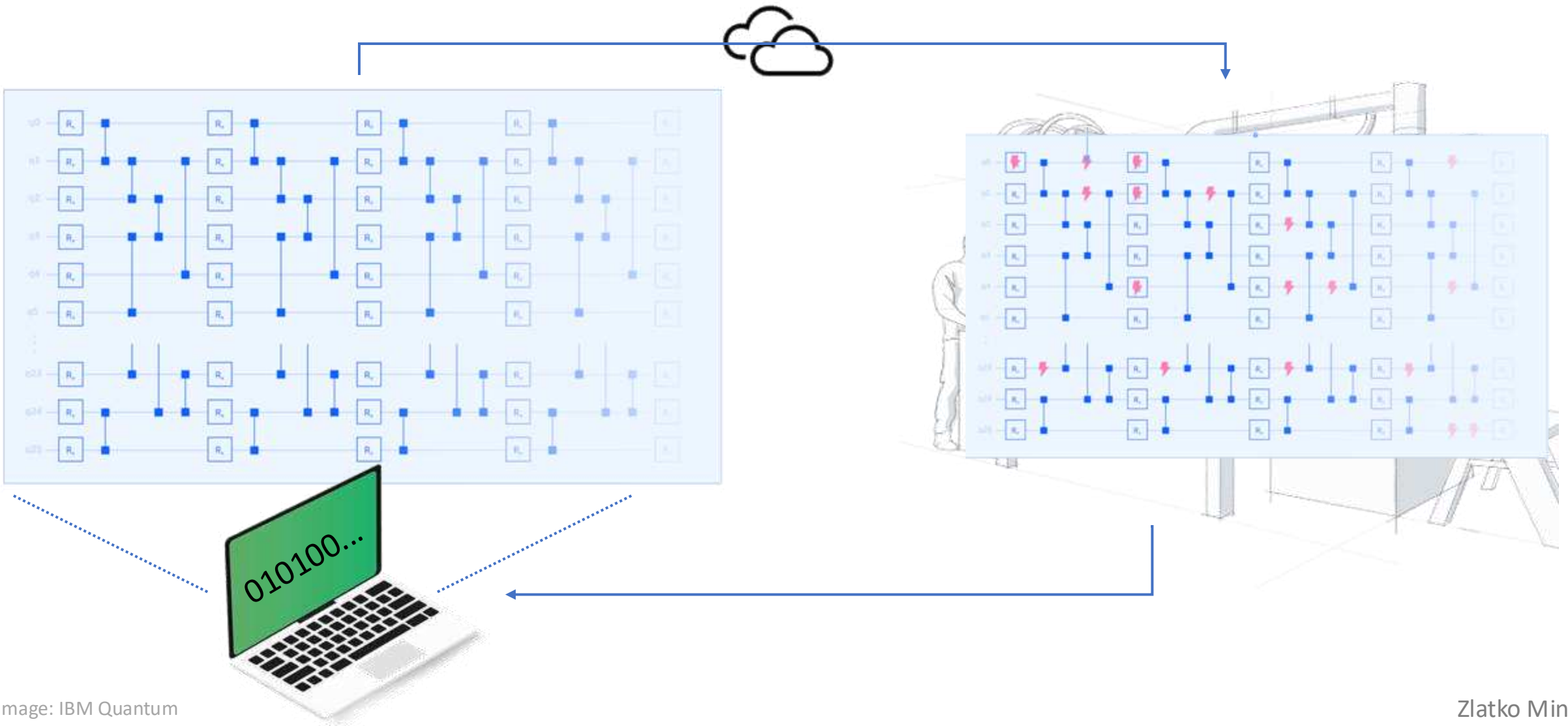
Asher Peres

Elements of 🤖 noise



Quantum simulation on a noisy quantum computer

Execute on a real quantum computer device and obtain results as classical data



Biggest challenge?

Please do share

hardware
development

decoherence

loss

stability

heat

algo
development

error correction
overheads

high error rates

scalability

Noise
(Errors)

Biggest challenge?

engineering

modularization

importance of N
in NISQ

hype

need CS/EE
talent

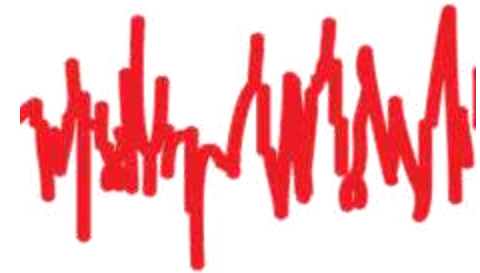
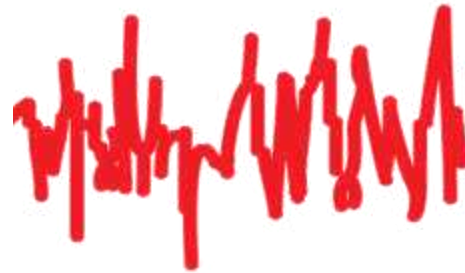
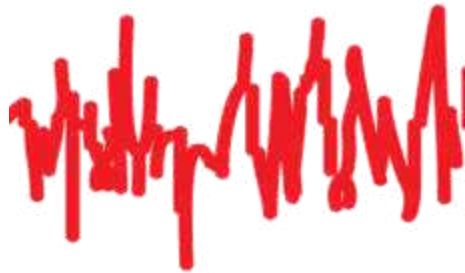
material
quality

gravity

expectations

Biggest challenge

Noise
(Errors)



How to deal with errors due to noise?

Monitor

Error occurs
Error detect

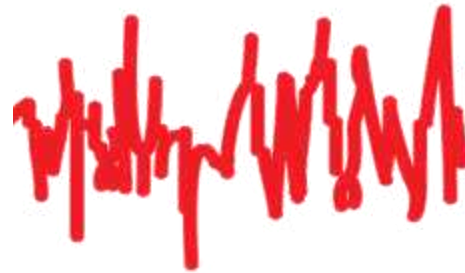


Quantum error correction

Shor, PRA (1995), ...

Monitor

Error anticipated
Tell signal detected

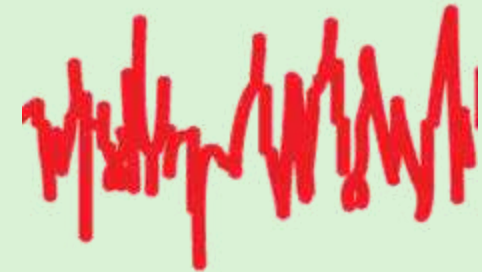


Catch and reverse

Minev, Nature (2019), ...

No monitor

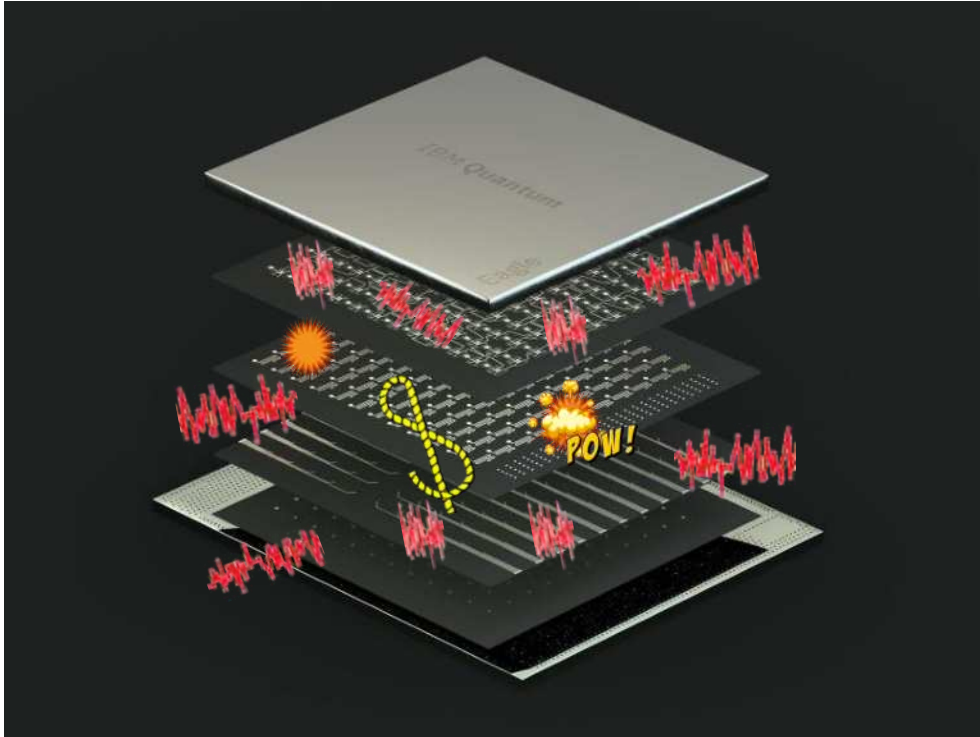
Error occurs
Error undetected



Error mitigation

... subject of today

Error mitigation and error correction



Error mitigation: working with what you have

- **benefit** suppress errors on classical results (expectation values)
- **q-cost** no extra qubits or hardware resources needed
- **c-cost** trades classical resources (post-processing) for lower error
- **limitation** bad asymptotic scaling: high number of samples & circuits

Error correction: protecting quantum information

- **benefit** suppress & correct errors to arbitrarily small level
- **q-cost** very large qubit and hardware overhead
- **c-cost** decoding and encoding can be classically costly
- **challenge** requires fault-tolerant operations and readout

Cancel quantum noise



High-level message

Learn

accurate, efficient, scalable



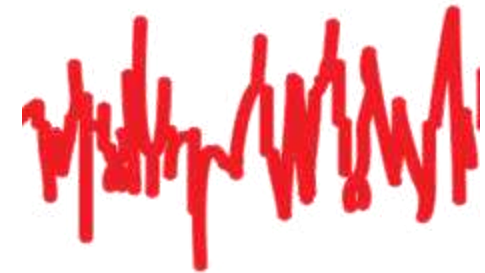
Cancel

noise with noise,
practical



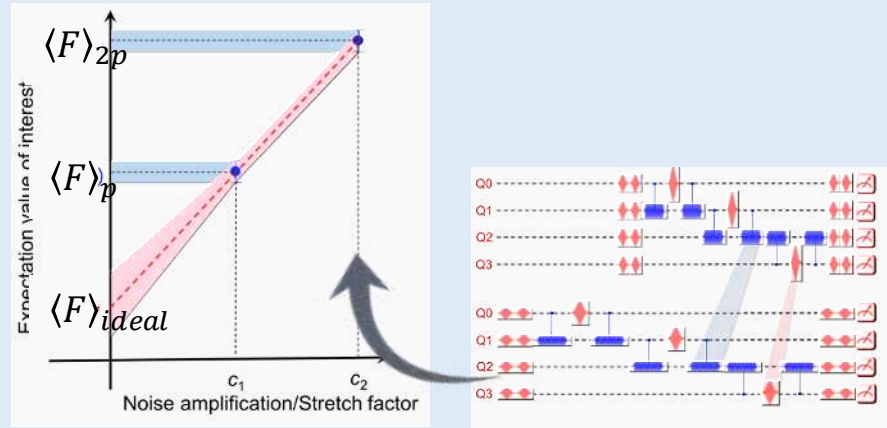
Cost

more noise more cost



Error mitigation landscape

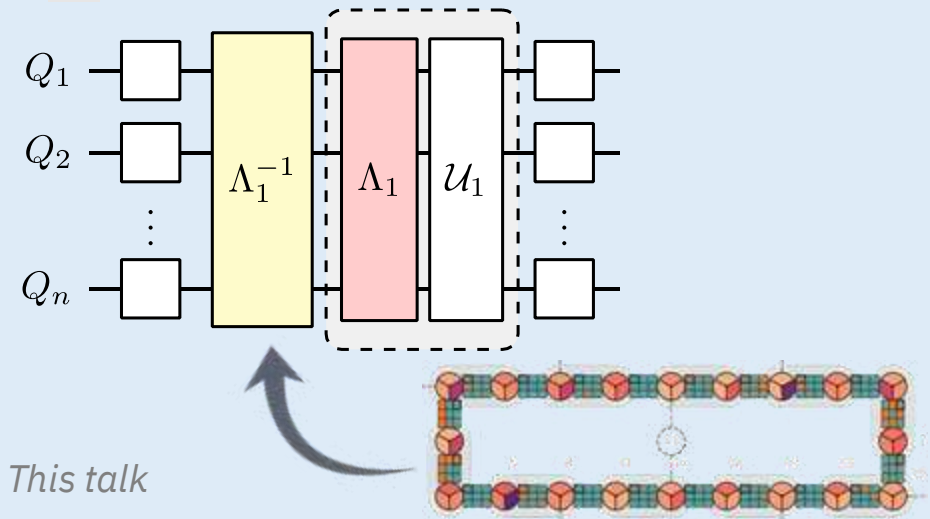
Zero-noise extrapolation (ZNE)



Nature 567, 491 (2019)

77

Probabilistic error cancelation (PEC)



more
speed

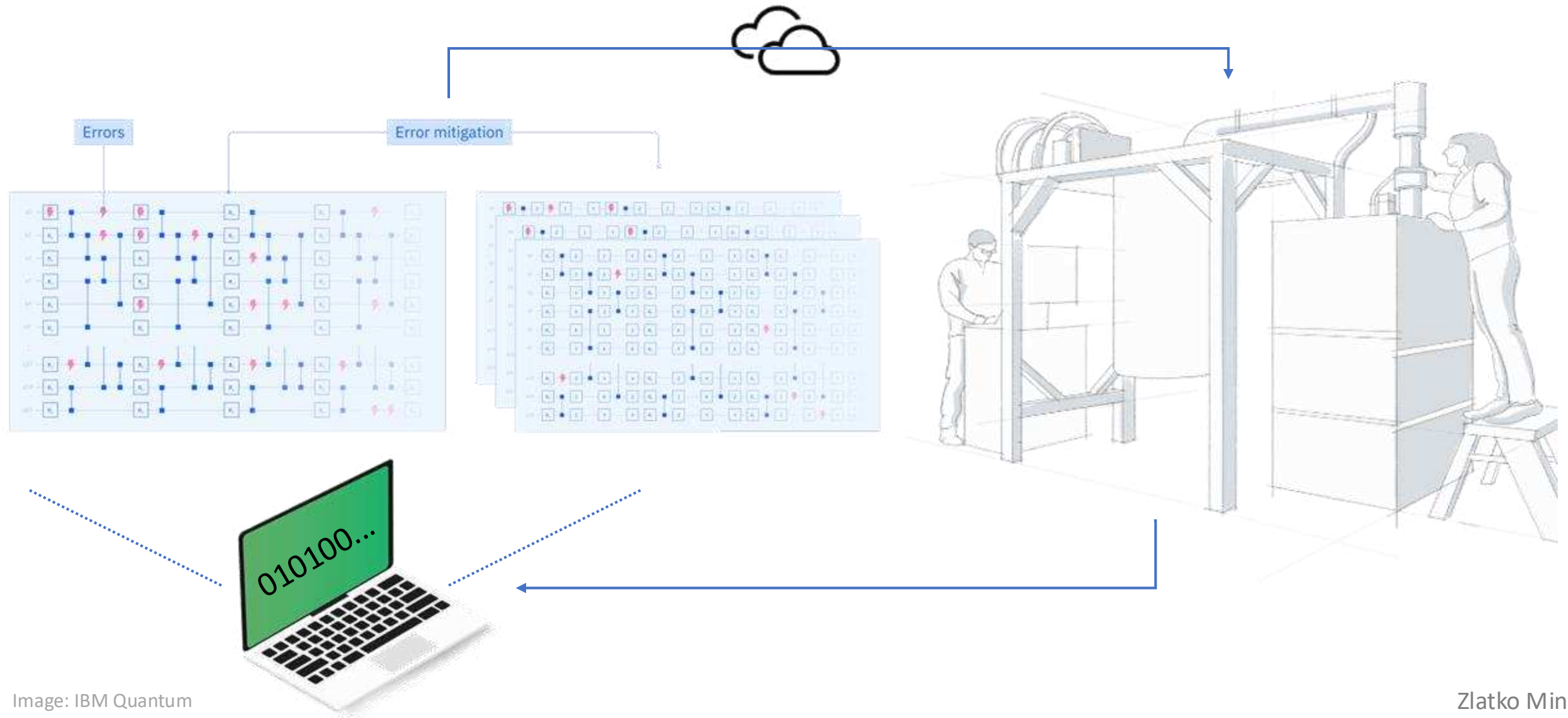


more information,
accuracy



Quantum error mitigation overview

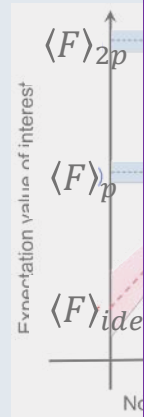
Execute on a real quantum computer device and obtain results as classical data



Error mitigation landscape

Zero-noise extrapolation (ZNE)

Probabilistic error cancelation (PEC)



 **Qiskit** **IBM Qiskit**

OverviewLearnCommunity

English

Get started

Overview

Getting Started

backend.run vs. Qiskit Runtime

Introduction to primitives

Tutorials

Get started with Estimator

Get started with error suppression and error mitigation

VQE with Estimator

CHSH with Estimator

Get started with Sampler

Qiskit Runtime IBM Client documentation > Error suppression and error mitigation with Qiskit Runtime

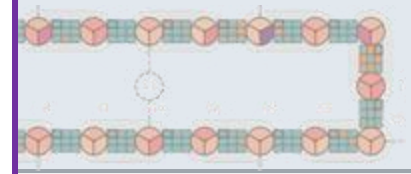
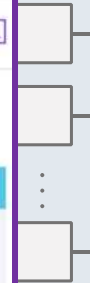
NOTE

This page was generated from [docs/tutorials/Error-Suppression-and-Error-Mitigation.ipynb](#).

Error suppression and error mitigation with Qiskit Runtime

```
[1]: import datetime
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt
plt.rcParams.update({"text.usetex": True})
plt.rcParams["figure.figsize"] = (6,4)
mpl.rcParams["figure.dpi"] = 200

from qiskit_ibm_runtime import Estimator, Session, QiskitRuntimeService,
Options
from qiskit.quantum_info import SparsePauliOp
from qiskit import QuantumCircuit
```

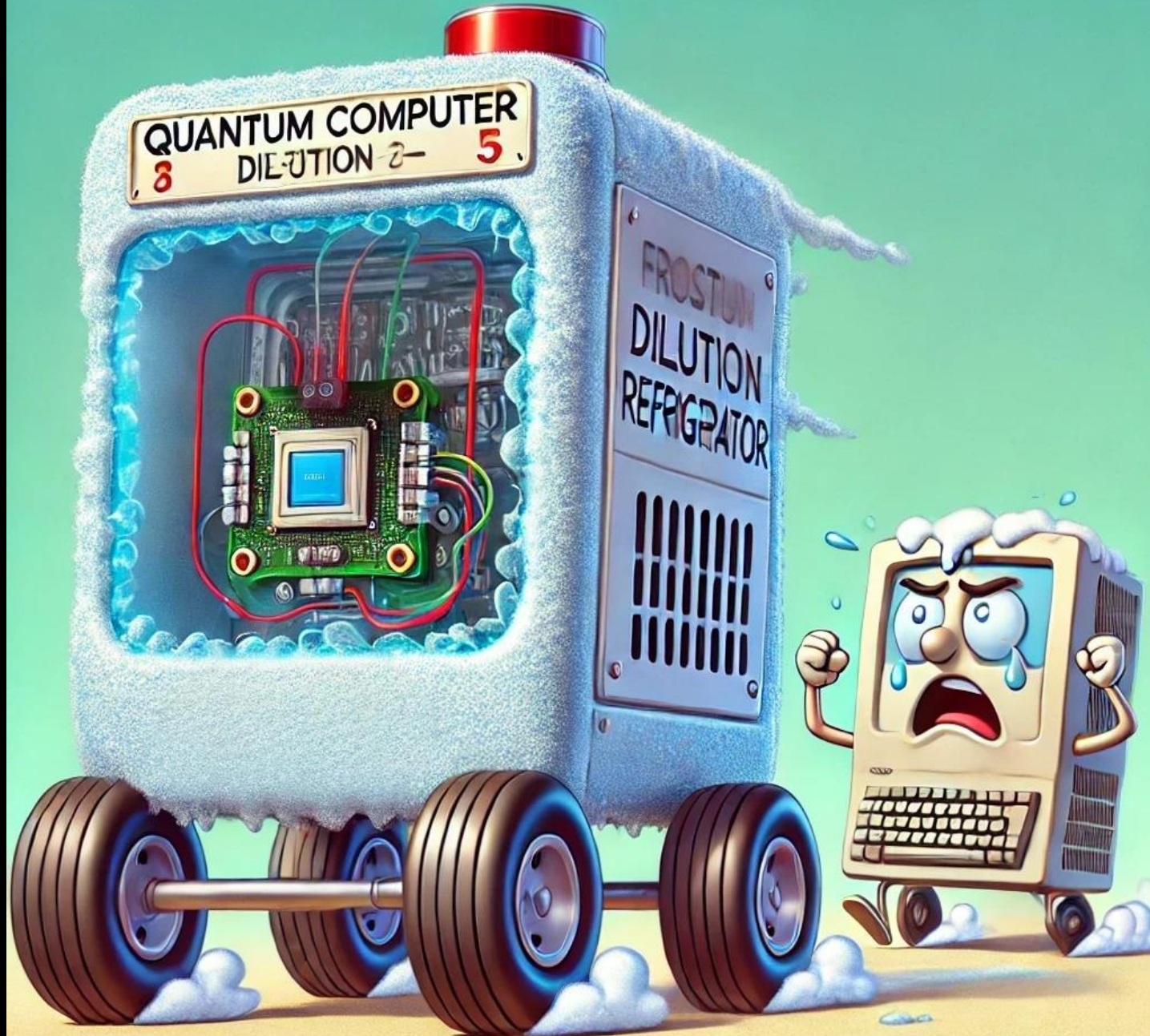


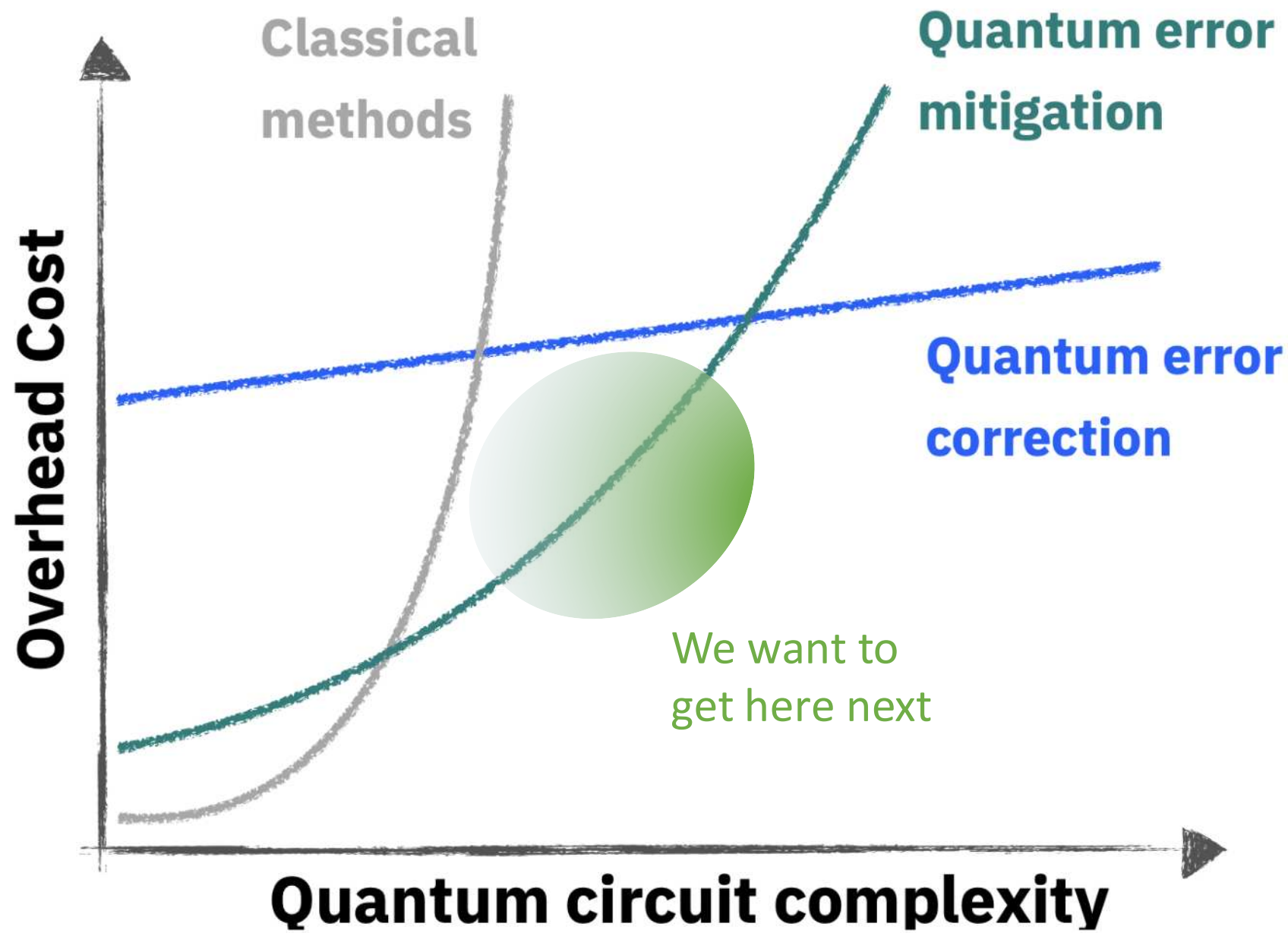
more
speed

more information,
accuracy



Quantum advantage





The important thing is not to stop questioning. Curiosity has its own reason for existence.

One cannot help but be in awe when they contemplate the mysteries of eternity, of life, of the marvelous structure of reality.

It is enough if one tries merely to comprehend a little of this mystery each day.

Albert Einstein

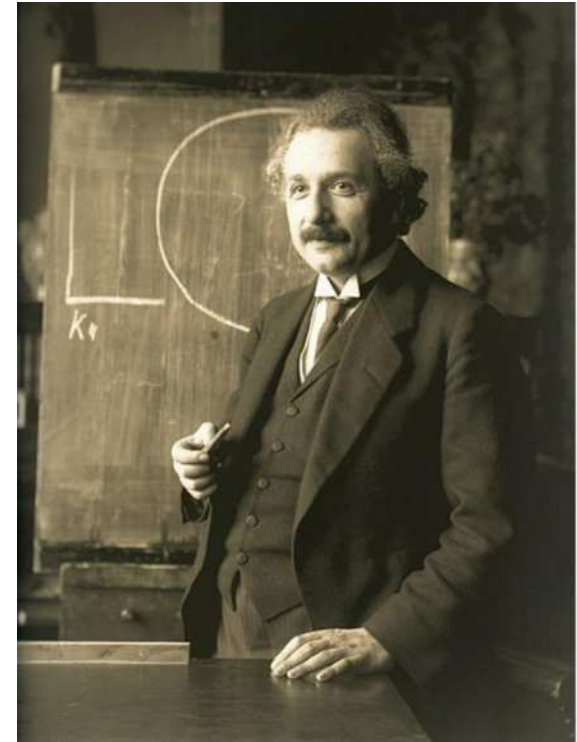


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